CAPSTONE PROJECT 2

Final Report

Machine Learning-Based Price Trend Prediction and Automated Trading System for Gold

by

LIM JIA XUAN

22110787

Bachelor of Computer Science

Supervisor: Ms Foo Jinny

Semester: September 2024

Date: 19 December, 2025

School of Computing and Artificial Intelligence

Faculty of Engineering and Technology

Sunway University

Abstract

Predicting short-horizon gold price movements is challenging due to market non-stationarity, macroeconomic sensitivity, and frequent regime shifts, which limit the effectiveness of traditional single-model approaches. This project proposes a Machine Learning-Based gold price trend prediction and automated trading system designed to generate reliable, risk-aware trading signals rather than maximize raw predictive accuracy.

The system follows the CRISP-DM methodology and adopts a precision-oriented ensemble learning strategy. A soft-voting ensemble consisting of XGBoost (38.2%), Logistic Regression (31.3%), and Gradient Boosting (30.5%) is trained using strict time-series cross-validation, probability calibration, and threshold optimization. Under a conservative optimized decision threshold of 0.57, the ensemble achieves a precision of 55.3%, recall of 61.4%, and an F1-score of 0.582, demonstrating a consistent statistical edge while remaining robust across varying market regimes. Real-time macroeconomic awareness is incorporated through FinBERT-based financial news sentiment, which dynamically adjusts trade entry thresholds without distorting learned price dynamics. To ensure operational transparency, the system is integrated with a real-time dashboard that visualizes AI confidence, sentiment context, and live portfolio states. Model outputs are executed through a fully automated trading engine featuring ATR-based risk management and a entry logic.

On unseen out-of-sample data, the system delivers strong trading performance, achieving a Sharpe Ratio of 3.05, a win rate of 58%, and a maximum drawdown of only −3.45%. These results validate a robust, explainable, and deployable Machine Learning framework for gold markets.

Contents

Abstract 2
Contents 3
List of Figures 8
List of Tables 10
1.0 Introduction 12
1.1 Background and Motivation 12
1.2 Problem Statement 14
1.3 Computing Challenges 14
1.4 Project Objectives 15
1.5 Scope of Work 16
1.6 Limitation 17
1.7 Research Significance 17
1.8 Summary of Methodology 18
1.9 Expected Outcomes 19
1.10 Project Timeline 19
2.0 Literature Review 22
2.1 Overview of Financial Market Gold Prediction 22
2.1.1 Forecasting in Gold Markets 22
2.1.2 Challenges Specific to Gold Forecasting 22
2.1.3 Key Drivers Emphasized for Predictive Modeling 22
2.1.3.1 Macroeconomic Indicators (Inflation and Real Interest Rates) 22
2.1.3.2 Energy Prices (Crude Oil) 22
2.1.3.3 Financial Markets and Geopolitical Sentiment 23
2.2 Machine Learning Models 23
2.2.1 Traditional Econometric Models 23
2.2.2 Rise of Machine Learning Approaches 24
2.2.3 Machine Learning Models in Financial Forecasting 24
2.2.3.1 Support Vector Machines (SVM) 25
2.2.3.2 Long Short-Term Memory (LSTM) 26
2.2.3.3 XGBoost (Extreme Gradient Boosting) 26
2.2.3.4 Comparative Analysis of ML Model Performance in Previous Studies 28

2.2.4 Model Training and Optimization Techniques........31
2.2.4.1 Train-Test Split and Cross-Validation........31
2.2.4.2 Hyperparameter Tuning........32
2.2.4.3 Regularization and Overfitting Prevention........32
2.2.4.4 Feature Scaling and Normalization........33
2.2.4.5 Evaluation During Training........33
2.3 Comparison of Machine Learning-Based Trading Platforms........34
2.3.1 Alpaca........34
2.3.2 QuantConnect........34
2.3.3 Identified Limitations of Existing Platforms........35
2.4 Technical Indicators as Features........35
2.5 Sentiment Analysis & News Integration........37
2.5.1 Sources of News Data for Sentiment Analysis........37
2.5.2 Advances in NLP Techniques for Financial Sentiment........38
2.5.3 Feature Engineering and Model Integration........38
2.6 Evaluation Metrics........39
2.6.1 Regression Metrics for Price Forecasting........39
2.6.2 Classification Metrics for Directional Prediction........39
2.6.3 Financial Metrics for Trading Strategy Evaluation........40
2.7 Trading Signal Generation and Strategy Design........41
2.7.1 Threshold-Based Trading Rules........41
2.7.2 Technical-Indicator Confirmation........41
2.7.3 Volatility-Sensitive Risk Controls........41
2.8 Limitations of Existing Studies and Project Gap Statement........42
2.9 Conclusion........42
3.0 Methodology........43
3.1 CRISP-DM Framework........43
3.2 Business Understanding........44
3.3 Data Understanding........45
3.3.1 Hourly Price Data (Yahoo Finance)........45
3.3.2 Real-Time News Headlines........46
3.3.3 Sentiment Scores and Market Regimes........46
3.3.4 Data Challenges........46

3.4 Data Preparation...47
3.4.1 Time Alignment and Cleaning 47
3.4.2 Feature Engineering...47
3.4.3 Target Variable Construction 49
3.4.4 Feature Cleaning and Dimensional Control 49
3.4.5 Data Splitting...49
3.4.6 Dynamic Sentiment Adjustment...50
3.5 Modelling 50
3.5.1 Base Models 51
3.5.2 Ensemble Construction..52
3.5.3 Probability Calibration 52
3.5.4 Threshold Optimization...52
3.6 Evaluation...53
3.6.1 Machine Learning Performance Evaluation 53
3.6.2 Trading Strategy and Backtesting Evaluation 54
3.6.3 Threshold-Based Decision Evaluation 54
3.6.4 Explainability and Transparency Evaluation..55
3.7 System Deployment 56
3.7.1 Microservices-Based Architecture 56
3.7.2 Performance and Reliability Considerations 57
3.7.3 Security and Ethical Considerations..57
3.7.4 Usability and Human-Centered Design...58
3.8 Risk Management and Ethical Considerations...58
3.8.1 Model Risk, Overfitting, and False Signals...58
3.8.2 Financial Risk and Simulation Constraints 59
3.8.3 Ethical Design and Human Oversight 59
3.8.4 Governance, Transparency, and Auditability 59
4.0 Work Plan & TimeLine 61
4.1 Activities & Milestones..61
4.1.1 Capstone 1 – Research, Design, and Planning 61
4.1.2 Capstone 2 – System Development & Evaluation..62
4.2 Proposed Timeline..64
4.2.1 Capstone 1 Timeline...64

4.2.2 Capstone 2 Timeline 65
4.3 Work Plans 66
4.3.1 Capstone 1 66
4.3.2 Capstone 2 67
5.0 Result and Discussion 68
5.1 Overview of System Outcomes 68
5.2 AI Model Development and Performance Results 69
5.2.1 Overview of Predictive Modeling Approach 69
5.2.2 Experimental Design and Data Architecture 69
5.2.3 Individual Model Evaluation and Selection 70
5.2.4 Soft Voting Mechanism and Weighting Strategy 71
5.2.5 Integrated Ensemble Performance 73
5.2.6 Feature Importance 74
5.2.6.1 Analysis of Key Predictive Drivers 74
5.2.6.2 Critical Interpretation & Methodological Considerations 76
5.3 Real-Time News Sentiment Integration Results 76
5.3.1 Role of News Sentiment in Gold Markets 76
5.3.2 Microservice Architecture & Implementation 76
5.3.3 Sentiment Extraction Using FinBERT 77
5.4 Trading Strategy Execution and Risk Control 80
5.4.1 Strategy Overview 80
5.4.2 Dynamic Position Sizing & Exit Management 80
5.5 Strategy Backtesting and Performance Evaluation 81
5.5.1 Simulation Setup (Out-of-Sample Testing) 81
5.5.2 Quantitative Performance Metrics 81
5.5.3 Equity Curve Analysis 83
5.5.4 Trade Logging & System Auditability 83
5.6 Implementation 84
5.6.1 Gold ML Auto-Trader Dashboard 84
5.6.1.1 KPI Dashboard 84
5.6.1.2 Live Trade Monitor 84
5.6.1.3 Market Analysis Chart 84
5.6.2 Automated Trade Execution Log 85

5.6.3 Real-Time News Intelligence Interface 86
5.6.4 AI Decision Core & Confidence Gauge 86
5.6.5 Raw Data Inspector 87
5.6.6 Model Validation & Performance Dashboard 87
5.6.7 Back Testing Results Panel 88
5.7 Functional Testing 89
5.8 System Stability 92
6.0 Evaluation 94
6.1 User Evaluation Session and System Walkthrough 94
6.2 Evaluation Questionnaire 94
6.3 Demographic 96
6.3.1 System Usability & Interface (UI/UX) 96
6.3.2 Functionality & Robustness 98
6.3.3 AI Model & Strategy Performance 100
6.3.4 Interpretability & Trust (Explainable AI) 102
6.3.5 Overall Impact & Value 104
6.3.6 Open-Ended Feedback 106
6.3.7 Discussion 109
7.0 Future Improvements 110
7.1 Enhanced Market Regime & Event Awareness 110
7.2 Domain-Specific Sentiment Refinement 110
7.3 Infrastructure for Production Reliability 110
7.4 Advanced Sequential Modeling 110
7.5 Towards Live Execution & Realism 111
8.0 Conclusion 112
8.1 Summary of Achievements 112
8.2 Evaluation Against Original Objectives 112
8.3 Limitations 112
8.4 Final Remarks 112
9.0 References 114
APPENDIX A: User Evaluation 121
APPENDIX B: Git Hub Link 127

List of Figures

Figure 1: Trend in Central Bank Gold Holdings 12
Figure 2 : Gold Price History 13
Figure 3 : Covid-19 Case No and Gold Prices Change 13
Figure 4 : SVM Formula 25
Figure 5 : Kernel Function 25
Figure 6 : Alpaca 34
Figure 7 : QuantConnect 34
Figure 8 : CRISP-DM Methodology 43
Figure 9 : Data Splitting 49
Figure 10 : Ensemble Architecture 50
Figure 11 : System Architecture 56
Figure 12 : Base Model Code 70
Figure 13 : Model Performance Battle Chart 71
Figure 14 : Smart Selection Logic Code 71
Figure 15 : Ensemble Composition Chart 72
Figure 16 : Load FinBert Code 77
Figure 17 : Latest News Headlines 78
Figure 18 : Threshold Adjustment Flowchart 78
Figure 19 : Strategy Flowchart 80
Figure 20 : Backtest Equity Curve 83
Figure 21 : Trade Logging 83
Figure 22 : Dashboard Interface 84
Figure 23 : Trade Log Tab Interface 85
Figure 24 : Live News Tab Interface 86
Figure 25 : AI Analysis Tab Interface 86
Figure 26 : Raw Data Tab Interface 87
Figure 27 : Model Training Insights Interface 87
Figure 28 : Model Training Insight Interface 2 87
Figure 29 : Back Testing Results Interface 88
Figure 30 : TC-01 (Minute Filter Logic) 89
Figure 31 : TC-02 (Weekend Handling Data) 89
Figure 32 : TC-02 (Weekend Handling Candle) 90

Figure 33 : TC-03 (Negative Sentiment) 90
Figure 34 : TC-04 (Positive Sentiment) 90
Figure 35 : TC-05 (Dynamic Threshold) 91
Figure 36 : TC-06 (Stop Loss Execution) 91
Figure 37 : TC-07 (Take Profit Execution) 91
Figure 38 : TC-11 (Process Isolation) 92
Figure 39 : TC-12 (Network Failure) 93
Figure 40 : User Evaluation 94
Figure 41 : Dashboard layout ratings 96
Figure 42 : Real-time price charts readability 96
Figure 43 : AI confidence gauge clarity 97
Figure 44 : Market closed/data updating feedback 97
Figure 45 : Overall UI effectiveness 98
Figure 46 : Data automation reliability 98
Figure 47 : News service relevance 99
Figure 48 : Handling weekends/holidays 99
Figure 49 : Simulation stability 100
Figure 50 : Strategy appropriateness 100
Figure 51 : Dynamic threshold logic 101
Figure 52 : FinBERT sentiment accuracy 101
Figure 53 : Backtest outputs credibility 102
Figure 54 : Factor analysis interpretability 102
Figure 55 : Feature importance clarity 103
Figure 56 : System transparency 103
Figure 57 : Trust in system signals 104
Figure 58 : Understanding ML in trading 104
Figure 59 : Value of unstructured news data 105
Figure 60 : End-to-end system implementation 105
Figure 61 : Overall system satisfaction 106
Figure 62 : Most impressive feature 106
Figure 63 : Bugs or confusing elements 107
Figure 64 : Suggested feature enhancements 108

List of Tables

Table 1 : Expected Deliverables of Project......19
Table 2 : Summary of CP1 Timeline......20
Table 3 : Summary of CP2 Timeline......21
Table 4 : ML Model Type......24
Table 5 : Difference Between ML Models......28
Table 6 : Performance of Different ML Model 1......29
Table 7 : Performance of Different ML Model 2......30
Table 8 : Performance of Different ML Model 3......31
Table 9 : Technical Indicator Comparison......37
Table 10 : Evaluation Metrics......40
Table 11 : Key Objectives and Guiding Principles......44
Table 12 : Price Data Comparison......46
Table 13 : Base Models Description......52
Table 14 : Model Performance Metric and Purpose......53
Table 15 : Backtest Evaluation Metrics and Objectives......54
Table 16 : Evaluation of explainability and transparency features......55
Table 17 : Trading system modules and their responsibilities......56
Table 18 : Activities & Milestones (CP1)......62
Table 19 : Activities & Milestones (CP2)......63
Table 20 : Proposed Timeline (CP1)......64
Table 21 : Proposed Timeline (CP2)......65
Table 22 : Work Plans (CP1)......66
Table 23 : Work Plans (CP2......67
Table 24 : Data Partitioning and Task Specification......70
Table 25 : Base Models Performance......71
Table 26 : Selected Models Evaluation......73
Table 27 : Ensemble Model Performance......74
Table 28 : Feature Importance......75
Table 29 : News Service Component Description......77
Table 30 : Impact of sentiment scores......79
Table 31 : Backtesting Strategy Performance Metrics......82
Table 32 : Trade Log Column Explanation......85

Table 33 : Functional Test Case..91
Table 34 : System Stability Test Case 93
Table 35 : Survey questions and labels used for user feedback evaluation 95

1.0 Introduction

1.1 Background and Motivation

Gold serves a dual role as both a chemically stable store of value and a critical hedge against economic uncertainty. As illustrated in Figure 1, emerging economies such as China, India, and Russia have significantly increased their gold reserves over recent decades, reflecting its enduring importance in global finance (Griffin, 2025).

Figure 1: Trend in Central Bank Gold Holdings

From a market perspective, gold has generally been seen as a safe haven currency that investors flock when the market is at its most volatile. Figure 2 illustrates that there has been consistent upward movement of global gold prices since 2015 up to 2025. This category has specific counter-cyclical characteristics, as the prices of this asset class sharply increase in case of great crises, like the 2008 financial disaster and the COVID-19 pandemic (Figure 3), which, in turn, strengthens its shield effect during the most severe uncertainty (Yousef and Shehadeh, 2020).

Figure 2 : Gold Price History

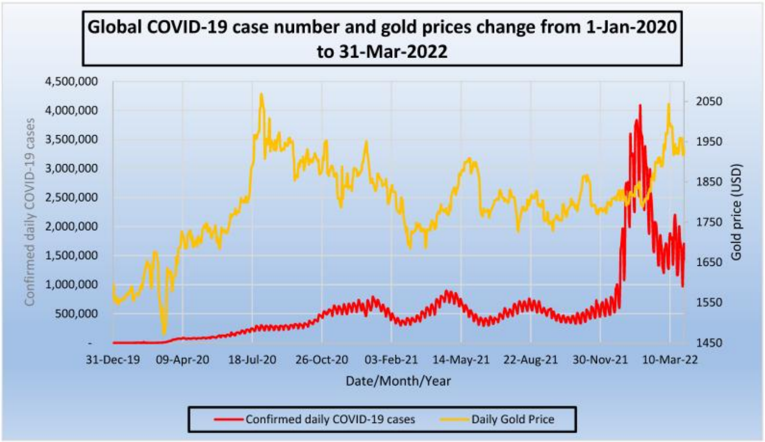


Figure 3 : Covid-19 Case No and Gold Prices Change

The prices of gold are affected by a complicated combination of macroeconomic factors, interest rates, and geopolitical activities (Sathyanarayana & Mohanasundaram, 2025). Conventional prediction methods tend to make linear assumptions that do not reflect the non-linear characteristics of this volatility (Ade, 2023), whereas contemporary research shows that machine learning models, namely the ensemble-based ones such as XGBoost, are far much better than conventional ones in turbulent markets (Sun and Deng, 2025).

Despite these computational advances, a critical gap persists: most existing systems operate as isolated prediction modules without integrated trading frameworks. This separation forces traders to manually interpret algorithmic signals and execute trades, introducing decision latency and increasing the risk of suboptimal decisions. This highlights a fundamental gap in end-to-end system design rather than purely predictive accuracy. This project addresses that integration challenge by developing a unified architecture that combines machine learning inference with automated trading logic within a simulated environment.

1.2 Problem Statement

The sensitivity of gold to many different related factors, therefore, makes forecasting the price of gold very hard. Most inaccurate forecasts are mainly caused by a lack of ability of some of the traditional forecasting tools to adjust to these fast changing variables. In addition, majority of the existing forecasting systems are not integrated to automated trading processes and only give predicted results. Because of this automation deficiency, users have to crack codes and trade manually, introducing some latency and making this potentially expensive in emergency situations.

Moreover, the structured numerical information is the one that is usually used in these systems, overlooking valuable unstructured information, which usually comes before changes in the market, which includes financial news, economic perspectives, and political utterances. There is some approach found in some of the recent research papers in this industry.

As a case in point, deep learning-time series prediction models with NLP-driven sentiment and event detectors can be used as hybrids to reveal concealed trends of structured and unstructured data sources (Gunasekaran, 2023). Nonetheless, not many systems successfully combine these functions in the form of real-time, automatized solution.

To bridge this gap, this project proposes the development of a comprehensive machine learning driven system that integrates real-time data ingestion, NLP-based news analysis, and autonomous trade execution within a simulation environment.

1.3 Computing Challenges

Gold price prediction and automated trading has a number of non-trivial challenges when it comes to computing. Financial time-series data are extremely noisy, non-stationary and non-linear in their relationship, which makes the use of standard statistical modeling inadequate. Moreover, the system should be able to combine heterogeneous data streams, such as structured numerical indicators and unstructured textual news data, which need other preprocessing and representation methods.

The other important issue is how to convert probabilistic machine learning results to solid real-time trading. This demands low-latency inference, strong decision thresholds, and algorithmic risk controls in order to avoid cascading mistakes. These issues require the use of more sophisticated principles of machine learning, feature engineering, and system-level design as opposed to single prediction models.

1.4 Project Objectives

There are few objectives for this project.

Objective 1: To develop a machine learning framework for predicting gold price movement

The first objective is to develop a robust machine learning framework for predicting short-term gold price direction rather than precise price values. The framework primarily employs tree-based models such as Extreme Gradient Boosting (XGBoost) and ensemble classification techniques, which are well-suited for capturing non-linear relationships and complex interactions among technical, macroeconomic, and cross-asset features in financial time-series data. Model performance is evaluated using classification-oriented metrics, including accuracy, precision, recall, and F1-score, to ensure balanced predictive capability under noisy market conditions.

Objective 2: To integrate financial news sentiment into the predictive and decision-making process

The second objective is to enhance predictive reliability by incorporating sentiment signals extracted from financial news. Natural language processing (NLP) techniques are used to transform unstructured news headlines into quantitative sentiment features that reflect market mood and event-driven risk. These sentiment signals are combined with technical and macroeconomic indicators to improve signal confidence and to reduce exposure during periods of heightened uncertainty.

Objective 3: To design and evaluate a risk-aware automated trading simulation based on model-driven signals

The third objective is to translate model predictions into actionable trading decisions through an automated trading strategy. The system generates Buy, Sell, or Hold signals using probability-based thresholds and predefined trading rules. A simulated trading environment is implemented to evaluate real-world feasibility, incorporating conservative risk management mechanisms such as position sizing limits, volatility-based stop-loss and take-profit levels, and strict drawdown control. The performance of the trading system is assessed using risk-adjusted financial metrics including total return, Sharpe Ratio, win rate, and maximum drawdown.

1.5 Scope of Work

- **Data Collection & Preprocessing**
  Collect structured market data (gold prices, macroeconomic indicators, cross-asset correlations) and unstructured financial news. The data will undergo cleaning, feature engineering, normalization, and temporal alignment to create a consistent time-series dataset suitable for machine learning (Omoseebi et al., 2025).
- **Model Development**
  Develop and compare multiple machine learning classifiers for predicting short-term gold price direction. Models will be evaluated using classification-specific metrics including accuracy, precision, recall, and F1-score to ensure balanced predictive performance (Srivastava & Sikroria, 2024).
- **Sentiment Analysis Integration**
  Implement natural language processing techniques to extract quantitative sentiment signals from financial news headlines. These sentiment features will be integrated with technical indicators to enhance prediction reliability during market-moving events.
- **Trading Strategy Design**
  Design rule-based trading logic that translates model predictions into actionable buy/sell/hold signals. The strategy will incorporate risk management components including position sizing limits, volatility-adjusted stop-loss, and take-profit mechanisms.
- **Automated Trading Simulation**
  Implement a simulation engine that executes trading decisions in a virtual environment, accounting for realistic factors such as execution delays and transaction costs. The system operates exclusively in simulation mode without connection to live trading platforms.
- **System Evaluation**
  Conduct comprehensive backtesting to evaluate both predictive accuracy and financial performance. Evaluation will include machine learning metrics (precision, recall, F1-score) and financial metrics (total return, Sharpe ratio, maximum drawdown, win rate) to assess practical trading viability.

1.6 Limitation

The work undertaken in this project is aimed at designing and testing an automated gold trading system under a simulated environment. Every trading is done using both real-time and past market data without any real financial transactions and thus zero risk of financial transactions.

The system is based on the historical price trends and sentiment derived out of the financial news headlines; hence its performance can suffer in unprecedented market circumstances like extreme geopolitical shock or change of regime structure. Also, sentiment analysis is restricted to the semantic interpretation of headlines, and does not involve the complete reasoning based on the article.

Although the framework is intended to have an extensible nature, in this study, only gold markets are covered: multi-asset portfolio optimization and integration with live brokers are not included.

1.7 Research Significance

Sustainability and Green AI

The project also focuses on computational efficiency, with lightweight ensemble models (XGBoost) being used instead of deep learning architectures, which are both resource-intensive. This green AI strategy greatly reduces the amount of energy that models need to be trained and inferred, and shows sustainable computing does not need to affect financial performance.

Ethical Safety and Risk Control

The system is under strict simulation environment to guarantee there is zero risk on the financial side. The execution logic contains hard-coded ethical safeguards, such as volatility-based stop-loss and hard position limits. This makes the system appear as a transparent decision support tool, as opposed to an uncontrollable autonomous agent.

Contribution to Computing

This project contributes to the existing knowledge of the area as it overcomes the problem of Inference-Execution Gap in financial computing. Whereas current literature is more concerned with the accuracy of prediction, this work offers a complete system architecture that helps to bridge the gap between the probabilistic forecasting and the automated execution. It provides a paradigm of processing non stationary, heterogeneous streams of data with low-latency environments.

Societal Impact

At the societal level, the project will be used to democratize financial intelligence. The system contributes to the reduction of the gap in information between institutional hedge funds and individual investors by automating the process of analyzing intricate geopolitical news and market trends. This also allows the user to make an objective and data-driven decision that reduces the emotional bias, which causes individuals to lose money personally.

1.8 Summary of Methodology

This project utilizes the CRISP-DM framework to structure the iterative development of the automated trading system (Schröer et al., 2021). The methodology consists of six phases:

- **Business Understanding**: Defines the project objective as building a trading system that prioritizes high-confidence, risk-controlled execution over high-frequency trading.
- **Data Understanding**: Aggregates heterogeneous data, including hourly market data (Gold, DXY) via Yahoo Finance and geopolitical news via RSS feeds, utilizing FinBERT for sentiment extraction.
- **Data Preparation**: Enforces strict temporal alignment to prevent look-ahead bias and constructs features such as cyclical time encodings and technical indicators (e.g., RSI, MACD).
- **Modeling**: Implements a probabilistic ensemble that outputs confidence scores for dynamic thresholding rather than simple binary signals (Vuong et al., 2021).
- **Evaluation**: Adopts a dual-layer strategy: TimeSeriesSplit cross-validation for ML performance (Precision, F1) and historical backtesting for financial viability (Sharpe Ratio, Max Drawdown) (Yousaf et al., 2025).
- **Deployment**: Utilizes a modular microservice architecture with file-based communication to link the data, inference, and dashboard components, ensuring crash isolation in a real-time simulation.

1.9 Expected Outcomes

Key deliverables of the project are outlined in Table 1.

Deliverable	Description
Predictive ML Model	An ensemble-based classification model (e.g., XGBoost) optimized to forecast short-term gold price direction with high precision.
NLP-Based Sentiment Module	A natural language processing pipeline that extracts quantitative sentiment scores from financial news headlines to augment price signals.
Automated Trading Simulation	A logic-based execution engine that converts model predictions into Buy, Sell, or Hold signals within a risk-controlled simulation environment.
Performance Evaluation Report	A comprehensive analysis of system performance using financial metrics (Sharpe Ratio, Max Drawdown, ROI) and statistical metrics (Accuracy, F1-Score).
Visualization Dashboard	An interactive interface displaying real-time price data, model confidence levels, sentiment trends, and simulated trade logs.
Modular Codebase	A documented, modular Python repository designed for reproducibility and potential adaptation to other financial assets.

Table 1 : Expected Deliverables of Project

1.10 Project Timeline

As shown in Table 2, the Capstone 1 timeline follows a clear flow, from planning to submission. Firstly ,it starts with choosing a project topic and getting the supervisor's approval. After that, a comprehensive literature review help me to understand the project idea clearly. As the weeks go by, key sections like the introduction, literature review, and methodology are written and refined based on supervisor suggestions. The CRISP-DM framework guides the design of the methodology. Along the way, activity logs and annotated bibliographies are kept up to date to track progress. In the final phase, the report is reviewed with the supervisor and submit.

Week Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Contact Supervisor	■													
Initial Meeting & Proposal		■												
Literature Review			■	■	■	■								
Write Introduction							■	■						
Complete Literature Review									■	■				
Methodology Design											■	■		
Review & Finalize Report													■	■
Activity Log Maintenance		■	■	■	■	■	■	■	■	■	■	■	■	■
Bibliography + Annotation											■	■	■	■
Supervisor Review (PD + AL)													■	■
Final Submission														■

Table 2 : Summary of CP1 Timeline

Table 3 illustrates that Capstone 2 schedule will be designed in 14 weeks so that it provides a systematic progression of the automated trading system. The timeline will start with the transition of the preliminary requirement verification and data pipeline creation into the model training, system integration, and the creation of final dashboard. To keep the consistent tracking of progress and the feedback implementation, continuous activities, including activity logging, consultation with supervisors, etc., are kept during the semester.

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Requirement Review & Setup	■													
Data Pipeline Implementation		■	■											
Feature Engineering (NLP + Tech)				■	■									
Model Training (Ensemble)						■	■	■						
Trading Logic & Backtesting									■	■				
System Integration (Microservices)											■	■		
Dashboard UI Development													■	

Final Report & Video Demo														
Activity Log Maintenance														
Supervisor Meetings / Review														
Final Submission														

Table 3 : Summary of CP2 Timeline

2.0 Literature Review

2.1 Overview of Financial Market Gold Prediction

2.1.1 Forecasting in Gold Markets

In old generation, gold has proven to be a dependable safe-haven investment, especially economic crisis. Man and Kynor (2023) found out that the U.S. Dollar Index and real interest rates are two macro-financial indicators that can affect the short-term gold price. Therefore, all these factor is can enhance the performance of the prediction model.

2.1.2 Challenges Specific to Gold Forecasting

Unlike stocks, which the prices is depend on the profits or cash flow, gold prices are affected by a mix of different factors (Taneva-Angelova et al., 2025). This cause some traditional model like ARIMA will not perform well in this kind of situation. Taneva-Angelova et al. (2025) suggest implement hybrid models to improve forecasting accuracy by combining machine learning, and sentiment-based techniques.

This statement is supported by Bunnag's (2023) research, which found that a hybrid ARIMA(2,1,3)–GARCH(1,1) model performed better than a normal ARIMA models. By incorporating volatility patterns, the hybrid model achieved lower MAE and RMSE, indicating that a hybrid model can capture the sudden change more effectively.

2.1.3 Key Drivers Emphasized for Predictive Modeling

2.1.3.1 Macroeconomic Indicators (Inflation and Real Interest Rates)

Kaufmann and Winters (1989) found a strong positive relationship between U.S. inflation and gold prices, but later studies, like Elfakhani et al. (2009), showed that this link weakened in the 1990s, and in some cases, reversed. This suggests that the relationship between inflation and gold is not stable over time and depends on broader economic conditions. Jakobsson (2022) confirmed that inflation still affects gold prices today, but the strength and direction of this effect can vary. Therefore, models should treat inflation as a nonlinear and time-sensitive feature.

2.1.3.2 Energy Prices (Crude Oil)

Oil prices impact gold through cost-push inflation and market sentiment. Simakova (2011) found a long-term connection between oil and gold prices, while Kiohos & Sariannidis (2010) noted that oil price shocks increase gold market volatility. However, energy prices generally have an indirect influence and are secondary compared to macroeconomic and financial factors.

2.1.3.3 Financial Markets and Geopolitical Sentiment

The U.S. Dollar Index (USDX) is the most reliable inverse predictor of gold prices (He et al., 2020). A stronger dollar makes gold more expensive for international buyers, reducing demand and lowering gold prices. Stock market indices and bond yields also impact gold. When interest rates or equity returns rise, investors move towards riskier or income-generating assets, reducing gold's appeal.

Additionally, gold acts as a haven during times of high uncertainty. Using a quantile-on-quantile regression, Bouoiyour et al. (2018) discovered that gold responds more strongly to events of extreme uncertainty, such as financial crises or geopolitical tensions. Uncertainty has an asymmetric and nonlinear impact on gold. This implies that incorporating sentiment indices (such as the VIX or news-based uncertainty measures) as interactive or nonlinear features into machine learning models can enhance forecasts, particularly in times of emergency.

In conclusion, a complex combination of financial, sentiment, and macroeconomic factors influence the prediction of the price of gold. It is advised to use hybrid models that combine machine learning with conventional techniques and sentiment analysis because gold prices are non-stationary and nonlinear.

2.2 Machine Learning Models

2.2.1 Traditional Econometric Models

Traditional time-series models like GARCH and ARIMA were used in the early days of gold price forecasting. Although ARIMA does a good job of modelling linear trends, it is unable to manage abrupt changes or volatility clustering. Although GARCH considers time varying volatility, the underlying process is still assumed to be linear. To improve performance, hybrid models like ARIMA–GARCH were introduced. For instance, Setyowibowo et al. (2022) used ARIMA(1,1,1)–GARCH(2,1) on daily gold prices and achieved better accuracy (RMSE ≈ 2.38, MAE ≈ 1.70). Still, these models faltered during extreme events due to their rigidity. As a result, they're now often used as benchmarks rather than the primary forecasting tools.

2.2.2 Rise of Machine Learning Approaches

Machine learning (ML) has gained popularity in recent years for the ability to handle noisy, high-dimensional, and nonlinear data. Unlike traditional methods, ML models learn directly from past patterns with fewer assumptions about how the data is distributed to generate a more accurate output. Table 4 compares different model types, including traditional models, deep learning models and so on.

Model Type	**Examples**	**Strengths**	**Limitations**
Traditional Models	ARIMA, GARCH, ARIMA-GARCH	Easy to implement; maintain effective under stability	Poor performance during high volatility or nonlinear regimes
Ensemble ML Models	Random Forest, XGBoost	Handles feature interactions and robust to noise	Requires careful tuning; less interpretable
Deep Learning Models	LSTM, GRU	Able to capture long-term dependencies (time series)	Data-hungry; sensitive to overfitting
Kernel-Based Models	Support Vector Machines (SVM)	Good at modeling nonlinearity (limited data)	Less effective when large datasets or time-dependencies

Table 4 : ML Model Type

2.2.3 Machine Learning Models in Financial Forecasting

ML models are now widely used in financial forecasting due to their ability to identify complex patterns. This project focuses on three practical ML models: SVM, LSTM, and XGBoost.

2.2.3.1 Support Vector Machines (SVM)

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\mathbf{w}^T\mathbf{w} + C\sum_{i=1}^{l}\xi_i \\ \text{subject to} \quad & y_i(\mathbf{w}^T\phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0. \end{aligned}$$

Figure 4 : SVM Formula

SVM is a supervised learning algorithm typically used for classification tasks, aiming to find the optimal separating hyperplane between data classes (Hsu et al., 2003). As shown in figure 4, for gold price forecasting, SVM can be used to predict price direction (upward or downward movement), converting continuous market data into actionable binary signals.

- **Kernel functions**, like the Radial Basis Function (RBF), allow SVM to handle non-linear feature spaces, improving its ability to model complex relationships between technical indicators and price direction as shown in figure 5.

linear	$k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1 \cdot \mathbf{x}_2$
polynomial	$k(\mathbf{x}_1, \mathbf{x}_2) = (\gamma\, \mathbf{x}_1 \cdot \mathbf{x}_2 + c)^d$
Gaussian or radial basis	$k(\mathbf{x}_1, \mathbf{x}_2) = \exp\left(-\gamma \lVert \mathbf{x}_1 - \mathbf{x}_2 \rVert^2\right)$
sigmoid	$k(\mathbf{x}_1, \mathbf{x}_2) = \tanh(\gamma\, \mathbf{x}_1 \cdot \mathbf{x}_2 + c)$

Figure 5 : Kernel Function

- **Input features** typically include moving averages, RSI, MACD, and past returns.

However, SVM does not inherently capture sequential dependencies. Time-related features (e.g., previous prices) must be manually engineered. Therefore, SVM is suggested to use as a directional classifier, supplementing regression-based models to support trading decisions. However, Yang and Dou (2014) proposed an enhanced SVM using Empirical Mode Decomposition (EMD) and Particle Swarm Optimization (PSO), improving directional gold prediction accuracy. Also, a study by Pillai and Mary (2024) showed that Least-Squares SVM (LS-SVM) performs better than many traditional methods when predicting gold prices.

2.2.3.2 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks are a specialized type of Recurrent Neural Network (RNN) designed to analyze sequential data, such as time-series price movements. Unlike traditional neural networks that treat each data point independently, LSTM networks retain historical context through memory cells and gating mechanisms:

- The input gate controls how much new information flows into the memory cell.
- The forget gate decides what historical information should be discarded.
- The output gate determines what information to pass forward to the next time step.

This architecture allows LSTM to model long-term dependencies as well as short-term patterns, making it suitable for predicting financial time series that exhibit temporal correlations (Hochreiter & Schmidhuber, 1997).

In gold price forecasting, LSTM networks are suited for modeling short- to medium-term price trajectories, analyzing sequences of past prices, technical indicators, and macroeconomic signals. For example, Nagata et al. (2025) analyzed Malaysian gold prices using LSTM. They found that the daily model's root mean squared error (RMSE) was approximately 21.27, indicating their strong adaptability to price changes. Directional accuracy, though sometimes modest (e.g., Gong, 2024, reported ~50.67%), can be improved when LSTM is combined with structured data models.

However, LSTMs are data-intensive and prone to overfitting on small datasets and computationally slower compared to tree-based models. Therefore, LSTM is suggested as the core time-series regression model, forecasting near-future gold price trends to support trend-based automated trading strategies.

2.2.3.3 XGBoost (Extreme Gradient Boosting)

Extreme Gradient Boosting (XGBoost) is a high-performance, ensemble-based machine learning algorithm designed for structured (tabular) data. It builds predictive models using gradient boosting decision trees (GBDT), where:

- Trees are built sequentially, with each new tree focusing on correcting errors made by prior trees.
- The model optimizes a regularized objective function to reduce overfitting while improving generalization.
- Parallel processing and memory-efficient techniques allow for fast model training and scalability.

In financial forecasting, XGBoost excels at modeling complex non-linear interactions among various market indicators without requiring extensive data preprocessing. Moreover, XGBoost ingests structured features such as technical indicators (RSI, MACD, moving averages), macroeconomic indicators (inflation rate, USD index, interest rates), sentiment features derived from financial news in order to predict future price values (regression) or price movement direction (classification). For example, Li (2024) combined economic indicators and technical data in XGBoost to produce stable and accurate gold price forecasts, even in volatile conditions.

Meanwhile, Jabeur et al. (2021) paired XGBoost with SHAP (SHapley Additive exPlanations) to enhance interpretability. SHAP values explain the contribution of each feature to a prediction, offering transparency over key factors (e.g., inflation, volatility, USD strength) influencing gold price changes, critical in financial decision-making. Therefore, XGBoost is suggested to function as the primary regression model, providing reliable trend predictions for automated trading execution. Table 5 summarize the difference between these 3 model.

Model	**Type**	**Data Type**	**Strengths**	**Limitations**
SVM	Classifier	Structured (Tabular)	Handles non-linear patterns; good for binary classification	Sensitive to noise; poor with sequential data
LSTM	Sequence Forecaster	Time-Series	Captures temporal patterns; models	Data-hungry; risk of overfitting; slower to train

			trends and reversals	
XGBoost	Ensemble Regressor / Classifier	Structured (Tabular)	High accuracy; robust to noise; interpretable via feature importance (e.g., SHAP)	Requires manual lag features for sequences

Table 5 : Difference Between ML Models

2.2.3.4 Comparative Analysis of ML Model Performance in Previous Studies

Zhang (2024) compared the performance of three ML models, which is Random Forest (RF), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGBoost) in predicting gold's relative returns by using historical data from Yahoo Finance. The models were evaluated using MSE, RMSE, MAE, R^2 and trend accuracy. Among the three, Random Forest delivered the best overall results, achieving the highest R^2 and trend accuracy, though it showed signs of overfitting. Even though SVR was less accurate overall but is predict extreme values more effectively and showed lower overfitting. In contrast, XGBoost underperformed due to underfitting during training and lagged in capturing sudden price movements. The result is shown in table 6.

Model	**Dataset**	**MSE**	**RMSE**	**MAE**	**R^2**	**Trend Accuracy**	**Notes**
RF	Train	1.31e-05	0.0036	0.0026690	0.89	0.88	Best overall fit; suffers from overfitting.
	Test	2.36e-05	0.0049	0.0032876	0.79		Higher R^2 and slightly lower error than SVR.
XGBoost	Train	3.38e-05	0.0058	0.0037	0.73	~0.853	Underfitting observed;

							poor trend prediction.
	Test	2.98e-05	0.0055	0.0038	0.72		Lags in detecting sharp price shifts.
SVR	Train	1.92e-05	0.0044	0.0030	0.84	0.86	Best at predicting extreme values. Less overfit.
	Test	2.21e-05	0.0047	0.0033	0.79		Slightly lower trend accuracy than RF.

Table 6 : Performance of Different ML Model 1

Next, Makala and Li (2021) compares the performance of ARIMA and Support Vector Machine (SVM) models in forecasting gold prices using historical data from 2015 to 2019. The methodology involved making the time series stationary (using the Augmented Dickey-Fuller test and differencing), selecting ARIMA parameters via AIC minimization, and testing SVM with three kernel functions: Linear, Polynomial, and Radial Basis Function (RBF). All models were evaluated using RMSE, MAPE, and R^2. As shown in Table 7, ARIMA performed moderately, while SVM-RBF also underperformed due to parameter sensitivity. This highlights that simpler SVM kernels like Linear and Polynomial provide better accuracy and generalization for gold price forecasting.

Model	**RMSE**	**MAPE (%)**	**R^2**	**Remarks**
SVM (Linear)	0.0276	2.49	0.9978	Best performance – accurate and consistent
SVM (Polynomial)	0.0275	2.49	0.9978	Best performance –

				nearly identical to SVM Linear
SVM (RBF)	10.8773	169.3	0.9875	Poorer performance – sensitive to parameters
ARIMA (2,1,2)(2,1,2,12)	36.1795	2897.59	0.8615	Weakest – limited accuracy due to non-stationary data

Table 7 : Performance of Different ML Model 2

As shown in table 8, the other study, conducted by Wang et al. (2025), analyzed gold futures price forecasting using both traditional and advanced machine learning models, including ARIMA, Random Forest, XGBoost, and the transformer-based Time-GPT model. Time-GPT was evaluated in two settings: univariate (using gold prices only) and multivariate (with exogenous variables such as the Dow Jones Index, crude oil prices, and the US Dollar Index). Models were trained on COMEX daily gold futures data from January 2008 to March 2024 and evaluated using MSE, RMSE, and MAE. The Time-GPT model, especially when incorporating exogenous variables, demonstrated the best overall performance, highlighting its superior ability to capture complex temporal and contextual patterns. XGBoost also showed strong results, particularly with the lowest MAE, making it a strong candidate for short-term forecasting. In contrast, ARIMA struggled significantly due to its limitations in handling non-stationary and volatile financial series.

Model	**MSE**	**RMSE**	**MAE**	**Performance Summary**
Time-GPT + Exogenous	345.84	18.60	15.33	Best overall; leverages macroeconomic variables for improved prediction

Time-GPT (Univariate)	422.87	20.56	16.80	Strong performance using only gold prices
XGBoost	311.99	17.66	11.82	Lowest MAE; excellent for short-term, non-linear dynamics
Random Forest	656.48	25.62	19.06	Moderate performance; benefits from non-linearity but less precise
ARIMA	1922.71	43.85	26.48	Weakest; struggles with volatility and complex patterns in gold futures prices

Table 8 : Performance of Different ML Model 3

2.2.4 Model Training and Optimization Techniques

Once the input features are prepared and the appropriate machine learning models are selected, the next critical phase involves model training and optimization. The way a model learns patterns from data through parameter adjustment, hyperparameter tuning, and overfitting prevention directly impacts its predictive accuracy and overall robustness.

2.2.4.1 Train-Test Split and Cross-Validation

Separating the dataset into three subsets (training, validation, and testing) is a basic step in the training of machine learning models. The model is fitted to the training set so that it can discover underlying patterns in the historical data. For example, the model learns optimal decision tree splits in XGBoost, whereas backpropagation is used in LSTM networks to update the model weights. However, performance on the training set alone does not reflect generalization ability.

The validation set serves as an internal checkpoint during training. It enables hyperparameter tuning, model selection, and early detection of overfitting by evaluating the model on unseen data during development. Importantly, no learning occurs directly from the validation set, it acts solely as a reference for tuning decisions.

The test set, in contrast, is reserved for final evaluation after the model has been fully trained and optimized. This set provides an unbiased estimate of the model's performance on entirely unseen data, simulating real-world deployment.

In financial time-series forecasting such as gold price prediction, maintaining the chronological order of data is critical. Unlike standard k-fold cross-validation, where data are shuffled randomly, time-series data must preserve temporal dependencies to avoid information leakage, where future data inadvertently inform past predictions. Therefore, methods such as walk-forward validation or rolling-origin evaluation are recommended. These techniques ensure that models are trained on past data and validated against future observations, reflecting actual market conditions (Cerqueira, Torgo, & Mozetič, 2019).

2.2.4.2 Hyperparameter Tuning

The performance of machine learning models is strongly influenced by the choice of hyperparameters, configurable settings that control the learning process. For instance, SVM requires tuning of kernel types and regularization strength, XGBoost involves adjusting tree depth and learning rates, while LSTM models depend on the number of neurons, layers, and dropout rates.

Common hyperparameter tuning methods like grid search and random search can be time-consuming. More efficient alternatives, such as Bayesian optimization, focus on the most promising areas of the search space (Bischl et al., 2021).

For XGBoost models, Randomized Hyperopt (a probabilistic tuning method) has been shown to outperform traditional approaches in both tuning speed and accuracy (Putatunda & Rama, 2020). In the case of LSTM models, El Ouardi et al. (2024) reported that gradient-based adaptive optimization led to faster convergence and lower prediction errors than others.

2.2.4.3 Regularization and Overfitting Prevention

Avoiding overfitting is key to making sure a model doesn't just memorize the training data but can also perform well on new, unseen data. Different models tackle this problem in different ways. In SVM, the C parameter balances between maximizing the margin and minimizing classification error (E.M. Jordaan & Smits, 2002). In XGBoost, the lambda (λ) and alpha (α) values act like guardrails, preventing the trees from becoming too complex and overfitting the training data (Ok & Emmanuel, 2025). With LSTM networks, a common trick is to use dropout layers, which randomly turn off some neurons during training. This forces the model to learn more general patterns instead of relying on specific paths (Cheng et al., 2017). Another helpful technique is early stopping, which simply pauses training when model stop improve performance when using validation set.

2.2.4.4 Feature Scaling and Normalization

Some models, like SVM and LSTM, are sensitive to the scale of input features. They require proper feature scaling to perform well. Techniques such as min-max normalization and z-score standardization ensure that all features contribute equally during training. This helps improve predictive accuracy and also speeds up convergence. De Amorim et al. (2022) found that the choice of scaling method can significantly affect model performance. In some cases, it may even have more impact than the model architecture itself. In contrast, tree-based models like XGBoost are typically less sensitive to feature scaling. This is because they split data based on thresholds rather than distance metrics.

2.2.4.5 Evaluation During Training

During training, models are evaluated using appropriate metrics to track their learning progress. These metrics also help identify signs of overfitting or underfitting. Key metrics that measure the accuracy of continuous price predictions for regression tasks like gold price prediction are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (Plevris et al., 2022). These calculate how close the predicted values are to the actual prices. For classification tasks, such as predicting whether the price will go up or down, metrics like accuracy, precision, recall, and F1-score are used to assess performance.

2.3 Comparison of Machine Learning-Based Trading Platforms

2.3.1 Alpaca

Figure 6 : Alpaca

Alpaca is a developer-oriented trading platform that provides REST and WebSocket APIs, allowing users to build and execute algorithmic trading strategies using Python or other programming languages. While Alpaca does not offer any built-in machine learning models or forecasting tools, it is widely used as an execution engine within custom machine learning pipelines. Developers can integrate ML models externally to fetch real-time or historical data, generate signals, and execute trades programmatically. Features like paper trading and multi-account support make Alpaca suitable for rapid prototyping and deployment of automated strategies. However, Alpaca lacks integrated tools for sentiment analysis, macroeconomic data integration, or explainability frameworks. Consequently, it functions as a low-level infrastructure tool requiring significant external development for building end-to-end intelligent trading systems.

2.3.2 QuantConnect

Figure 7 : QuantConnect

QuantConnect is a cloud-based algorithmic trading platform built on the LEAN open-source engine. It provides extensive support for machine learning by allowing integration with libraries like TensorFlow, PyTorch, scikit-learn, and XGBoost. QuantConnect enables researchers and traders to design, backtest, and deploy custom ML-based strategies across various asset classes, including gold (XAU/USD) through CFD contracts. It provide some built-in macroeconomic and sentiment datasets, but user still need to integrate it manually. Additionally, it does not have built-in explainability tools like SHAP to assist users in interpreting model predictions. Users are required to manually create complete data pipelines, oversee the ingestion of external data, and put model transparency mechanisms into place. Although QuantConnect's flexibility and strong backtesting infrastructure are its main advantages, implementing complex AI-driven systems calls for highly skilled technical personnel.

2.3.3 Identified Limitations of Existing Platforms

Despite supporting automated trading and machine learning integration, platforms such as Alpaca and QuantConnect are devoid of essential features needed for specialised gold price forecasting. Existing method lack end-to-end pipelines that able to handle everything from data collection to prediction and automated trade execution. They also don't provide gold-specific models or an easy way to integrate data from multiple sources. On top of that, there's no built-in support for explainability tools like SHAP to help users understand model decisions. This project addresses those gaps by combining price data, macroeconomic indicators, and market sentiment. It aims to build a specialized machine learning system for short- to mid-term gold forecasting that delivers real-time trading signals along with clear, interpretable predictions.

2.4 Technical Indicators as Features

Technical indicators are valuable for capturing short- to mid-term price movements in automated gold trading. Derived from historical price and volume data, they reveal trends, volatility, momentum, and potential reversals. Integrating these features into models like XGBoost and LSTM helps the system detect patterns more effectively and generate timely trading signals.

The Moving Average (MA), one of the frequently used indicators, is helps identify the overall trend by smoothing out short-term price fluctuations. For instance, the 20-day MA (MA20) can signal bullish or bearish sentiment based on whether the price stays above or below it. The Relative Strength Index (RSI) measures momentum by comparing recent gains and losses. Values above 70 suggest the market is overbought, while values below 30 suggest it's oversold (Chong et al., 2014). The MACD is often used to time entries and exits. It tracks the difference between the 12-day and 26-day EMAs and uses a signal line to detect momentum shifts (Chio, 2022).

Beyond improving model interpretability, technical indicators also support rule-based auto-trading strategies. For example, a MACD crossover or RSI threshold breach can trigger automatic buy or sell actions.When combined with machine learning predictions, these signals enhance decision-making and improve the overall reliability of the trading system. Table 9 summarizes and compare different technical indicator.

Indicator	**Category**	**Purpose**	**Typical Interpretation**
Moving Average (MA)	Trend-Following	Identify trend direction	Price > MA → Uptrend; Price < MA → Downtrend
Relative Strength Index (RSI)	Momentum Oscillator	Detect overbought/oversold conditions	RSI > 70 → Overbought; RSI < 30 → Oversold
MACD	Trend & Momentum	Detect trend reversals and momentum shifts	MACD > Signal → Bullish; MACD < Signal → Bearish
Bollinger Bands	Volatility Indicator	Gauge price volatility and breakout risk	Price breaks upper/lower band → Possible reversal
Stochastic Oscillator	Momentum Oscillator	Identify short-term reversals	%K > %D → Buy; %K < %D → Sell
ADX (Average Directional Index)	Trend Strength	Measure the strength of a trend	ADX > 25 → Strong trend ADX < 20 → Weak/no trend
OBV (On-Balance Volume)	Volume-Based	Combine price and volume to confirm trends	Rising OBV confirms uptrend;

			falling OBV confirms downtrend
ATR (Average True Range)	Volatility Indicator	Measure market volatility	Higher ATR → Higher volatility; useful for stop-loss

Table 9 : Technical Indicator Comparison

2.5 Sentiment Analysis & News Integration

Traditional financial forecasting models have primarily relied on structured quantitative data such as price and volume. However, recent studies highlight the growing significance of unstructured textual information, including financial news, social media, and analyst reports, in influencing commodity prices. Gold, in particular, as a recognized safe-haven asset, demonstrates heightened sensitivity to macroeconomic events such as inflation, geopolitical tensions, and monetary policy announcements. For example, Smales (2014) examined the role of news sentiment in the gold futures market, finding that negative news events significantly increase price volatility, especially during times of financial uncertainty. In a more recent study, Liu et al. (2024) examined sentiment gleaned from Chinese-language headlines and found that adding sentiment features to conventional technical indicators improved the precision of short-term gold futures price forecasting.

2.5.1 Sources of News Data for Sentiment Analysis

The diversity and timeliness of news data sources have a direct impact on the precision and resilience of sentiment analysis in financial applications. These usually consist of unstructured sources like social media sites (Twitter, Reddit), news aggregators (like Google News RSS feeds), and structured financial news APIs (like Bloomberg, Reuters). Social media offers instantaneous but frequently noisy sentiment signals, whereas APIs and aggregators provide consistency and credibility. For sentiment model training, annotated corpora such as the Financial PhraseBank and FinBERT Sentiment are still essential.

A lexicon-based method that applies domain-specific sentiment lexicons to carefully selected news articles was presented by Taj et al. (2019). Their research highlighted the significance of linguistic nuance and news quality, showing that when combined with a strong vocabulary, traditional financial news can still be a trustworthy source of sentiment.

In the meantime, Shahare (2017) concentrated on enhancing coverage and responsiveness by fusing sentiment from social media with news data. Their research demonstrated that, despite the difficulties associated with noise and data preprocessing, integrating Twitter and other platforms with news feeds can improve the detection of sentiment that moves the market.

Collectively, these studies show that a hybrid approach offers a more thorough and efficient basis for financial sentiment analysis by integrating structured news, social sentiment, and annotated datasets.

2.5.2 Advances in NLP Techniques for Financial Sentiment

Financial-specific Natural Language Processing (NLP) techniques have advanced significantly, allowing for more precise topic identification and sentiment extraction. Using extensive corpora of financial texts, domain-specific pretrained models like FinBERT (Araci, 2019) have outperformed general-purpose language models in financial sentiment classification tasks. Models like BERTopic (Grootendorst, 2022) use transformer-based embeddings and sophisticated clustering algorithms to find logical and understandable themes in financial news data, further advancing topic modelling approaches beyond traditional methods like Latent Dirichlet Allocation (LDA). These subjects frequently correlate with the main factors that influence the dynamics of the gold price, such as central bank policy announcements, inflation trends, and geopolitical risks.

2.5.3 Feature Engineering and Model Integration

To improve forecasting models, a number of studies have converted textual sentiment and thematic insights into quantitative features. Among the methods are producing topic probability distributions, binary indicators for particular events, and encoding sentiment scores as continuous variables. To create enriched datasets for machine learning algorithms, these characteristics are frequently coupled with technical indicators and macroeconomic variables. Recurrent neural networks, including Long Short-Term Memory (LSTM) networks, and models like XGBoost have been widely used to identify intricate patterns and temporal dependencies in the aggregated data, improving the forecasting accuracy of gold prices.

2.6 Evaluation Metrics

In automated trading systems and gold price forecasting, assessing the effectiveness of machine learning models is essential. Regression (forecasting future prices) and classification (predicting the direction of price movement) are both involved in commodity price prediction, so choosing the right evaluation metrics is crucial to determining the model's quality and practical efficacy. According to earlier studies, depending just on one metric can result in inaccurate conclusions, especially in erratic financial markets like gold trading (Botchkarev, 2018).

2.6.1 Regression Metrics for Price Forecasting

There are some commonly used measures include:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual prices. Lower MAE indicates better prediction stability.
- **Root Mean Squared Error (RMSE):** Similar to MAE but penalizes larger errors more severely. Suitable when large deviations carry higher risk.
- **Mean Absolute Percentage Error (MAPE):** Expresses errors as percentages, enhancing interpretability in financial contexts. However, MAPE can become unstable and in accurate when actual values are almost zero.
- **R-squared (R^2):** Represents the proportion of variance explained by the model. A value closer to 1 indicates a stronger fit.

In order to obtain a more thorough evaluation, Botchkarev (2018) stressed the importance of applying multiple regression metrics simultaneously. He highlighted additional measures such as sMAPE (Symmetric Mean Absolute Percentage Error) and Median Absolute Error (MdAE) as helpful for handling skewed or volatile datasets like gold prices.

2.6.2 Classification Metrics for Directional Prediction

As shown in table 10, Classification metrics are used when the model is used to forecast the direction of price movement, such as "up" or "down":

Metric	Description	Use Case
Accuracy	Ratio of correct directional predictions to total predictions.	Simple indicator of overall performance.
Precision	Ratio of true positive predictions to all predicted positives.	Important when false positives lead to costly trades.
Recall	Ratio of true positive predictions to all actual positives.	Measures how many real price increases/drops were correctly predicted.
F1-Score	Harmonic mean of precision and recall.	Balances both metrics when trade-offs exist.

Table 10 : Evaluation Metrics

2.6.3 Financial Metrics for Trading Strategy Evaluation

In financial trading, prediction accuracy alone does not ensure profitability. Therefore, evaluating models using financial performance metrics is essential. Cumulative Return measures total profit or loss over the trading period, directly showing strategy profitability. The Sharpe Ratio assesses risk-adjusted returns by comparing excess returns to return volatility, a higher value indicates better performance relative to risk (Schäfer et al., 2025). Maximum Drawdown (MDD) highlights the largest historical loss from peak to trough, revealing potential risk exposure (Goldberg & Mahmoud, 2014). Finally, Win Rate reflects the percentage of profitable trades, indicating the consistency of the strategy's signals. These metrics collectively provide a realistic assessment of trading system performance.

2.7 Trading Signal Generation and Strategy Design

In automated gold trading systems, raw model predictions must be rigorously transformed into actionable Buy, Sell, or Hold decisions. This signal generation layer is vital for translating forecasting into profitable trades; without it, prediction accuracy alone cannot ensure trading success.

2.7.1 Threshold-Based Trading Rules

Threshold-based strategies convert continuous forecast outputs into discrete signals using set criteria. For example, a Buy signal is issued if the model forecasts a price increase above +0.5%, while a Sell occurs if a decline below –0.5% is predicted; otherwise, the position is held. However, Qiu et al. (2010) cautioned that fixed thresholds often fail in regimes of fluctuating volatility, leading to missed trades or premature exits. These limitations highlight the need for more adaptable threshold mechanisms.

2.7.2 Technical-Indicator Confirmation

To reduce false signals, many studies advocate for combining model predictions with traditional technical indicators. For instance, Chong and Ng (2014) analyzed the application of RSI and MACD across multiple OECD stock markets. They discovered that signals confirmed by both predictive models and indicators such as a Buy when predicted upward trend coincides with RSI<30, duplicate considerably fewer false positives and increase overall strategy reliability. Their empirical evidence suggests that this layered confirmation significantly enhances signal validity.

2.7.3 Volatility-Sensitive Risk Controls

Effective trading systems also incorporate adaptive risk management using volatility-based stop-loss and take-profit levels. For example, adjusting stop thresholds to 1.5× ATR (Average True Range) allows systems to dynamically adapt to market volatility. This approach mitigates excessive losses during high volatility and locks in profits during calmer periods. Fu et al. (2025) demonstrated that ATR-based dynamic rules significantly reduce drawdowns and improve return consistency in commodity markets.

2.8 Limitations of Existing Studies and Project Gap Statement

Despite the promising results of previous studies in gold price forecasting using machine learning and time series models, several limitations persist. Many existing models, including ARIMA, SVM, Random Forest, and XGBoost, suffer from overfitting, especially when handling volatile financial data or lacking proper feature selection and cross-validation. Furthermore, most studies focus solely on predictive performance without integrating the results into actionable trading strategies. There is a noticeable gap in the incorporation of real-time automated trading mechanisms, simulation environments, and feedback loops to validate predictions under realistic market conditions. This project aims to bridge these gaps by combining robust predictive modeling with automated trade execution and simulated backtesting, providing a more comprehensive and practical framework for real-world application.

2.9 Conclusion

The literature emphasizes the complexity of gold price forecasting due to its sensitivity to macroeconomic variables, market sentiment, and structural volatility. Traditional econometric models, while interpretable, struggle to capture nonlinear dynamics and abrupt regime shifts. As a result, recent studies favor hybrid and machine learning-based approaches, especially LSTM, SVM, and XGBoost models, augmented by sentiment and technical indicators. Drawing on these insights, this project adopts the CRISP-DM methodology to systematically integrate diverse data sources, build predictive models, and deploy an end-to-end gold trading simulation platform. Each phase of the CRISP-DM cycle supports best practices from financial forecasting literature, ensuring both methodological rigor and practical relevance.

3.0 Methodology

3.1 CRISP-DM Framework

This project employs the **CRISP-DM** (Cross Industry Standard Process for Data Mining) methodology to ensure a well-structured, iterative, and best-practice-oriented approach to the development of a machine learning-based gold price prediction and automated trading system (Shimaoka et al., 2024). As shown in figure 8, the methodology supports the integration of various system components across six well-defined phases.

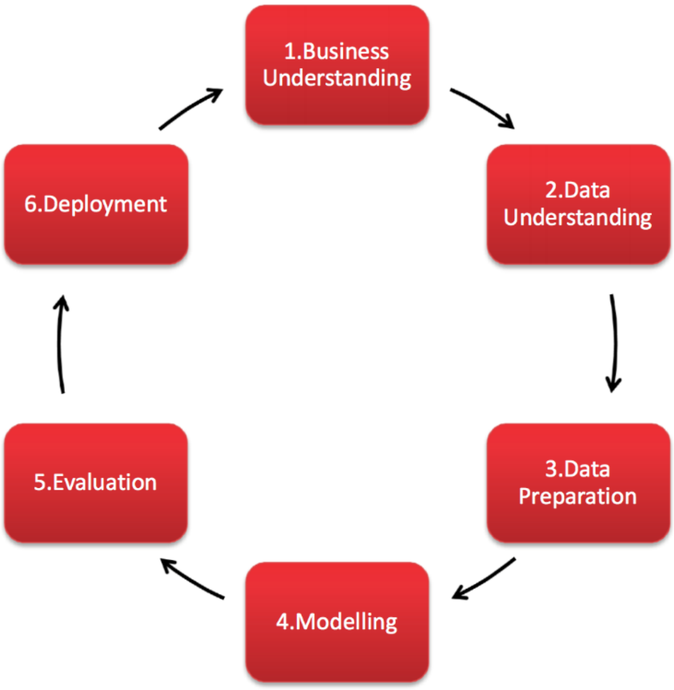


Figure 8 : CRISP-DM Methodology

3.2 Business Understanding

This project aims to develop a Machine Learning Based Gold Price Trend Prediction and Automated Trading System that generates real-time, explainable buy/sell guidance with strong risk control. The market forces, macroeconomic, geopolitical and the sentiment of the investor all affect gold prices. In order to increase the quality of decisions, the system incorporates real-time news sentiment, which gives background to high-probability, trend-aligned trades. Rather than trying to predict the precise price of gold per spot, the system tries to predict the directional trend within a multi-hour horizon to enable a stronger and statistically significant trading strategy. Table 11 is a summary of the most important objectives and guiding principles of the system.

Aspect	Description
Trading Objective	Deliver precise guidance prioritizing directional trend expectancy over exact price forecasting, ensuring the system executes only when AI confidence and sentiment signals converge.
Risk Control	Limit position sizes, apply stop-loss/take-profit logic, and adapt thresholds based on volatility and news sentiment.
News Sentiment	Influences decision thresholds dynamically to respond to breaking macro or geopolitical events.
Success Metrics	Cumulative profit, win rate, max drawdown, Sharpe ratio, and prediction reliability.
Deployment Principles	Microservice architecture for modularity and crash isolation, low-latency operation, and transparent, responsible AI guidance.

Table 11 : Key Objectives and Guiding Principles

By leveraging news sentiment and structured risk principles, the system aims to produce actionable and interpretable trading decisions in a volatile market environment.

3.3 Data Understanding

In this study, heterogeneous data sources have been used to balance between quantitative market behaviour, as well as the qualitative information that influences the movement of gold prices. The system is based on three fundamental data groups, which include hourly market prices, real time news headline, and derived sentiment and market regime indications.

3.3.1 Hourly Price Data (Yahoo Finance)

As shown in table 12, hourly OHLCV data are retrieved via the *yfinance* API using a high-frequency data pipeline with automatic fallback handling. Gold Futures (GC=F) serve as the primary price series, while selected macro-correlated assets are incorporated to capture broader market risk, liquidity, and investor sentiment effects. All assets are aligned to a unified hourly timeline to ensure temporal consistency.

Asset	Ticker	Frequency	Data Fields	Justification
Gold Futures	GC=F	Hourly	OHLCV	Serves as the dependent variable for directional forecasting.
USD Index	DX-Y.NYB	Hourly	Close	To evaluate the inverse relationship between USD strength and dollar-denominated gold prices.
S&P 500	^GSPC	Hourly	Close	Included to model capital flows between traditional risk assets and safe-haven gold.
VIX Index	^VIX	Hourly	Close	To capture "fear-driven" demand spikes during periods of heightened market uncertainty.
NASDAQ 100	^IXIC	Hourly	Close	Used to observe gold's behavior relative to high-growth, interest-rate-sensitive equity sectors.
Bitcoin	BTC-USD	Hourly	Close	To test whether crypto-market sentiment acts as a complementary or competing alternative to gold.

Ethereum	ETH-USD	Hourly	Close	Provides broader context on speculative risk-appetite within the decentralized finance ecosystem.

Table 12 : Price Data Comparison

3.3.2 Real-Time News Headlines

The News Service of the system fetches headlines of edited Google News RSS feeds on the gold market, announcements by the central bank, inflation, the US dollar, and geopolitical matters. All the headlines consist of the timestamp (UTC), source, title and the article link. RSS feeds offer lightweight and fast text feeds which can be used in real-time trading. Preliminary filtering eliminates the irrelevant domains (e.g., lifestyle, entertainment). There is no tokenization or lemmatization done as FinBERT is meant to take raw financial text in input.

3.3.3 Sentiment Scores and Market Regimes

Every incoming headline will be rated by FinBERT, which will label it as Positive, Neutral, or Negative with an approximation of +1 to -1. Headlines that have very low scores are disregarded to minimize noise. The system sums up sentiment every hour and saves the output as JSON which is consumed in the Inference Engine. Simultaneously, another Market Regime Detector classifies each time step as Bull, Bear, or Sideways depending on SMA crossovers to provide a further structural market context.

3.3.4 Data Challenges

The gathered data has a number of practical problems such as inconsistent time stamps in the various markets, flat-line price segments on weekends, quick changes in sentiment due to breaking news and high number of engineered features by the technical pipeline. The following constraints were used to design a well-planned data preparation process as is outlined in Section 3.4.

3.4 Data Preparation

The data preparation stage ensures that all inputs to the machine learning pipeline are temporally coherent, numerically stable, and free from information leakage, while reflecting realistic trading constraints. The process consists of time alignment, data cleaning, feature engineering, and target construction.

3.4.1 Time Alignment and Cleaning

Because financial assets trade on different schedules, strict temporal alignment is enforced across all datasets. The preparation pipeline retains only exact hourly observations to guarantee uniform spacing and reliable indicator computation. For example, records with irregular minute offsets are excluded to prevent uneven time gaps:

df = df[df['Date'].dt.minute == 0]

Non-trading periods (e.g., weekends) are excluded to avoid flat-line price behaviour that may distort volatility-sensitive indicators. In addition, flat candles with negligible price movement are filtered out to prevent learning from stale values.

df = df[df['Date'].dt.dayofweek < 5]

To synchronize external macro-assets with gold prices without introducing look-ahead bias, all datasets are aligned using a backward-looking nearest timestamp strategy. This ensures that only information available at the decision time is accessible to the model. All cleaned datasets are stored under version control to preserve reproducibility and auditability.

merged = pd.merge_asof(gold, dxy, on="Date", direction="backward")

3.4.2 Feature Engineering

Feature engineering is designed to capture key dimensions of gold price behaviour, including trend momentum, volatility dynamics, macroeconomic pressure, and intraday seasonality.

1. **Technical Indicators**
   The system employs a suite of widely adopted technical indicators computed using *ta-lib*, including RSI, MACD and its signal line, moving averages, Bollinger Bands, and the Average True Range (ATR). These indicators collectively describe momentum strength, trend persistence, and market volatility.
2. **Cross-Asset Macro Proxies**

To encode macroeconomic relationships and relative market strength, the system constructs engineered ratios such as Gold/DXY and Gold/BTC. These ratio-based features often provide more stable and interpretable signals than raw external prices, as they normalize gold movements against broader market conditions.

- **Dynamic Correlations**: In addition to ratios, the system computes 24-hour rolling correlations between Gold and macro-benchmarks. This allows the model to detect shifts in market sensitivity, such as when Gold begins to decouple from the US Dollar or move in lockstep with equity volatility.

3. **Cyclical Time Features**

To model intraday behavioural patterns, hour-of-day information is encoded using sinusoidal transformations that preserve cyclical structure:

*df['Hour_Sin'] = np.sin(2 * np.pi * hour / 24)*

*df['Hour_Cos'] = np.cos(2 * np.pi * hour / 24)*

4. **Temporal Causality (Feature Lagging)**

df[features] = df[features].shift(1)

To ensure the system remains realistic and free from look-ahead bias, a strict one-hour backward shift ($T-1$) is applied to all technical and macro features. This ensures that at any decision point, the model only has access to finalized data from the previous hour, mirroring real-world execution constraints.

5. **Market Regime Awareness (Environment Detection)**

Beyond local price action, the system is designed to incorporate a Market Regime Detector to provide high-level structural context. This module classifies the gold market into three distinct states: Bull, Bear, or Neutral.

- **Detection Logic**: The regime is determined by evaluating the relationship between a fast moving average (MA-20) and a slow moving average (MA-50), combined with the Slope of the trend to measure momentum strength.

By labeling the market environment, the system can differentiate between "Trending" and "Sideways" periods, allowing the inference engine to apply different confidence thresholds depending on the volatility regime.

These engineered features collectively form a multi-dimensional feature space, providing the ensemble models with the necessary inputs to differentiate between high-probability trends and market noise.

3.4.3 Target Variable Construction

The predictive task is formulated as a short-horizon directional classification problem. Forward returns are computed over a fixed future window, and a minimum return threshold is applied to filter out micro-fluctuations that are unlikely to be economically meaningful. Observations that do not exceed this threshold are excluded from training to reduce label noise:

df['target_bin'] = (future_return > threshold).astype(int)

The final horizon length and threshold values are selected empirically based on noise characteristics and stability observed during exploratory analysis.

3.4.4 Feature Cleaning and Dimensional Control

To reduce sensitivity to long-term price drift, raw OHLCV variables are excluded from the final input space. The model relies exclusively on engineered, normalized, and ratio-based features with stronger stationarity properties, improving generalization and out-of-sample robustness.

3.4.5 Data Splitting

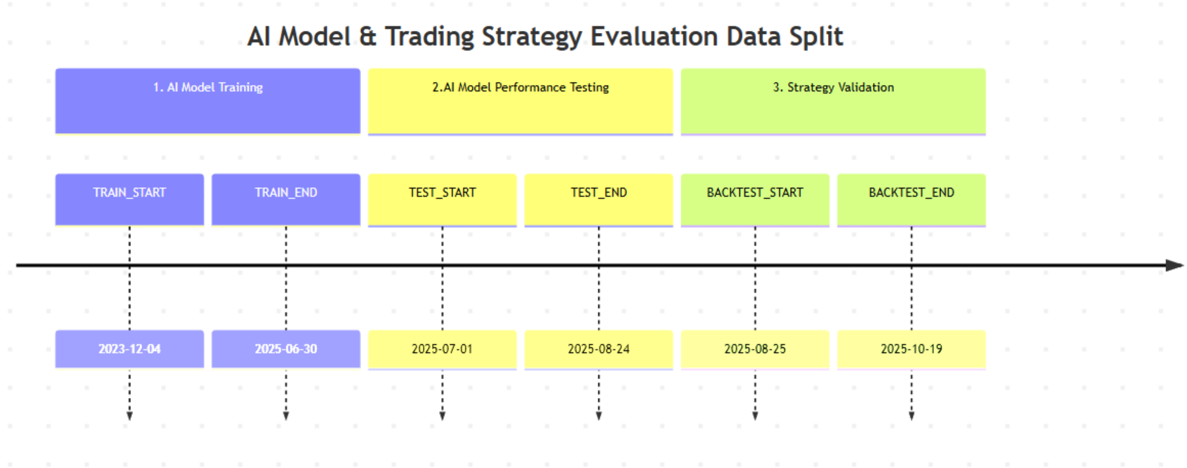


Figure 9 : Data Splitting

As shown in figure 9, the dataset is partitioned using a Non-Shuffled Temporal Split. A chronological division (Training → Testing → Out-of-Sample Validation) is strictly maintained to ensure the model is always evaluated on data it has never encountered during training, mirroring a true live trading scenario.

3.4.6 Dynamic Sentiment Adjustment

Sentiment scores are not used as static features but as tactical modifiers. At runtime, the AI probability threshold is shifted by a sentiment-derived delta , allowing macro-news to act as a final "filter" before trade execution.

3.5 Modelling

The modelling stage defines how predictive intelligence is constructed and operationalized within the trading system. Rather than generating direct buy/sell decisions, the problem is formulated as a probabilistic time-series classification task, where the model outputs calibrated confidence scores. This design decouples prediction from execution, allowing downstream components to apply adaptive risk controls, dynamic thresholds, and external signals without retraining the model.

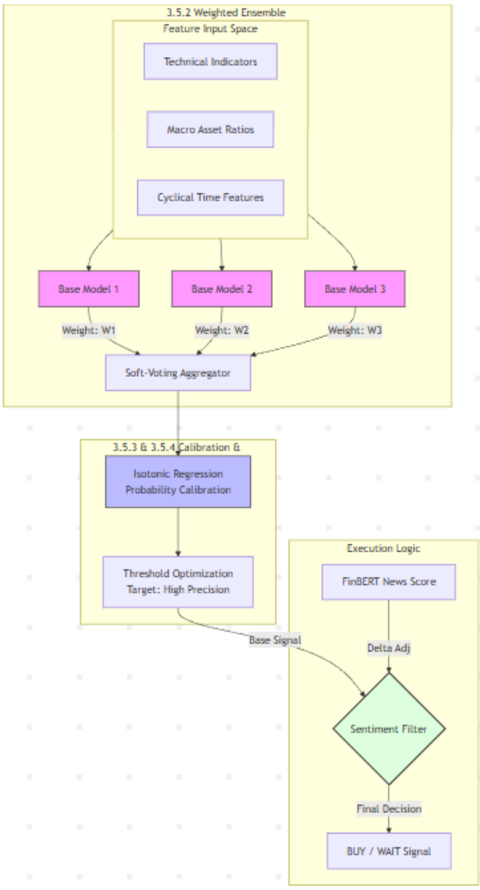


Figure 10 : Ensemble Architecture

As shown in figure 10, the modelling pipeline is composed of four sequential stages: base model selection, ensemble construction, probability calibration, and threshold optimization.

Each stage is designed to address a specific challenge associated with financial time-series prediction, including non-linearity, non-stationarity, and decision reliability under uncertainty.

3.5.1 Base Models

Multiple supervised learning algorithms are evaluated as candidate base learners to capture the diverse structural characteristics of the gold market. Due to the non-linear, non-stationary, and regime-dependent nature of financial time-series data, no single model architecture is assumed to be universally optimal. Therefore, a set of complementary models is selected to balance expressiveness, generalization capability, interpretability, and computational efficiency, which are critical requirements for near–real-time trading inference. Table 13 shows different base models use in this project.

Model	Methodological Purpose
Logistic Regression	Provides a linear probabilistic baseline with stable and interpretable confidence outputs, serving as a reference model under low complexity assumptions.
Support Vector Classifier (RBF Kernel)	Captures non-linear decision boundaries through kernel transformation, enabling the modelling of complex feature relationships beyond linear separability.
Random Forest	Models non-linear feature interactions using an ensemble of decision trees, improving robustness to noise and reducing variance through bagging.
Gradient Boosting Classifier	Learns complex patterns by sequentially correcting residual errors, allowing the model to adapt to gradual market shifts and structured dependencies.
Multi-Layer Perceptron	Learns non-linear representations across multiple hidden layers, capturing interactions between technical, temporal, and macro-level features without explicit feature engineering.
XGBoost	An optimized gradient-boosted tree model designed for efficiency and scalability, particularly effective at

	modelling regime shifts and higher-order feature interactions in volatile markets.

Table 13 : Base Models Description

3.5.2 Ensemble Construction

The system uses soft-voting ensemble strategy instead of using one predictive model. All base models add a probability estimate and the ensemble output is calculated with a weighted average of the probability estimates.

This approach is designed to:

- Reduce model-specific bias
- Improve robustness across different market regimes (trending, ranging, volatile)
- Mitigate over-reliance on any single hypothesis about market behaviour

The weights of the models are allocated according to their stability during cross-validation and not to the model with the highest accuracy, but to the model with the best stability through time. This design option is based on the characteristics of the system that prioritizes reliability and the management of risks over aggressive short term optimization.

3.5.3 Probability Calibration

Raw probability outputs from machine learning models are often poorly aligned with real-world likelihoods, particularly in imbalanced and non-stationary financial datasets. To address this limitation, the ensemble output is calibrated using Isotonic Regression.

Calibration transforms raw confidence scores into probabilities that better reflect empirical outcome frequencies. This step is essential because downstream trading logic relies directly on probability thresholds. A calibrated output ensures that a confidence score (e.g., $P = 0.55$) has consistent semantic meaning across time, improving decision interpretability, operational safety, and user trust.

3.5.4 Threshold Optimization

Rather than fixing a single arbitrary decision threshold, the system performs threshold optimization during validation to identify an operating region that balances sensitivity and precision. This allows the model to focus on high-confidence predictions while suppressing low-quality signals during uncertain market conditions.

In deployment, the optimized threshold serves as a baseline reference point. It can be dynamically adjusted by external components, such as real-time news sentiment analysis, without modifying or retraining the predictive model. This separation between prediction generation and execution logic enhances system flexibility and supports adaptive behaviour under changing macroeconomic conditions.

3.6 Evaluation

The assessment plan will be structured in such a way that it would help to test both the technical validity of the machine learning models and the practical functionality of the trading system in the conditions of a real market. Instead of concentrating on predictive accuracy, the evaluation framework employs a multi dimensional evaluation approach that includes AI performance, trading behaviour, risk control and decision transparency. This guarantees the system to be dependable, readable and appropriate to use in a real time financial setting.

3.6.1 Machine Learning Performance Evaluation

The predictive models are assessed with a series of standard classification measures as seen in table 14, which are reasonable in time-series financial data.

Metric	Purpose in Trading Context
Accuracy	Measures overall directional correctness of predictions
Precision	Evaluates how often predicted buy signals are correct, minimizing false entries
Recall	Measures the ability to capture profitable upward price movements
F1-score	Balances Precision and Recall, suitable for imbalanced market conditions
TimeSeriesSplit CV	Ensures temporal robustness and realistic model validation

Table 14 : Model Performance Metric and Purpose

Precision and F1-score are emphasized over raw accuracy, as false positive signals in trading can directly translate into financial losses. This metric selection reflects the system's focus on high-confidence, selective trade execution rather than frequent predictions.

3.6.2 Trading Strategy and Backtesting Evaluation

To evaluate whether model predictions translate into economically meaningful outcomes, a custom Python-based backtesting framework is used. As shown in table 15, the backtester simulates the complete trading pipeline, including model inference, probability thresholding, position execution, stop-loss and take-profit rules, and capital tracking.

Metric	Evaluation Objective
Total Return	Measures overall profitability of the strategy
Equity Curve	Tracks cumulative capital growth over time
Win Rate	Indicates trade-level prediction effectiveness
Max Drawdown	Quantifies worst-case capital loss for risk control
Sharpe Ratio	Evaluates risk-adjusted return
Total Trades	Reflects strategy selectivity and execution frequency

Table 15 : Backtest Evaluation Metrics and Objectives

Rather than optimizing for raw return alone, the evaluation prioritizes drawdown control and risk-adjusted performance, ensuring capital preservation during volatile or news-driven market conditions. This aligns with the system's design philosophy as a conservative, precision-focused trading strategy.

3.6.3 Threshold-Based Decision Evaluation

Since trading decisions are triggered by probability thresholds, the evaluation also examines how effectively these thresholds regulate trade quality. Threshold values are selected through validation and are dynamically adjusted at runtime based on real-time sentiment signals.

The evaluation verifies that:

- Higher thresholds reduce false-positive trades during uncertain or bearish sentiment regimes
- Lower thresholds allow participation during stable, bullish market conditions
- Threshold logic maintains consistent behaviour across different volatility environments

- This assessment ensures that the AI confidence score is actionable, not merely predictive.

3.6.4 Explainability and Transparency Evaluation

Beyond performance, the system is evaluated on its ability to provide transparent and interpretable decision support. Explainability tools are integrated into the dashboard to allow inspection of model behaviour during both historical and live inference.

Component	Evaluation Purpose
Feature Importance	Identifies dominant technical and macro drivers
AI Confidence Curve	Visualizes probability stability over time
Sentiment Indicators	Shows external news influence on decisions
Decision Context View	Explains why trades are executed or rejected

Table 16 : Evaluation of explainability and transparency features

As shown in table 16, this evaluation dimension supports ethical AI deployment by ensuring that trading decisions remain understandable to users and auditable during system review.

3.7 System Deployment

The system is deployed as a modular, fault-tolerant microservice architecture designed for continuous real-time operation in a resource-constrained environment. The deployment strategy prioritizes stability, isolation, scalability, and transparency, ensuring that failures in one component do not compromise the entire trading system.

3.7.1 Microservices-Based Architecture

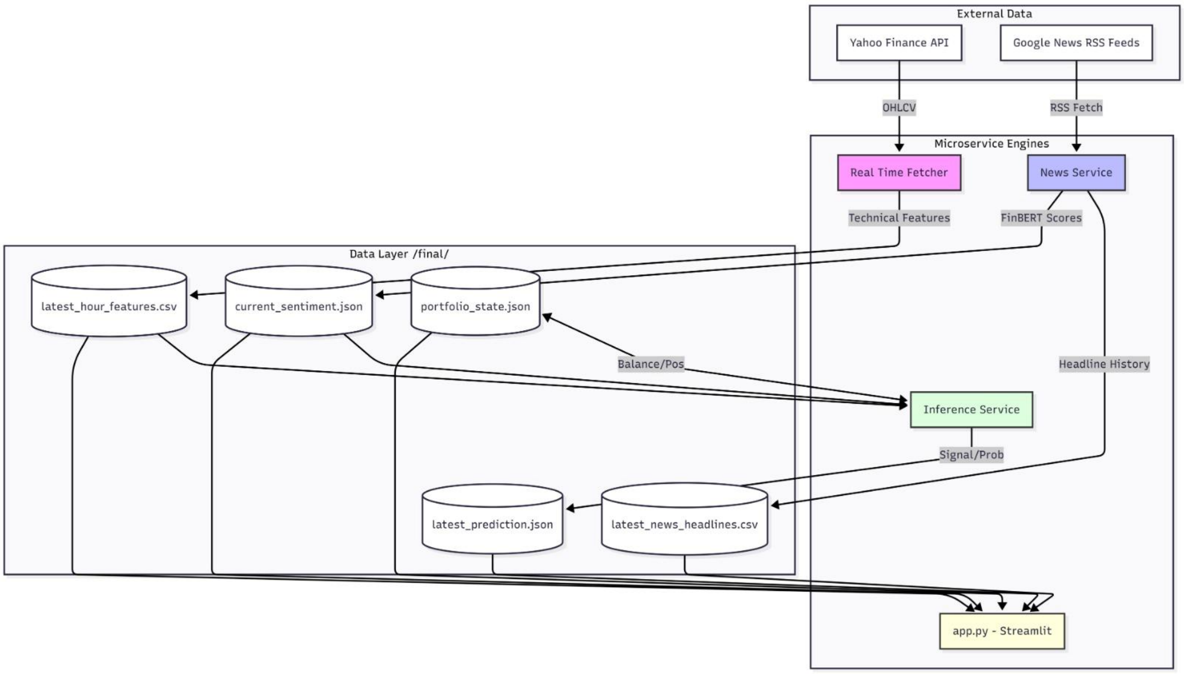


Figure 11 : System Architecture

Figure 11 shows that the system is decomposed into four independent services, each responsible for a clearly defined function within the trading pipeline.

Module	Responsibility
Fetcher	Collects, cleans, and aligns hourly market data
News Service	Retrieves RSS headlines and computes sentiment scores
Inference Engine	Performs ML inference, threshold adjustment, and risk control
Dashboard	Visualizes predictions, sentiment, and system state

Table 17 : Trading system modules and their responsibilities

As shown in table 17, each module runs as a separate Python process and communicates exclusively through lightweight CSV and JSON files. This file-based communication strategy was intentionally chosen over direct inter-process messaging to:

- eliminate tight coupling between services,
- simplify debugging and auditing,
- allow safe restarts without state corruption.

This architecture enables crash isolation, for example, a failure in the News Service does not interrupt price fetching or model inference, thereby improving system reliability and uptime.

3.7.2 Performance and Reliability Considerations

To ensure predictable performance, each service operates on a fixed execution schedule aligned with the hourly data frequency. Computationally intensive tasks such as model inference and sentiment classification are isolated from data collection to prevent blocking or latency propagation.

Key deployment design choices include:

- lightweight processes instead of heavy containers to reduce overhead,
- persistent state storage using JSON files to support seamless recovery after restarts,
- deterministic execution order to maintain temporal consistency.

These measures ensure the system remains responsive even during high market volatility or rapid news cycles.

3.7.3 Security and Ethical Considerations

From a security perspective, the system processes only publicly available market and news data and stores no personal, confidential, or user-identifiable information. As a result, it remains compliant with common data protection principles such as GDPR by design.

Ethical safeguards are explicitly incorporated:

- The system functions as decision support, not as a guarantee of financial returns.
- Model confidence scores and sentiment influence are exposed transparently through the dashboard.
- Known limitations, particularly sentiment misclassification during extreme geopolitical events, are acknowledged rather than concealed.

These design choices promote responsible AI usage and reduce the risk of misleading users.

3.7.4 Usability and Human-Centered Design

The dashboard is designed to support both technical and non-technical users by presenting complex AI outputs in an interpretable and intuitive manner. Key usability features include:

- real-time candlestick charts with technical overlays,
- a sentiment indicator reflecting macro news context,
- an AI confidence gauge showing probabilistic model output,
- clear visual distinction between BUY, HOLD, and WAIT states.

The interface emphasizes clarity over complexity, ensuring that users can quickly understand why a decision is made without needing to inspect raw model outputs or code.

3.8 Risk Management and Ethical Considerations

The project is created in a completely simulated and academic setting, but it is also created to give a reflection of the structural, ethical, and risk-related aspects of a real-world algorithmic trading system. The approach to the financial risk, model uncertainty, and ethical responsibility explicitly targets addressing these three areas, thus aligning with the ideas of responsible AI development and gets the system ready to be extended in the future.

3.8.1 Model Risk, Overfitting, and False Signals

Overfitting is the main methodological risk in financial machine learning a model that fits historical noise as opposed to the generalizable market structure. This risk is enhanced in gold price forecasting by regime changes that are caused by macroeconomic policy, geopolitical events and sentiment shocks.

To mitigate this, the system is designed with:

- strict temporal separation of training and testing data,
- walk-forward validation using TimeSeriesSplit,
- conservative target labeling to suppress micro-noise,
- ensemble learning to reduce single-model bias.

These are design decisions intended to enhance robustness in the fluctuating market conditions. However, the methodology also clearly admits that no predictive model can correctly predict future price movement in full and all the outputs are considered to be probabilistic rather than deterministic cues.

3.8.2 Financial Risk and Simulation Constraints

The entire trading in this project is carried out in a non-commercial and simulated setting. No actual capital is used and all the performance indicators such as Sharpe Ratio, drawdown and returns are created by using historical backtesting. Nevertheless, the methodology clearly takes into account real world financial risks that include: capital loss, volatility clustering, execution latency, extreme black swan events. Conservative position sizing, ATR-based stop-losses and threshold based signal filtering are risk-conscious mechanisms that are included at the design level to capture best practices found in professional trading systems.

3.8.3 Ethical Design and Human Oversight

Algorithmic trading systems can significantly influence market behavior if deployed irresponsibly. Although this project remains academic, ethical responsibility is treated as a core design constraint.

The system is intentionally structured as a decision-support framework, not a fully autonomous trading agent. Human oversight is assumed in any future deployment scenario, ensuring that final trading decisions remain accountable and interpretable.

Furthermore, the system avoids reliance on easily manipulated sentiment sources (e.g., social media), instead using curated financial news feeds to reduce the risk of adversarial or misleading inputs.

3.8.4 Governance, Transparency, and Auditability

To support ethical governance and traceability, the system incorporates multiple transparency mechanisms at the methodological level:

- **Trade Transparency and Audit Trail**

  All predictions, thresholds, sentiment adjustments, and trade decisions are logged in structured CSV and JSON formats, enabling post-hoc inspection and analysis.

- **Explainability and Model Accountability**

A combination of feature importance and performance measures are incorporated into the design to explain the effect of input features on the predictions to minimize black-box behaviour.

- **Model Version Control and Reproducibility**

  Trained models and configurations are versioned and stored to ensure reproducibility and controlled experimentation.

4.0 Work Plan & TimeLine

4.1 Activities & Milestones

4.1.1 Capstone 1 – Research, Design, and Planning

Phase	Task Description	Duration (Week)	Status
Phase Planning	Select project topic	Week 1	**Done**
	Confirm topic and submit SAF form	Week 2	**Done**
	Initial Meeting with Supervisor	Week 3	**Done**
Requirement Analysis	Draft initial work plan and prepare documentation template	Week 3–4	**Done**
	Conduct literature review	Week 4–6	**Done**
	Analyze market to identify gaps and opportunities	Week 6	**Done**
	Review academic papers related to gold price prediction and trading	Week 6	**Done**
	Review relevant model documentation (e.g., LSTM, XGBoost)	Week 7	**Done**
	Define scope of the project	Week 7	**Done**
Design	Prepare list of functional and non-functional requirements	Week 7	**Done**
	Meeting & Proposal – Define objectives, scope, and feasibility. Finalize title and research angle with supervisor.	Week 8-9	**Done**
	Literature Review – Gather, analyze, and summarize journal articles on gold prediction models, indicators, NLP, automated trading, etc.	Week 8-9	**Done**
	Write Introduction – Draft background, motivation, problem statement, objectives, and scope.	Week 10	**Done**
Phase Planning	**Complete Literature Review** – Consolidate critical reviews of key technologies, models, and gaps. Structure full Chapter 2.	Week 10-11	**Done**

Requirement Analysis	**Methodology Design** – Define CRISP-DM-based methodology, tools, models, pipeline, and diagrams.	Week 11-13	**Done**
	Review & Finalize Report – Incorporate supervisor feedback. Final polish before submission.	Week 13–14	**Done**

Table 18 : Activities & Milestones (CP1)

4.1.2 Capstone 2 – System Development & Evaluation

Phase	Task Description	Duration (Week)	Status
Development	**Data Pipeline:** Implement Fetcher for Yahoo Finance & News RSS; integrate FinBERT for sentiment scoring.	Week 1–3	**Done**
	Model Training: Train and calibrate Ensemble models (XGBoost, SVC); perform probability threshold optimization.	Week 4–7	**Done**
	System Core: Develop file-based microservices (Inference Engine, Signal Generator) to replace monolithic backend.	Week 7–9	**Done**
	Dashboard: Build visualization interface using Streamlit to display real-time signals and charts.	Week 10–11	**Done**
Testing & Debugging	**Unit Testing:** Validate individual modules (Data Fetcher, Sentiment Scorer) for crash resilience.	Week 8–9	**Done**
	Backtesting: Run historical simulations to calculate Sharpe Ratio, Max Drawdown, and Win Rate.	Week 9–11	**Done**
	Integration Testing: Verify end-to-end data flow between microservices (Fetcher → Inference → Dashboard).	Week 11–12	**Done**
Deployment	**Simulation Environment:** Finalize local simulation loop (no cloud deployment required for this scope).	Week 12	**Done**
	Live Monitoring: Execute "Forward Test" mode to validate real-time decision stability.	Week 12	**Done**

Finalization	**Supervisor Review:** Present preliminary results and dashboard demo for feedback.	Week 13	**Done**
	Final Report & Video: Complete dissertation and record final system demonstration.	Week 14	**Done**

Table 19 : Activities & Milestones (CP2)

4.2 Proposed Timeline

4.2.1 Capstone 1 Timeline

Week / Task Description	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Status
Select project topic															**Done**
Confirm topic and submit SAF form															**Done**
Initial Meeting with Supervisor															**Done**
Draft initial work plan and documentation template															**Done**
Conduct literature review															**Done**
Analyze market gaps & opportunities															**Done**
Review academic papers (gold prediction, trading)															**Done**
Review model documentation (LSTM, XGBoost)															**Done**
Define scope of the project															**Done**
Prepare list of functional & non-functional requirements															**Done**
Meeting & Proposal – Finalize title and research angle with supervisor															**Done**
Literature Review – Summarize prediction models, NLP, indicators															**Done**
Write Introduction – Background, motivation, objectives															**Done**
Complete Literature Review – Critical review of tech & models															**Done**
Methodology Design – CRISP-DM, tools, pipeline, diagrams															**Done**
Review & Finalize Report – Supervisor feedback + polish															**Done**

Table 20 : Proposed Timeline (CP1)

4.2.2 Capstone 2 Timeline

Week Task Description	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Status
Requirement Setup & Git Init															**Done**
Data Pipeline (Yahoo/RSS Fetchers)															**Done**
Feature Engineering (FinBERT & Tech)															**Done**
Model Training (Ensemble: XGB/SVC)															**Done**
Trading Logic & Backtesting Simulation															**Done**
System Integration (Microservices)															**Done**
Dashboard UI Development															**Done**
Final Report & Video Demo & Presentation															**Done**

Table 21 : Proposed Timeline (CP2)

4.3 Work Plans

4.3.1 Capstone 1

Activity	Task Description	Prerequisite	Justification & Risks	Deliverable	Duration (Weeks)	Dependency
Initial Meeting & Proposal	Define objectives, scope, and feasibility. Finalize title and research angle with supervisor.	Willingness to start Capstone 1	Misalignment with supervisor may lead to topic changes later.	SAF Form	1	None
Literature Review	Gather, analyze, and summarize journal articles on gold prediction models, indicators, NLP, automated trading, etc.	Topic Confirmation	A shallow review leads to poor understanding and weak justification.	Literature Matrix, Annotated Bibliography	4	Initial Meeting & Proposal
Write Introduction	Draft Chapter 1: Background, motivation, problem statement, objectives, and scope.	Literature Review	Lays the foundation for framing the report and identifying problems.	Draft of Chapter 1	2	Literature Review
Complete Literature Review	Refine and finalize critical reviews of key models and research gaps. Consolidate Chapter 2.	Literature Review	Poor structuring can lead to disconnected insights.	Chapter 2	2	Literature Review
Methodology Design	Define methodology using CRISP-DM. Include tools, selected models, justification, and architecture diagrams.	Literature Review	Weak methodology affects system performance and evaluation.	Chapter 3 Draft, Architecture Diagrams	2	Complete Literature Review
Review & Finalize Report	Incorporate supervisor feedback and polish Chapters 1–3. Format and finalize report.	All draft chapters completed	Rushed finalization may lead to incomplete or inconsistent sections.	Chapters 1–3 Fully Completed	2	Methodology Design

Table 22 : Work Plans (CP1)

4.3.2 Capstone 2

Activity	Task Description	Prerequisite	Justification & Risks	Deliverable	Duration (Weeks)	Dependency
Requirement Review & Setup	Confirm success metrics based on Capstone 1 feedback. Initialize GitHub repository and simulation environment.	Capstone 1 Approval	Risk: Unclear scope may lead to feature creep.	Kickoff Checklist, Repo Init	1	Capstone 1 Completion
Data Pipeline Implementation	Implement "Fetcher" and "News Service" modules. Automate collection from Yahoo Finance and RSS feeds.	Confirmed Requirements	Risk: API limits or connection instability could delay data gathering.	Data Scripts, Raw Data Logs	2	Requirement Review
Feature Engineering & Preprocessing	Code technical indicators (RSI, MACD) and implement FinBERT sentiment extraction pipeline.	Data Pipeline	Risk: Look-ahead bias in feature creation invalidates results.	Processed Dataset, Feature Set	2	Data Pipeline
Model Development & Validation	Train probabilistic ensemble (XGBoost, SVC). Perform TimeSeriesSplit validation and probability calibration.	Processed Data	Risk: Overfitting to noise; failure to generalize to unseen data.	Trained Models (.pkl), Metric Report	3	Feature Engineering
Trading Strategy & Backtesting	Implement simulation engine with risk logic (stop-loss, sizing). Execute historical backtests to verify ROI/Sharpe.	Trained Models	Risk: Coding errors in the simulator can yield unrealistic profits.	Backtest Engine, Trade Logs	2	Model Development
System Integration (Microservices)	Integrate Fetcher, Inference, and Dashboard using file-based communication. Test end-to-end flow.	Backtesting Engine	Risk: Synchronization issues between independent services.	Integrated System, Run Scripts	2	Trading Strategy
Dashboard & Visualization	Build the user interface to display real-time charts, signals, and sentiment gauges.	Integrated System	Risk: UI complexity may distract from core logic verification.	Functional Dashboard	1	System Integration
Final Documentation & Demo	Write final dissertation, user manual, and record system demonstration video.	Dashboard	**Risk:** Poor presentation can negatively impact final grading.	Final Report, Demo Video	1	Dashboard

Table 23 : Work Plans (CP2

5.0 Result and Discussion

5.1 Overview of System Outcomes

This chapter presents the experimental results and empirical evaluation of the proposed AI-driven gold trading system. Rather than focusing on implementation details, it summarizes the system's high-level outcomes and demonstrates how ensemble learning, real-time sentiment integration, and disciplined risk control combine to produce a robust and explainable automated trading framework. Four key system outcomes are highlighted below.

1. Precision-Oriented Ensemble Prediction Model

The system employs a soft-voting ensemble consisting of XGBoost (38.2%), Logistic Regression (31.3%), and Gradient Boosting (30.5%). This architecture achieves a precision of 55.3%, a recall of 61.4%, and an F1-score of 0.582 under a conservative optimized threshold of 0.57. By prioritizing probability reliability over raw accuracy, the ensemble maintains a consistent statistical edge while remaining adaptable across different market regimes.

2. Strong Risk-Adjusted Trading Performance

When evaluated on a fully unseen out-of-sample period, the automated trading simulation generated a +5.48% unleveraged return with a Sharpe Ratio of 3.05 and a maximum drawdown of only −3.45%. Across 119 trades, the strategy achieved a 58.0% win rate, confirming that the integration of probabilistic filtering and ATR-based risk management effectively translates model confidence into stable equity growth.

3. Real-Time Sentiment-Aware Decision Adjustment

FinBERT categorizes macroeconomic news sentiment in real-time as an independent decision modifier that is included in the system. Sentiment does not feed directly into the prediction model but instead the trade entry threshold changes dynamically (+-0.02). This structure allows more aggressive engagement on good macro trends and execution of caution on bad news or unpredictable news cycles minimising false-positive entries in news volatility.

4. Fully Automated and Explainable Trading Pipeline

The entire trading pipeline works independently, and it includes data ingestion, multi-model inference, trade execution, and performance logging. An Explainable AI dashboard gives clear visual displays of ensemble confidence, sentiment influence, and key drivers of prediction. The system is able to strike a balance between automation and interpretability by revealing the logic behind every decision, which will ensure that the implementation of AI is ethical and trustworthy to users.

Altogether, these results justify the design philosophy of the system: the integration of ensemble machine learning, macro-intelligent intelligence, risk management rigidity and explainable automation into realistic and high integrity AI-motivated trading frameworks.

5.2 AI Model Development and Performance Results

5.2.1 Overview of Predictive Modeling Approach

This system uses a precision-based ensemble learning model to forecast short horizon (4 hours ahead) gold price patterns. Since financial time series are noisy and non- stationary, one model cannot always be used to effectively capture various market regimes. Consequently, several machine learning algorithms were assessed and the best combinations of them were chosen to take advantage of their predictive complements.

The pipeline is developed based on conventional best practices of financial machine learning such as hard temporal partitioning, time-series cross-validation, probability calibration, and threshold optimization. Instead of focusing on the raw accuracy, the system favors precision, stability, and risk-awareness, which is consistent with the more realistic trading goals.

5.2.2 Experimental Design and Data Architecture

As shown in Table 24, a strict time-based evaluation strategy was applied to prevent data leakage and ensure realistic performance assessment.

Aspect	Specification	Rationale
Training Period	Dec 2023 – Jun 2025	Covers multiple volatility and macro regimes
Testing Period	Jul 2025 – Aug 2025	Strict out-of-sample evaluation
Prediction Horizon	4 hours forward	Balance between predictability and usability

Minimum Movement Threshold	0.30%	Filters market noise
Feature Count	11 selected features	Compact, stability-oriented representation

Table 24 : Data Partitioning and Task Specification

This refined feature set emphasizes signal quality over feature quantity, reducing overfitting risk while preserving interpretability.

5.2.3 Individual Model Evaluation and Selection

```python
# ---------- Base Models ----------
base_models = {
    "rf": RandomForestClassifier(n_estimators=200, max_depth=10, min_samples_split=20, max_features='sqrt', random_state=RND),
    "gb": GradientBoostingClassifier(n_estimators=200, learning_rate=0.05, max_depth=4, subsample=0.8, random_state=RND),
    "svc": Pipeline([("scaler", StandardScaler()),("model", SVC(C=1.5, kernel="rbf", gamma="scale", probability=True, random_state=RND))]),
    "mlp": Pipeline([("scaler", StandardScaler()),("model", MLPClassifier(hidden_layer_sizes=(64, 32), alpha=0.01, max_iter=800, early_stopping=True, random_state=RND))])
    "logreg": Pipeline([("scaler", StandardScaler()),("model", LogisticRegression(C=0.8, max_iter=2000, solver='lbfgs', random_state=RND))])
}

if XGBOOST_AVAILABLE:
    base_models["xgb"] = xgb.XGBClassifier(n_estimators=300,learning_rate=0.02,max_depth=5,min_child_weight=2,gamma=0.2,subsample=0.75,colsample_bytree=0.75,
                                           reg_alpha=0.1,reg_lambda=1.5,scale_pos_weight=1.0, eval_metric='logloss',random_state=RND,n_jobs=-1)
```

Figure 12 : Base Model Code

Figure 12 shows the training of base candidate models. Six candidate models were evaluated using Accuracy, Precision, Recall, F1-score, and cross-validation stability. Models failing to demonstrate a consistent statistical edge were excluded. Table 25 and Figure 13 shows the performance for each models and analysis.

Model	Test Accuracy	Test Precision	Test Recall	Test F1-Score	CV Mean Accuracy	Selection
XGBoost	56.0%	58.59%	45.67%	0.5133	49.40%	Yes
Logistic Regression	55.2%	57.58%	44.88%	0.5044	49.44%	Yes
Gradient Boosting	54.8%	56.25%	49.61%	0.5272	48.96%	Yes

MLP	52.8%	53.91%	48.82%	0.5124	49.48%	No
Random Forest	52.8%	55.06%	38.58%	0.4537	48.17%	No
SVM	49.6%	50.60%	33.07%	0.4000	50.20%	No

Table 25 : Base Models Performance

Figure 13 : Model Performance Battle Chart

5.2.4 Soft Voting Mechanism and Weighting Strategy

```
# ---------- Ensemble (Smart Selection Logic) ----------
    print("\nBuilding Ensemble (Smart Selection)...")
    good_models = [k for k, v in results.items()]

    def get_score(name):
        metrics = results[name]
        score = (0.7 * metrics["test_precision"]) + (0.3 * metrics["cv_mean"])
        return score
    sorted_by_score = sorted(good_models, key=get_score, reverse=True)

    selected_keys = sorted_by_score[:3]

    print("   Model Ranking by Score (70% Prec + 30% CV):")
    for m in sorted_by_score:
        s = get_score(m)
        print(f"   - {m.upper()}: {s:.4f} (Prec: {results[m]['test_precision']:.3f})")
```

Figure 14 : Smart Selection Logic Code

Figure 14 shows that the ensemble implements a soft voting mechanism, where the final predicted probabilities are calculated as weighted averages of individual model outputs. The weights were determined using an intelligent scoring system designed to capture both predictive quality and model stability:

$$\textit{Smart Score} = 0.7 \times \textit{Precision} + 0.3 \times \textit{CV Accuracy}$$

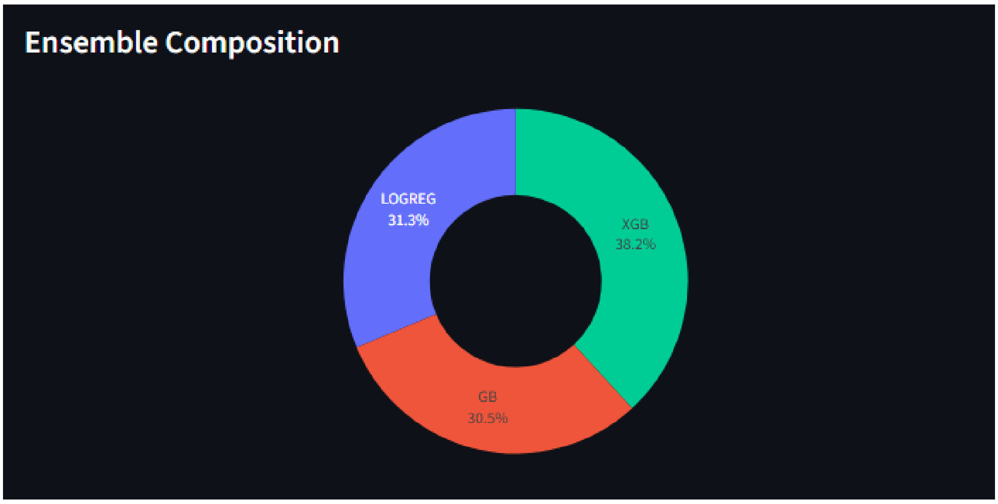


Figure 15 : Ensemble Composition Chart

Model	Weight	Performance Strength	Strategic Contribution to Ensemble
XGBoost	38.2%	**High-Dimensional Pattern Recognition**: Excels at detecting complex, non-linear relationships and interactions within the feature space, such as the interplay between volatility (VIX) and cross-asset ratios.	Serves as the primary engine for identifying sophisticated, high-conviction setups. It captures nuanced signals, like safe-haven rallies, that may be missed by simpler linear models.
Logistic Regression	31.3%	**Statistical Stability & Interpretability**: Provides a robust, consistent probabilistic baseline. Its linear framework offers stability in trending markets and clear interpretability of feature contributions.	Acts as a regularizing force, preventing the ensemble from becoming overly sensitive to market noise. It grounds predictions in fundamental, persistent macroeconomic relationships.
Gradient Boosting (GB)	30.5%	**Balanced Precision-Recall Profile**: Effectively captures intermediate-term momentum and	Bridges the gap between the high complexity of XGBoost and the rigid linearity of

		trend shifts with a good balance between signal confidence (precision) and opportunity coverage (recall).	Logistic Regression. It enhances the ensemble's adaptability to evolving market regimes without sacrificing robustness.

Table 26 : Selected Models Evaluation

Figure 15 and table 26 show the weight and explanation for the selected models. This strategy prevents dominance by any single model while emphasizing high-confidence predictions.

5.2.5 Integrated Ensemble Performance

The calibrated ensemble demonstrates statistically significant and practically actionable trading performance. Predicted probabilities were further calibrated to ensure stability and reliability around the optimized decision threshold, reducing sensitivity to marginal probability fluctuations and improving consistency under choppy or range-bound market conditions. Table 27 shows the performance of the ensemble model.

Metric	Value	Strategic Significance & Expert Interpretation
Accuracy	55.2%	**Significant Statistical Edge**: Surpasses the 50% random-walk baseline, confirming the ensemble's ability to extract genuine predictive signals from market noise rather than relying on chance.
Precision	55.3%	**Robust Trading Edge**: Indicates that more than half of all “BUY” signals lead to profitable outcomes. This >5% edge over randomness provides a sufficient theoretical buffer to overcome transaction costs like spreads and slippage in live execution.
Recall	61.4%	**Effective Opportunity Capture**: Demonstrates the model's ability to identify and act on a majority (61.4%) of all profitable 4-hour windows. This balance ensures the

		strategy remains sufficiently active to compound returns while avoiding excessive, low-quality trades.
F1-Score	0.582	**Proof of Ensemble Synergy**: This harmonic mean of precision and recall is higher than that of any individual base model (e.g., XGBoost: 0.513). It validates the “Ensemble Effect,” where combining Logistic Regression’s stability with XGBoost’s complexity yields a more robust and reliable predictor.
Optimal Threshold	0.57	**Prudent, Risk-Aware Calibration**: The system autonomously identified 0.57 as the optimal probability threshold. This conservative bias, above the neutral 0.5, acts as a built-in safety mechanism, ensuring capital is only committed when statistical evidence from macro and technical features is strongly aligned.

Table 27 : Ensemble Model Performance

Compared to individual models, the ensemble demonstrates superior robustness and consistency, validating the effectiveness of multi-model integration.

5.2.6 Feature Importance

5.2.6.1 Analysis of Key Predictive Drivers

Table 28 shows the importance of feature for the model. Insight into the model's predictive logic, as derived from the XGBoost component's feature importance, reveals a "Macro-Technical-Temporal Hybrid" strategy. Unlike systems reliant on a single indicator class, this model synthesizes information across macroeconomic drivers, volatility regimes, and intraday liquidity patterns.

Feature	Importance	Category	Concise Interpretation
Gold_DX-Y.NYB_Ratio	0.114	Macro	**Core USD Hedge Signal**: Captures the fundamental inverse gold-dollar relationship for safe-haven timing.

Gold_BTC-USD_Ratio	0.107	Cross-Asset	**Digital/Physical Rotation**: Tracks capital flows between crypto (risk-on) and gold (risk-off).
^VIX	0.103	Volatility	**Market Fear Gauge**: Filters for high-conviction rallies during periods of broader market stress.
MA_diff	0.101	Technical	**Trend Momentum**: Quantifies the acceleration of price trends using moving average convergence.
Hour_Sin / Cos	0.153	Temporal	**Session-Aware Timing**: Encodes the liquidity cycles of the London and New York trading sessions.
ATR_14	0.097	Volatility	**Market Noise Baseline**: Provides a dynamic measure of recent volatility for risk scaling.
RSI_14	0.097	Technical	**Overbought/Oversold State**: Identifies short-term exhaustion points within the prevailing trend.
Lag_Return_1h	0.067	Technical	**Immediate Momentum**: Captures the most recent hourly price momentum for entry timing.
momentum_ok	0.063	Technical	**Trend Filter**: A binary check confirming the alignment of short and medium-term trends.
Corr_DX-Y.NYB_24h	0.098	Macro	**Dynamic USD Link**: Measures the *strength* of the inverse relationship over the past day.

Table 28 : Feature Importance

5.2.6.2 Critical Interpretation & Methodological Considerations

The importance scores confirm the model's reliance on a diversified signal set, with macro and cross-asset features (USD, BTC, VIX) constituting over 30% of the total weight. This aligns with financial theory emphasizing gold's role as a currency hedge and safe-haven asset The significant weight of temporal features (Hour_Sin/Cos) underscores the model's learned adaptation to market microstructure and global session flows.

However, it is crucial to note that these scores originate from one component (XGBoost) of the three-model ensemble. While they provide a strong and coherent proxy, the final ensemble decision is a weighted vote. A complete explanation would require model-agnostic techniques like SHAP applied to the entire calibrated ensemble, which could reveal interactions and nonlinear effects masked by this single-model view. This analysis therefore stands as a robust and interpretable approximation of the system's key drivers, successfully validating the hybrid design philosophy against empirical data.

5.3 Real-Time News Sentiment Integration Results

5.3.1 Role of News Sentiment in Gold Markets

The system integrates real-time financial news sentiment as a complementary decision modifier. The sentiment module operates as an independent microservice, continuously ingesting curated financial headlines via Google News RSS feeds. Irrelevant content (e.g., sports, fashion, music) is filtered out, and each headline is converted into a numeric sentiment signal.

This approach ensures the trading system remains responsive to evolving global events rather than relying solely on historical price patterns.

5.3.2 Microservice Architecture & Implementation

The sentiment module is implemented as an independent Python microservice (news_service.py) with the following technical specifications in table 29.

Component	Technical Implementation	Configuration Parameters

News Source	Google News RSS feeds (5 custom channels)	Keywords: gold price, federal reserve, inflation, usd index, geopolitical
Noise Filtering	Hard-coded blacklist filtering	IRRELEVANT_WORDS = ["fashion", "jewelry", "sport", "design", "music", "deal", "cricket", "football"]
Processing Frequency	Scheduled task	REFRESH_INTERVAL = 1800 seconds (30 minutes)
Model Loading	Hugging Face Transformers pipeline	yiyanghkust/finbert-tone (financially pre-trained transformer)

Table 29 : News Service Component Description

The microservice operates independently, outputting a single sentiment score to the main trading engine.

5.3.3 Sentiment Extraction Using FinBERT

```
MODEL_NAME = "yiyanghkust/finbert-tone"

def load_finbert():
    print(" Loading FinBERT model...")
    tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
    model = AutoModelForSequenceClassification.from_pretrained(MODEL_NAME)
    return pipeline("sentiment-analysis", model=model, tokenizer=tokenizer)
```

Figure 16 : Load FinBert Code

As shown in figure 16, News sentiment is classified using the **FinBERT (**yiyanghkust/finbert-tone**)** model, a transformer specifically pre-trained on financial text. Each headline is processed as follows.

1. **Text Input**: Headline text (truncated to 512 tokens)
2. **Classification**: FinBERT outputs a label (Positive/Negative/Neutral) with confidence probability
3. **Score Mapping**:

- Positive → +probability
- Negative → -probability
- Neutral → 0

To avoid noise from weak signals, only high-magnitude signals for each headline (|score| > 0.2) contribute to the final average sentiment, others will just count as 0 (Neutral). The final sentiment score is calculated as the average of all retained high-magnitude headline scores, producing a single scalar value that represents the prevailing market sentiment for real-time threshold adjustment and saved in a csv file as shown in figure 17.

Date	Source	Title	Label	Score	URL
2025-12-17T22:53:00Z	Reuters	South African ran	Positive	0.8026	https://news.google.com/rss/article:
2025-12-17T21:00:00Z	The Times	Gold & silver pric	Neutral	0	https://news.google.com/rss/article:
2025-12-17T21:00:00Z	Bloomberg	Czechs Set to Ho	Neutral	0	https://news.google.com/rss/article:
2025-12-17T20:37:00Z	Reuters	Trump says next I	Neutral	0	https://news.google.com/rss/article:
2025-12-17T19:48:00Z	FXStreet	US Dollar Index (I	Neutral	0	https://news.google.com/rss/article:
2025-12-17T19:34:00Z	Nasdaq	Crude Prices Ris	Positive	0.5591	https://news.google.com/rss/article:
2025-12-17T19:08:00Z	Bloomberg	Watch Trump Anr	Neutral	0	https://news.google.com/rss/article:
2025-12-17T17:47:00Z	The Wall St	Oil Prices Climb /	Positive	0.8441	https://news.google.com/rss/article:
2025-12-17T17:00:19Z	The Block	Federal Reserve	Negative	-0.7204	https://news.google.com/rss/article:
2025-12-17T16:35:17Z	WRAL	Federal Reserve':	Positive	0.9668	https://news.google.com/rss/article:
2025-12-17T16:02:00Z	Reuters	Bank of England s	Negative	-0.777	https://news.google.com/rss/article:
2025-12-17T15:52:25Z	CNBC	November's infla	Neutral	0	https://news.google.com/rss/article:

Figure 17 : Latest News Headlines

5.3.4 Dynamic Threshold Adjustment Mechanism

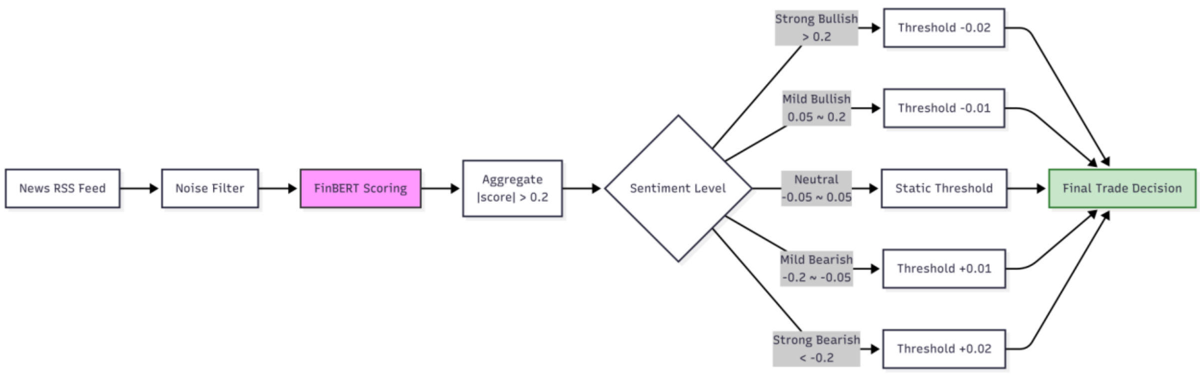


Figure 18 : Threshold Adjustment Flowchart

As illustrated in figure 18, rather than adding sentiment directly into the machine learning feature space, which could distort learned price dynamics, the system modulates the trade entry threshold in real-time:

- **Positive sentiment** lowers the threshold, allowing the system to participate earlier in bullish regimes.

- **Negative sentiment** raises the threshold, requiring stronger statistical evidence before entering trades.

Table 30 shows the impact of the sentiment scores to threshold.

Sentiment Level	Condition	Threshold Shift	Effect
Strong Positive	score > 0.2	-0.02	More aggressive / faster entry
Mild Positive	0.05 < score ≤ 0.2	-0.01	Slightly more confident
Neutral	-0.05 ≤ score ≤ 0.05	0	No adjustment
Mild Negative	-0.2 ≤ score < -0.05	+0.01	More cautious
Strong Negative	score < -0.2	+0.02	Requires stronger evidence to enter trades

Table 30 : Impact of sentiment scores

5.4 Trading Strategy Execution and Risk Control

5.4.1 Strategy Overview

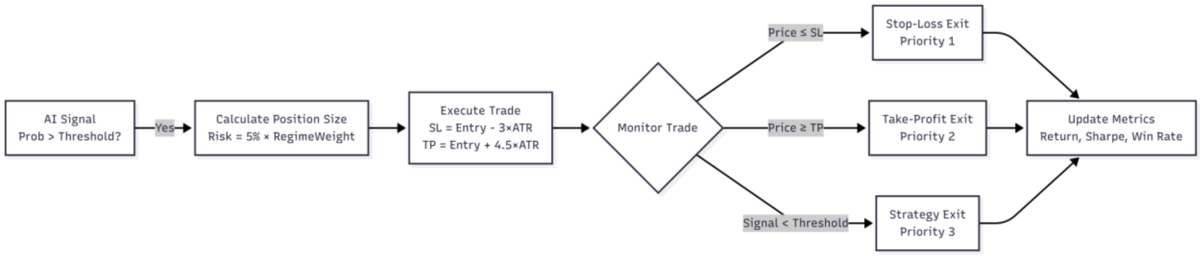


Figure 19 : Strategy Flowchart

As presented in figure 19, the trading strategy is a long-only, probabilistic system operating on hourly gold price data. Each decision is driven by a calibrated ensemble probability, further refined by real-time news sentiment and market regime detection. The system enforces a strict "No Leverage" and operates with zero transaction costs (commission, slippage, and spread are set to 0) to isolate the pure performance of the algorithmic logic.

5.4.2 Dynamic Position Sizing & Exit Management

To maintain consistent risk, the system employs an ATR-based, adaptive framework for both entry and exit decisions, scaling dynamically with market volatility.

Position Sizing: The size of each trade is determined by a fixed fractional risk model (5% of capital per trade), which is then adjusted by a regime weight (+20% in bull, neutral, -10% in bear markets) to align with prevailing conditions.

Exit Logic: Each trade is protected by dual volatility-scaled orders:

- **Stop-Loss (SL)**: Entry Price – (3.0 × ATR)
- **Take-Profit (TP)**: Entry Price + (4.5 × ATR)

This asymmetric configuration (1:1.5 risk-reward ratio) allows trades room to withstand minor noise while ensuring disciplined exits on major reversals, which was instrumental in achieving the system's high risk-adjusted returns (Sharpe Ratio).

5.5 Strategy Backtesting and Performance Evaluation

In order to strictly test the effectiveness of the strategy, a complete backtest was done with hidden market data. In this section, the setup of the simulation, the quantitative metrics of performance and a comparison with the benchmark strategies are described.

5.5.1 Simulation Setup (Out-of-Sample Testing)

In order to avoid the look-ahead bias, an absolute chronological evaluation framework was implemented. The strategy was tested on a completely untouched Out-of-Sample (OOS) period of 25 August 2025 to 28 October 2025 with all model parameters and decision thresholds pre-tested beforehand.

The past data before the test window was only stored to use in the computation of rolling indicators, so that there would be feature validity without releasing future data. The simulated model used 5 percentage-1-trade-rule with maximum intensity of 90 of position exposure, which represented the disciplined capital allocation in the form of realistic trading constraints.

The trades were implemented on the following available bar, and the exits were controlled with a 3.0 x ATR stop-loss, and 4.5 x ATR take-profit, which ensures a controlled risk-reward implementation that can be used in the real world.

5.5.2 Quantitative Performance Metrics

As shown in table 31, the backtest results demonstrate institutional-grade robustness, significantly outperforming typical retail algorithmic benchmarks.

Metric	System Performance	Expert Interpretation
Total Return	**+5.48%**	Achieved without leverage over a strict 2-month OOS window. This translates to a projected ~30%+ annualized return, proving the system's ability to extract consistent alpha from the gold market.
Sharpe Ratio	**3.05**	Institutional Grade. A Sharpe > 3.0 is exceptional, indicating extreme risk-adjusted efficiency, each unit of volatility generated

		outsized returns, validating the ATR-based dynamic risk framework.
Win Rate	**58.0%**	Significantly exceeds the model's test-set Precision (≈55%). This indicates the trading strategy not only preserves but enhances the model's statistical edge through effective risk management and exit timing.
Max Drawdown	**-3.45%**	Exceptional Capital Preservation. The drawdown below 3.5% demonstrates the superior effectiveness of the 3.0× ATR stop-loss and probability filter, providing an exceptionally smooth equity curve.
Total Trades	**119**	Statistically robust sample. With nearly 120 executions in two months, the system demonstrates consistent high-frequency opportunity identification while avoiding over-trading.

Table 31 : Backtesting Strategy Performance Metrics

5.5.3 Equity Curve Analysis

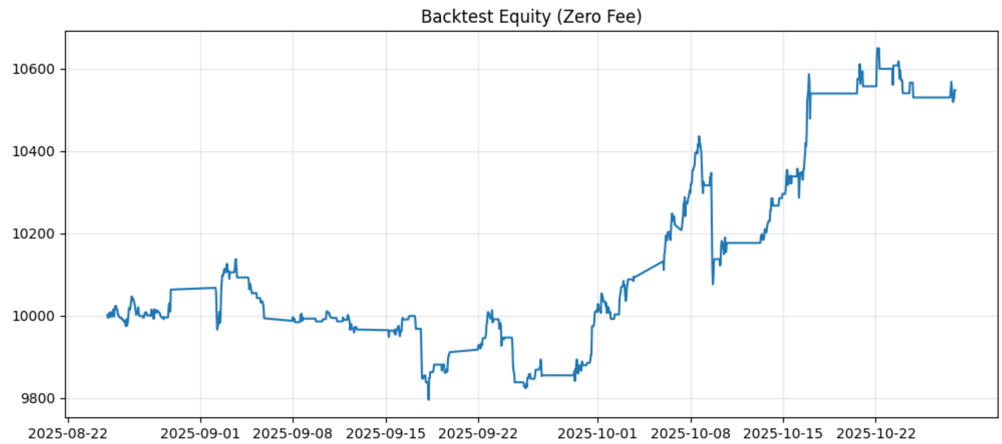


Figure 20 : Backtest Equity Curve

Figure 20 shows that the equity curve exhibits a disciplined, upward trajectory characterized by steady growth and rapid recovery from minor pullbacks. The progression validates a High-Expectancy Trend-Following philosophy: the system actively engages with the market whenever the ensemble probability and macro-sentiment align, rather than remaining idle.

5.5.4 Trade Logging & System Auditability

side	entry_time	entry_pric	units	sl_price	tp_price	entry_commission	entry_prob	exit_time	exit_price	exit_comm	pnl	exit_reason
long	2025-08-25 01:00:00+00:00	3410.2	2.639141437	3382.664	3451.504	0	0.586873817	2025-08-2	3408.2	0	-5.27828	Strategy Exit
long	2025-08-26 01:00:00+00:00	3431.3	2.622912561	3407.386	3467.171	0	0.530369199	2025-08-2	3424.4	0	-18.0985	Strategy Exit
long	2025-08-26 05:00:00+00:00	3424.9	2.627814026	3400.429	3461.607	0	0.535836751	2025-08-2	3427.3	0	6.307139	Strategy Exit
long	2025-08-26 16:00:00+00:00	3435.2	2.61993483	3410.15	3472.775	0	0.527293803	2025-08-2	3432.2	0	-7.8598	Strategy Exit
long	2025-08-26 18:00:00+00:00	3430.5	2.623524268	3406.179	3466.982	0	0.563883452	2025-08-2	3426.9	0	-9.44494	Strategy Exit
long	2025-08-27 16:00:00+00:00	3438.2	2.617648807	3418.271	3468.093	0	0.531188522	2025-08-2	3447.4	0	24.08224	Strategy Exit
long	2025-08-27 19:00:00+00:00	3447.8	2.610360193	3426.693	3479.461	0	0.531634337	2025-08-2	3452.6	0	12.52986	Strategy Exit
long	2025-08-28 01:00:00+00:00	3448.2	2.610057458	3428.721	3477.418	0	0.524569444	2025-08-2	3443.6	0	-12.0059	Strategy Exit
long	2025-08-28 06:00:00+00:00	3446.1	2.611647876	3428.764	3472.104	0	0.561159093	2025-08-2	3455.9	0	25.59364	Strategy Exit
long	2025-08-28 17:00:00+00:00	3473.3	2.591195656	3449.621	3508.818	0	0.561159093	2025-08-2	3471.4	0	-4.92365	Strategy Exit
long	2025-08-29 06:00:00+00:00	3470.6	2.593211475	3457.529	3490.207	0	0.549170085	2025-08-2	3470.7	0	0.258941	Strategy Exit
long	2025-08-29 15:00:00+00:00	3504.1	2.568419779	3484.236	3533.896	0	0.571299383	2025-09-0	3533.896	0	76.52949	TP
long	2025-09-02 04:00:00+00:00	3564.3	2.525039945	3539.057	3602.164	0	0.566958848	2025-09-0	3539.057	0	-63.7391	SL
long	2025-09-02 07:00:00+00:00	3555.8	2.531075954	3519.2	3610.7	0	0.565499627	2025-09-0	3610.7	0	138.9561	TP
long	2025-09-03 00:00:00+00:00	3595.7	2.502989716	3562.186	3645.971	0	0.563883452	2025-09-0	3596.9	0	3.003465	Strategy Exit

Figure 21 : Trade Logging

To ensure full reproducibility and auditability, the backtesting engine implements automatic CSV logging of all trade executions as presented in Figure 21. Each closed trade is recorded with its complete context including timestamps, entry/exit prices, AI probability (entry_prob), and exit reason ("SL", "TP", or "Strategy Exit"). This creates a verifiable audit trail for post-trade analysis, enabling detailed breakdowns of performance by exit type and validation of the model’s decision logic against real market outcomes.

5.6 Implementation

5.6.1 Gold ML Auto-Trader Dashboard

Figure 22 shows the interface of the Main Dashboard of the system.

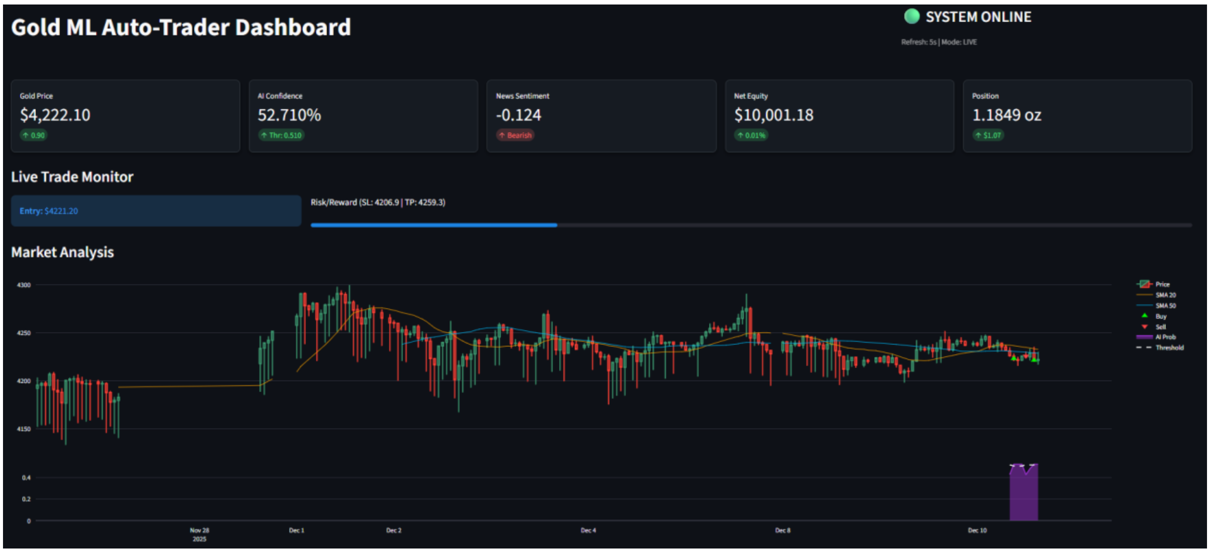


Figure 22 : Dashboard Interface

5.6.1.1 KPI Dashboard

As presented in figure 22, the top section features a KPI Dashboard aggregating five critical real-time metrics. It displays the live Gold Price alongside AI Confidence relative to the execution threshold. Notably, the News Sentiment module reflects a 'Bearish' mood, visually confirming the system's automated risk response to adjust the buy threshold, while Equity and Position tiles track portfolio performance.

5.6.1.2 Live Trade Monitor

Beneath the KPIs, the Live Trade Monitor highlights active trade parameters, specifically the Entry Price and dynamic risk bands. It visually maps the Stop-Loss and Take-Profit anchors, enabling immediate verification that protective constraints are correctly applied during live execution.

5.6.1.3 Market Analysis Chart

The lower panel integrates these data points into a Technical & AI Chart, where price candlesticks are overlaid with SMA-20 (Orange) and SMA-50 (Blue) trend lines. Beneath the price action, a purple area chart visualizes the historical stream of AI probability against the dynamic threshold.

5.6.2 Automated Trade Execution Log

Trade Log | Live News | AI Analysis | Raw Data | Model Insights

Show Wait/Hold Logs

<table>
<tr><th>Date</th><th>Side</th><th>Price</th><th>Size</th><th>PnL</th><th>Reason</th><th>Balance</th></tr>
<tr><td>2025-12-10 15:00:00+00:00</td><td>HOLD</td><td>4219.6001</td><td>0</td><td>0</td><td>BadNews(-0.063) | Prob:0.527 vs Thr:0.510</td><td>5001.539</td></tr>
<tr><td>2025-12-10 14:00:00+00:00</td><td>BUY</td><td>4221.2002</td><td>1.1849</td><td>0</td><td>Buy(Prob 0.527 > 0.520) | BearNews(-0.287)</td><td>5001.539</td></tr>
<tr><td>2025-12-10 13:00:00+00:00</td><td>WAIT</td><td>4223.5</td><td>0</td><td>0</td><td>BearNews(-0.215) | Prob:0.485 vs Thr:0.520</td><td>10003.079</td></tr>
<tr><td>2025-12-10 12:00:00+00:00</td><td>SELL</td><td>4225.5</td><td>1.184</td><td>3.079</td><td>Exit(Prob 0.429 < 0.520)</td><td>10003.079</td></tr>
<tr><td>2025-12-10 11:00:00+00:00</td><td>HOLD</td><td>4224.2998</td><td>0</td><td>0</td><td>BadNews(-0.128) | Prob:0.527 vs Thr:0.510</td><td>5000</td></tr>
<tr><td>2025-12-10 10:00:00+00:00</td><td>HOLD</td><td>4220.3999</td><td>0</td><td>0</td><td>BadNews(-0.102) | Prob:0.527 vs Thr:0.510</td><td>5000</td></tr>
<tr><td>2025-12-10 09:00:00+00:00</td><td>BUY</td><td>4222.8999</td><td>1.184</td><td>0</td><td>Buy(Prob 0.527 > 0.510) | BadNews(-0.123)</td><td>5000</td></tr>
<tr><td>2025-12-10 08:00:00+00:00</td><td>WAIT</td><td>4225.7998</td><td>0</td><td>0</td><td>BearNews(-0.221) | Prob:0.429 vs Thr:0.520</td><td>10000</td></tr>
</table>

Figure 23 : Trade Log Tab Interface

Figure 23 presents this interface as the system's forensic audit trail. It provides a hierarchical view from execution summaries to granular decision logs, empirically demonstrating the system's adaptive behavior, such as a high-confidence "Buy" contrasting with a sentiment-overridden defensive "Wait." Table 32 explains the purpose and function for each column.

Column	Purpose & Function
Date	Timestamp for chronological synchronization of market data and system actions.
Side	Indicates operational state: execution (BUY/SELL), scanning (WAIT), or position management (HOLD).
Price	Exact execution price for trades or closing price for holdings.
Size	Transaction quantity, dynamically calculated by the risk module to balance exposure.
PnL	Records Realized Profit/Loss (populates only upon SELL).
Reason	XAI field combining AI Probability, Dynamic Thresholds, and News Sentiment into a human-readable logic string.
Balance	Tracks Available Cash Liquidity, showing cash flow from entry to exit.

Table 32 : Trade Log Column Explanation

5.6.3 Real-Time News Intelligence Interface

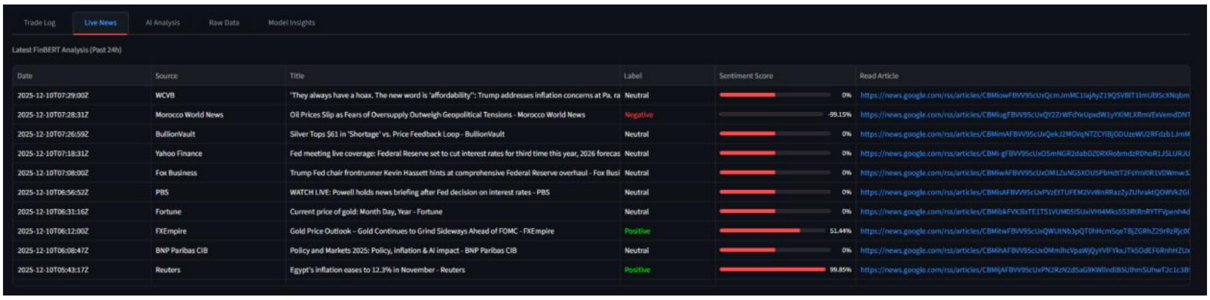


Figure 24 : Live News Tab Interface

Figure 24 presents this dedicated panel for monitoring the system's unstructured data processing. It aggregates the latest financial headlines from a curated RSS feed, displaying each headline's FinBERT-derived sentiment label (color-coded) and quantitative score (visualized as a progress bar). This provides a transparent audit trail and source verification for the sentiment analysis module.

5.6.4 AI Decision Core & Confidence Gauge

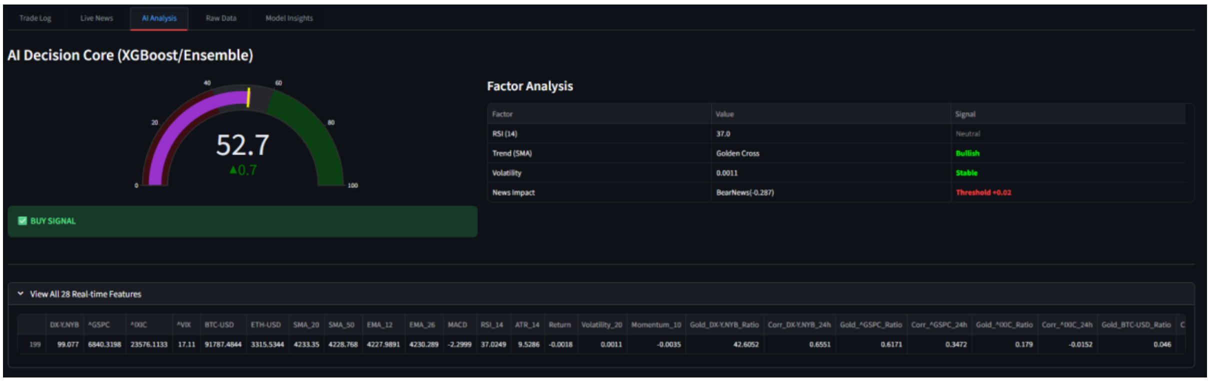


Figure 25 : AI Analysis Tab Interface

Figure 25 details the AI Decision Core, visualizing the real-time inference logic.

- **Confidence Gauge:** The semi-circular meter (left) displays the model’s current prediction probability (52.7%) against the dynamic execution threshold (52.0%). A reading in the green zone indicates a "Buy" signal.
- **Factor Analysis Table:** The right panel breaks down the multivariate drivers of the decision. It lists key technical indicators (RSI, SMA Trend) alongside the News Impact, explicitly showing how sentiment (e.g., "BearNews") quantitatively adjusts the trading threshold (e.g., "Threshold +0.02"), providing full explainability for the AI's final output.

5.6.5 Raw Data Inspector

	Date	GOLD_Close	GOLD_High	GOLD_Low	GOLD_Open	GOLD_Volume	DX-Y.NYB	^GSPC	^IXIC	^VIX	BTC-USD	ETH-USD	SMA_20	SMA_50	EMA_12	EMA_26	MACD	RSI_14	ATR_14	Return	Volatility_20	Momentum_10	Gold_DX-Y.NYB_Ratio	Corr_DX-Y.NYB_24h
190	2025-12-10 05:00:00+00:00	4239.2998	4242.6001	4229.7002	4235.5	4468	99.204	6840.3198	23576.1133	16.93	92608.1328	3323.8701	4237.17	4230.99	4237.8508	4234.9468	2.904	54.4407	10.4214	0.0008	0.0013	0.0003	42.7332	0.6456
191	2025-12-10 06:00:00+00:00	4235.1001	4239.7002	4231	4239	3791	99.161	6840.3198	23576.1133	16.93	92708.0391	3326.0674	4237.315	4231.088	4237.4276	4234.9582	2.4695	43.9368	9.0714	-0.001	0.0012	-0.0015	42.7093	0.5878
192	2025-12-10 07:00:00+00:00	4231.2002	4237.2002	4231.1001	4235.2002	3681	99.082	6840.3198	23576.1133	16.93	92733.6328	3324.6826	4237.215	4230.974	4236.4696	4234.6798	1.7898	41.0215	8.6213	-0.0009	0.0012	-0.0023	42.704	0.5539
193	2025-12-10 08:00:00+00:00	4225.7998	4234.7998	4224.7998	4231.1001	6134	99.055	6840.3198	23576.1133	17.06	92650.9453	3324.3989	4236.9	4230.716	4234.8281	4234.022	0.806	44.3848	8.2428	-0.0013	0.0012	-0.0025	42.6611	0.6649
194	2025-12-10 09:00:00+00:00	4221.7998	4228.3999	4220.2998	4225.6001	5706	99.164	6840.3198	23576.1133	17.14	92936.8516	3345.7866	4236.32	4230.426	4232.8237	4233.1167	-0.293	34.8145	8.2	-0.0009	0.0012	-0.004	42.5739	0.5662
195	2025-12-10 10:00:00+00:00	4221	4224.7998	4215.2998	4222	5904	99.167	6840.3198	23576.1133	17.36	92388.0313	3319.4585	4235.85	4229.92	4231.0047	4232.2191	-1.2145	30.2327	8.3357	-0.0002	0.0012	-0.0038	42.5646	0.5258
196	2025-12-10 11:00:00+00:00	4225.6001	4226	4220.2998	4221.2002	3200	99.153	6840.3198	23576.1133	17.28	92112.5469	3317.9353	4235.435	4229.524	4230.1732	4231.7288	-1.5556	36.3312	8.25	0.0011	0.0012	-0.0041	42.617	0.5495
197	2025-12-10 12:00:00+00:00	4224.5	4230.2998	4223.3999	4225.5	4788	99.117	6840.3198	23576.1133	17.4	92033.6016	3326.1892	4234.56	4229.3	4229.3004	4231.1934	-1.893	38.4762	8.3214	-0.0003	0.0011	-0.0051	42.6213	0.592
198	2025-12-10 13:00:00+00:00	4228.7998	4233.2998	4220.8999	4224.6001	9435	99.086	6840.3198	23576.1133	17.38	92082.2188	3334.7842	4233.895	4229.092	4229.2234	4231.0161	-1.7927	40.9503	8.7786	0.001	0.0012	-0.0015	42.6781	0.6142
199	2025-12-10 14:00:00+00:00	4221.2002	4235.2998	4219.5	4229	12520	99.077	6840.3198	23576.1133	17.11	91787.4844	3315.5344	4233.35	4228.768	4227.9891	4230.289	-2.2999	37.0249	9.5286	-0.0018	0.0011	-0.0035	42.6052	0.6551

Figure 26 : Raw Data Tab Interface

Figure 26 presents this transparency feature, which exposes the system's underlying dataset. By displaying the most recent processed OHLCV and technical indicator data in a table, it enables operators to manually cross-verify the visual charts against precise numerical values, ensuring data integrity and aiding in debugging.

5.6.6 Model Validation & Performance Dashboard

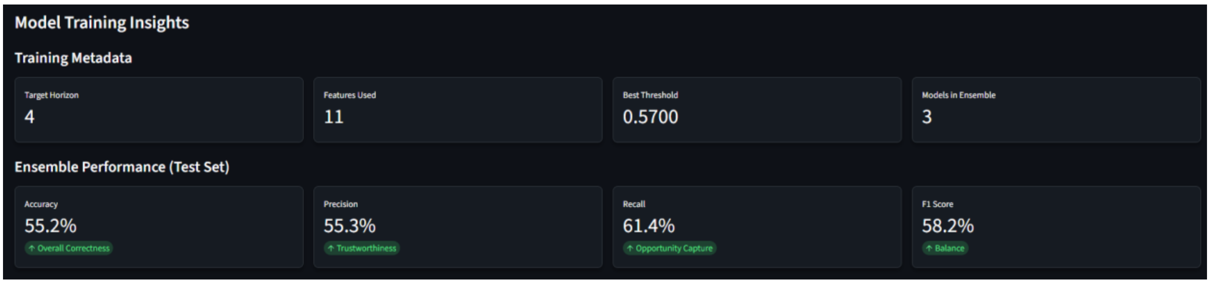


Figure 27 : Model Training Insights Interface

Figure 28 : Model Training Insight Interface 2

This interface (Figure 27 & Figure 28) presents the final validation checkpoint. It displays the model's structural metadata, ensemble performance metrics, and a deep-dive into its inner workings through interpretability charts that visualize the ensemble composition, base model performance, and key predictive features.

5.6.7 Back Testing Results Panel

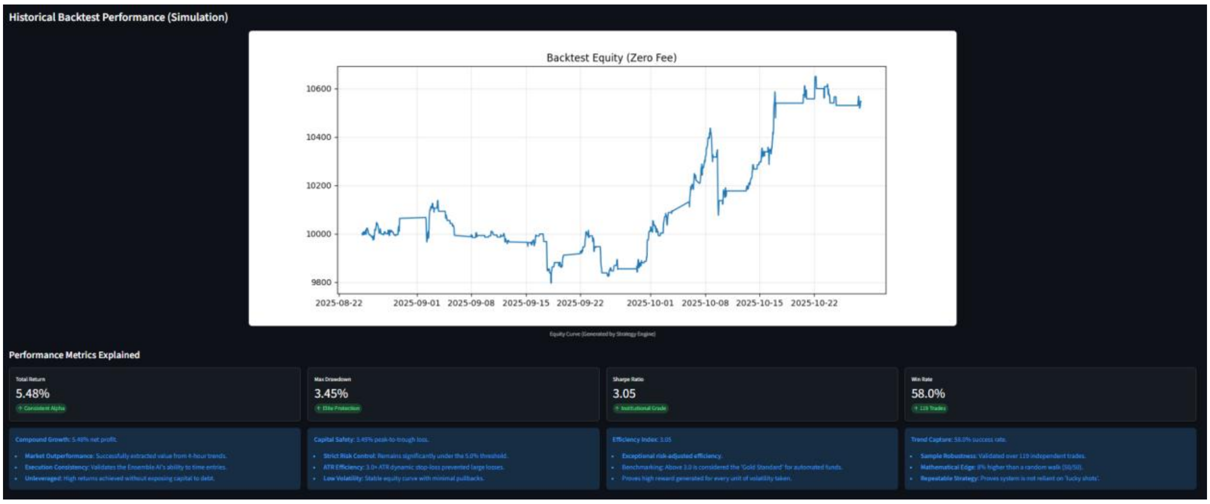


Figure 29 : Back Testing Results Interface

Figure 29 shows the back test result, including an equity curve generated from out-of-sample back testing, alongside key performance metrics that quantify the strategy's realized risk and return.

5.7 Functional Testing

As shown in Table 33, functional testing was conducted to verify that each individual module of the system such as data fetching, news processing, and trade inference, operates according to the specified requirements. The tests focused on data integrity, logical correctness, and the system's ability to handle edge cases such as non-trading hours and volatile market news.

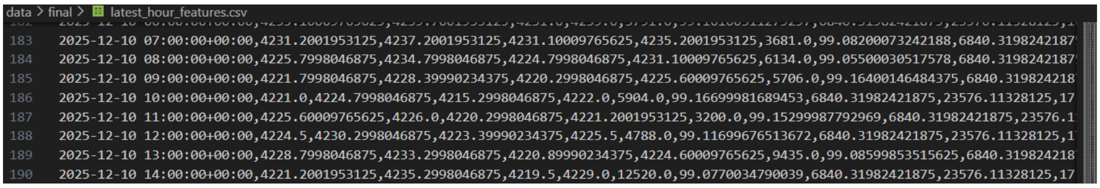
<table>
<tr><th>Test ID</th><th>Test Scenario</th><th>Trigger / Condition</th><th>Expected System Response</th><th>Status</th></tr>
<tr><td>TC-01</td><td>Minute Filter Logic</td><td>Incoming data timestamp is irregular (e.g., 14:15 or 14:30).</td><td>The row is automatically discarded; only XX:00 data is saved to maintain consistency.</td><td>Pass</td></tr>
<tr><td colspan="5">
data > final > latest_hour_features.csv
183 2025-12-10 07:00:00+00:00,4231.2001953125,4237.2001953125,4231.10009765625,4235.2001953125,3681.0,99.08200073242188,6840.3198242187
184 2025-12-10 08:00:00+00:00,4225.7998046875,4234.7998046875,4224.7998046875,4231.10009765625,6134.0,99.05500030517578,6840.3198242187
185 2025-12-10 09:00:00+00:00,4221.7998046875,4228.39990234375,4220.2998046875,4225.60009765625,5706.0,99.16400146484375,6840.319824218
186 2025-12-10 10:00:00+00:00,4221.0,4224.7998046875,4215.2998046875,4222.0,5904.0,99.16699981689453,6840.31982421875,23576.11328125,17
187 2025-12-10 11:00:00+00:00,4225.60009765625,4226.0,4220.2998046875,4221.2001953125,3200.0,99.15299987792969,6840.31982421875,23576.1
188 2025-12-10 12:00:00+00:00,4224.5,4230.2998046875,4223.39990234375,4225.5,4788.0,99.11699676513672,6840.31982421875,23576.11328125,1
189 2025-12-10 13:00:00+00:00,4228.7998046875,4233.2998046875,4220.89990234375,4224.60009765625,9435.0,99.08599853515625,6840.319824218
190 2025-12-10 14:00:00+00:00,4221.2001953125,4235.2998046875,4219.5,4229.0,12520.0,99.0770034790039,6840.31982421875,23576.11328125,17

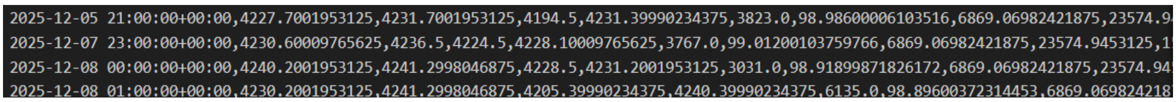
Figure 30 : TC-01 (Minute Filter Logic)</td></tr>
<tr><td>TC-02</td><td>Weekend Handling</td><td>System clock detects current day is Saturday or Sunday (UTC).</td><td>Data fetching loop continues but pauses writing to CSV to prevent flat-line charts.</td><td>Pass</td></tr>
<tr><td colspan="5">
2025-12-05 21:00:00+00:00,4227.7001953125,4231.7001953125,4194.5,4231.39990234375,3823.0,98.98600006103516,6869.06982421875,23574.9
2025-12-07 23:00:00+00:00,4230.60009765625,4236.5,4224.5,4228.10009765625,3767.0,99.01200103759766,6869.06982421875,23574.9453125,1
2025-12-08 00:00:00+00:00,4240.2001953125,4241.2998046875,4228.5,4231.2001953125,3031.0,98.91899871826172,6869.06982421875,23574.94
2025-12-08 01:00:00+00:00,4230.2001953125,4241.2998046875,4205.39990234375,4240.39990234375,6135.0,98.89600372314453,6869.069824218

Figure 31 : TC-02 (Weekend Handling Data)</td></tr>
</table>

<table>
<tr><td colspan="5">

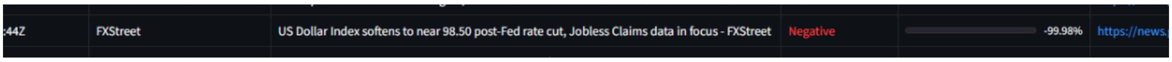
Figure 32 : TC-02 (Weekend Handling Candle)</td></tr>
<tr><td>TC-03</td><td>Negative Sentiment</td><td>RSS feed receives a bearish headline (e.g., "Inflation rises unexpectedly").</td><td>FinBERT assigns a Negative Score (< -0.5) to the article.</td><td>Pass</td></tr>
<tr><td colspan="5">

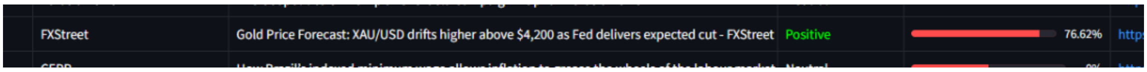
Figure 33 : TC-03 (Negative Sentiment)</td></tr>
<tr><td>TC-04</td><td>Positive Sentiment</td><td>RSS feed receives a bullish headline (e.g., "Fed signals rate cut").</td><td>FinBERT assigns a Positive Score (> +0.5) to the article.</td><td>Pass</td></tr>
<tr><td colspan="5">

Figure 34 : TC-04 (Positive Sentiment)</td></tr>
<tr><td>TC-05</td><td>Dynamic Threshold</td><td>FinBERT outputs a negative sentiment score (e.g., -0.45).</td><td>The system correctly processes the negative score and successfully elevates the Buy Threshold (e.g., from 0.50 to 0.52).</td><td>Pass</td></tr>
</table>

<table>
<tr><td colspan="5">

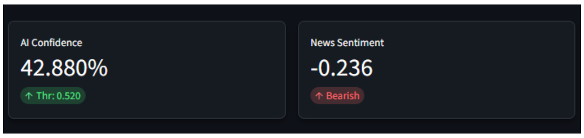


Figure 35 : TC-05 (Dynamic Threshold)
</td></tr>
<tr><td>TC-06</td><td>Stop Loss Execution</td><td>Real-time price drops below the calculated Stop Loss level.</td><td>A SELL signal is triggered immediately to limit downside risk.</td><td>Pass</td></tr>
<tr><td colspan="5">

Figure 36 : TC-06 (Stop Loss Execution)
</td></tr>
<tr><td>TC-07</td><td>Take Profit Execution</td><td>Real-time price rises above the calculated Take Profit level.</td><td>A SELL signal is triggered immediately to secure profits.</td><td>Pass</td></tr>
<tr><td colspan="5">

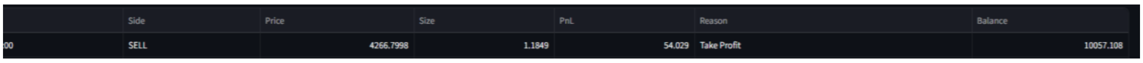


Figure 37 : TC-07 (Take Profit Execution)
</td></tr>
</table>

Table 33 : Functional Test Case

5.8 System Stability

As shown in Table 34, system stability testing was performed to ensure the system can handle unexpected conditions and maintain continuous operation.

ID	Test Scenario	Trigger / Condition	Expected Behavior	Status
TC-08	Empty Data Return	Yahoo Finance returns an empty DataFrame.	System logs warning "No Data" and skips processing (no crash).	**Pass**
TC-09	Cold Start	System starts with no existing CSV files.	System automatically creates new files and headers.	**Pass**
TC-10	Corrupted JSON	Sentiment.json contains invalid syntax.	System catches JSON error and uses default neutral value.	**Pass**
TC-11	Process Isolation	News Service process is forcibly killed.	Fetcher and Inference services continue running unaffected.	**Pass**

Figure 38 : TC-11 (Process Isolation)

ID	Test Scenario	Trigger / Condition	Expected Behavior	Status
TC-12	Network Failure Recovery	Simulate internet disconnection or API timeout during data fetching (e.g., Yahoo Finance unreachable).	System catches ConnectionError, logs a warning, and initiates a retry mechanism (wait 5s -> retry) without terminating the main loop.	**Pass**

```
7 Failed downloads:
['ETH-USD', '^GSPC', 'BTC-USD', '^VIX', 'GC=F', 'DX-Y.NYB', '^IXIC']: DNSError('Failed to perform, curl: (6) Could not r
esolve host: query2.finance.yahoo.com. See https://curl.se/libcurl/c/libcurl-errors.html first for more details.')
 No data returned.
 Fetch failed or empty.
 Connecting to Yahoo Finance... (23:49:56)
```

Figure 39 : TC-12 (Network Failure)

Table 34 : System Stability Test Case

6.0 Evaluation

6.1 User Evaluation Session and System Walkthrough

Figure 40 : User Evaluation

As shown in figure 40, to evaluate the system's usability, a structured survey was conducted via Google Forms. Participants were guided through the system's features, either through recorded demonstrations or direct interaction to observe the live dashboard, AI confidence gauges, and sentiment indicators in action.

6.2 Evaluation Questionnaire

In the evaluation phase, user feedback for the system was gathered using a USE-standard questionnaire distributed via Google Forms. A total of 51 responses were collected and used to assess the system's performance. Table 35 shows the questionnaire items with its respective labels.

Question	Label
The dashboard layout (Tabs, Columns) was intuitive and easy to navigate.	A1
The real-time price charts and technical indicators (SMA, RSI) were clear and easy to read.	A2
The AI Confidence Gauge (dashboard meter) helped me quickly understand the model's current stance.	A3
The system provided clear visual feedback when the market was closed or data was updating.	A4
Overall, the user interface effectively presented complex financial data.	A5

The system successfully automated the data fetching and feature engineering processes without manual intervention.	B1
The News Service updated consistently and provided relevant headlines.	B2
The system handled special market conditions (e.g., weekends, holidays) logically by pausing or waiting.	B3
The simulation ran smoothly without crashing or freezing during the test period.	B4
The model's high-precision strategy felt appropriate for gold trading.	C1
The Dynamic Thresholding mechanism (adjusting buy limits based on news sentiment) seemed logical.	C2
The integration of FinBERT (NLP) effectively captured market sentiment (Bullish/Bearish).	C3
The backtest results (e.g., Equity Curve, Sharpe Ratio) provided convincing evidence of the strategy's potential.	C4
The "Factor Analysis" table helped me understand why the AI made a specific decision.	D1
The "Feature Importance" chart clearly showed which technical indicators the model values most.	D2
I felt the system was transparent, not a "black box" (i.e., I understood the logic behind the trades).	D3
I would trust this system's signals to assist in making trading decisions.	D4
This system improved my understanding of how Machine Learning is applied in algorithmic trading.	E1
The inclusion of unstructured data (News) significantly added value compared to a pure technical analysis system.	E2
This project demonstrates a complete, end-to-end implementation of a financial trading system.	E3
Overall, I am satisfied with the performance and features of this AI trading system.	E4
What was the most impressive feature of the system?	F1
Did you encounter any bugs or confusing elements? If yes, please describe.	F2
If you could add one feature to this system, what would it be?	F3

Table 35 : Survey questions and labels used for user feedback evaluation

6.3 Demographic

6.3.1 System Usability & Interface (UI/UX)

1. **The dashboard layout (Tabs, Columns) was intuitive and easy to navigate.**

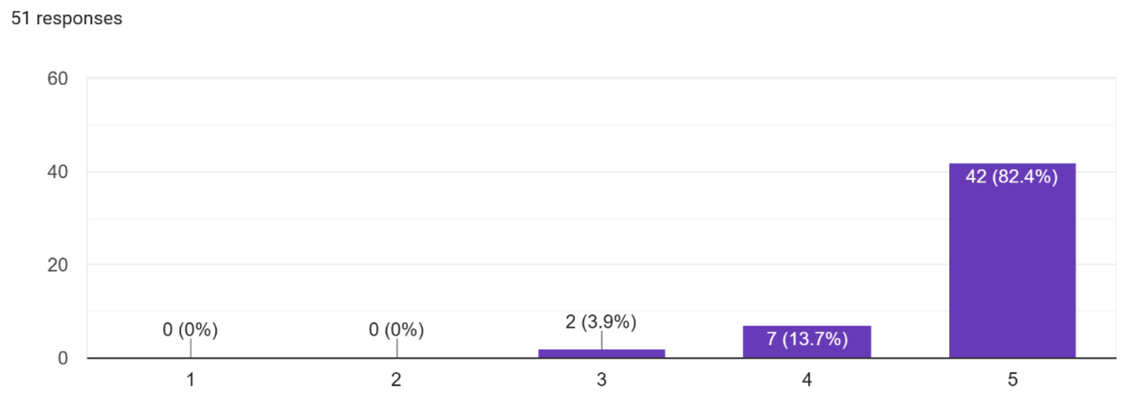


Figure 41 : Dashboard layout ratings

The layout was rated as 5/5 by 82.4% of the participants, and 4/5 by another 13.7% of the participants; 3.9% of the participants were neutral. This means that the overall organization as well as the flow of navigation were very effective. There was no difficulty in locating features and this implies the layout is usability-friendly.

2. **The real-time price charts and technical indicators (SMA, RSI) were clear and easy to read.**

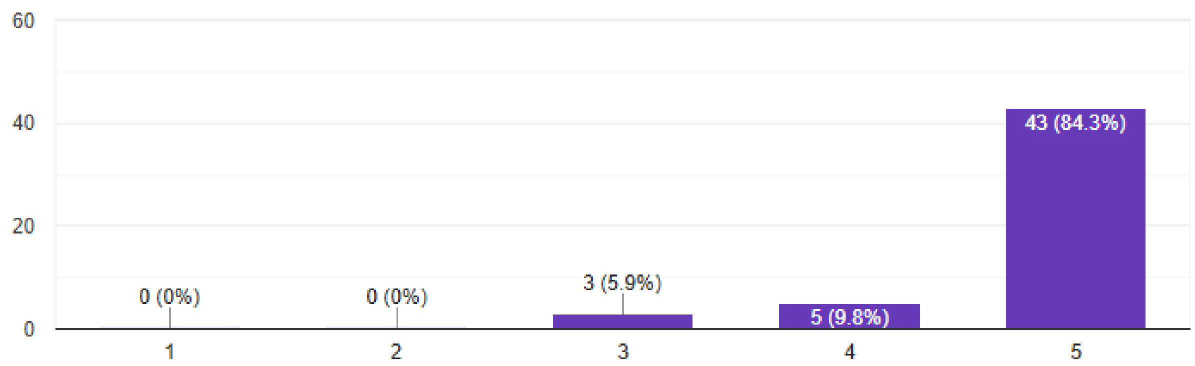


Figure 42 : Real-time price charts readability

The percentage of those who rated the charts 5/5 was 84.3 and those who rated them 4/5 was 9.8 as compared to only 5.9 who said they were neutral. The findings demonstrate that the visualization of financial indicators is very readable and easily accessible. The chart design was also able to communicate the complex data as users could easily interpret the market trends.

3. **The AI Confidence Gauge (dashboard meter) helped me quickly understand the model's current stance.**

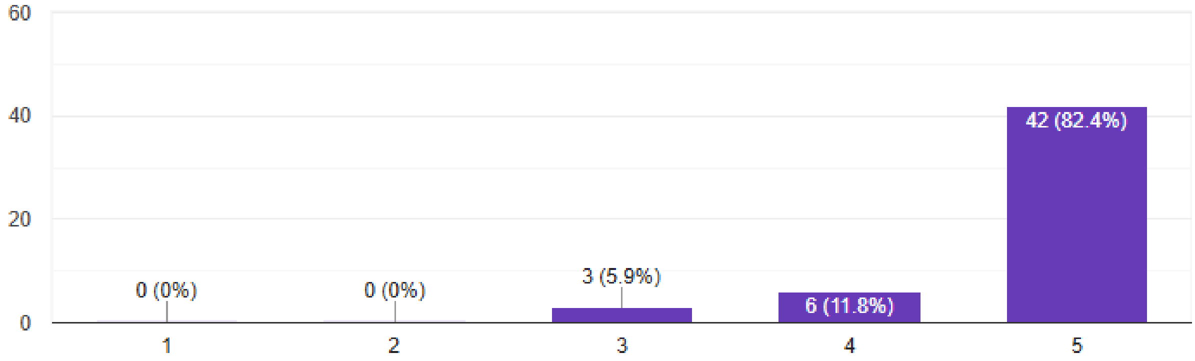


Figure 43 : AI confidence gauge clarity

82.4% gave the gauge a 5/5 rating, 11.8% selected 4/5, and 5.9% chose neutral. This is indicative that the gauge is useful in increasing the interpretability since the confidence level of the model is well expressed. The increased transparency benefited users and lessened cognitive load in the process of making judgments on predictions.

4. **The system provided clear visual feedback when the market was closed or data was updating.**

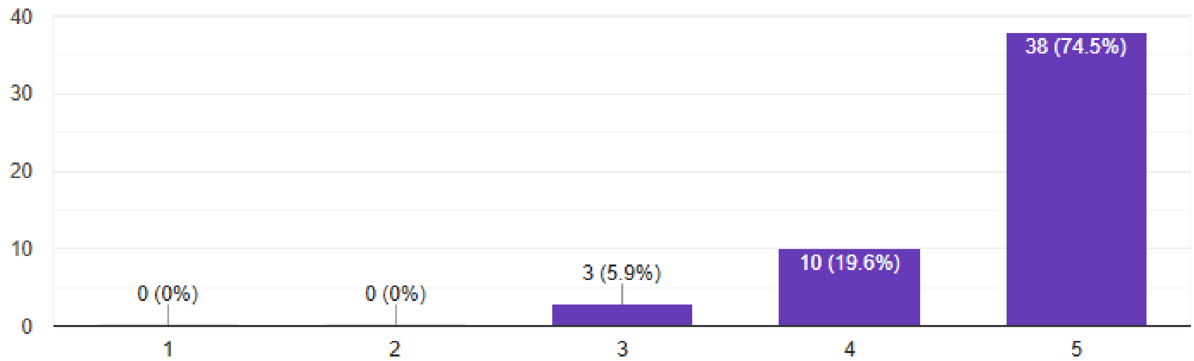


Figure 44 : Market closed/data updating feedback

This feature was rated 5/5 by 74.5% of the participants, 4/5 by 19.6% and neutral by 5.9%. These findings prove that the status indicators of the system are high-quality and noticeable. The provision of clear feedback decreased user confusion in the non-trading periods or data refresh events, which led to an easy interaction.

5. **Overall, the user interface effectively presented complex financial data**

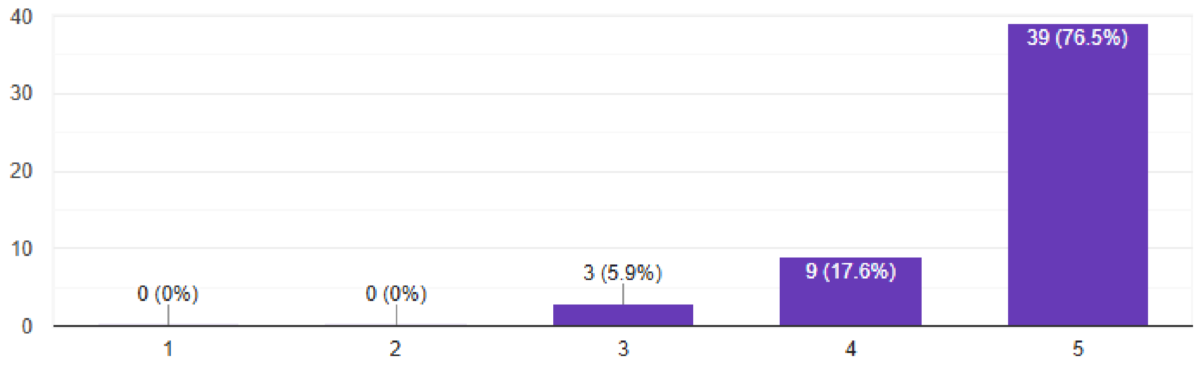


Figure 45 : Overall UI effectiveness

76.5% rated the UI 5/5, 17.6% rated 4/5, with only 5.9% neutral. This means that the system is effective in making complex information on financial matters to be simplified into digestible visual materials. The interface design enables effective interpretation of data and positively influences the level of decision-making among users.

6.3.2 Functionality & Robustness

6. The system successfully automated the data fetching and feature engineering processes without manual intervention.

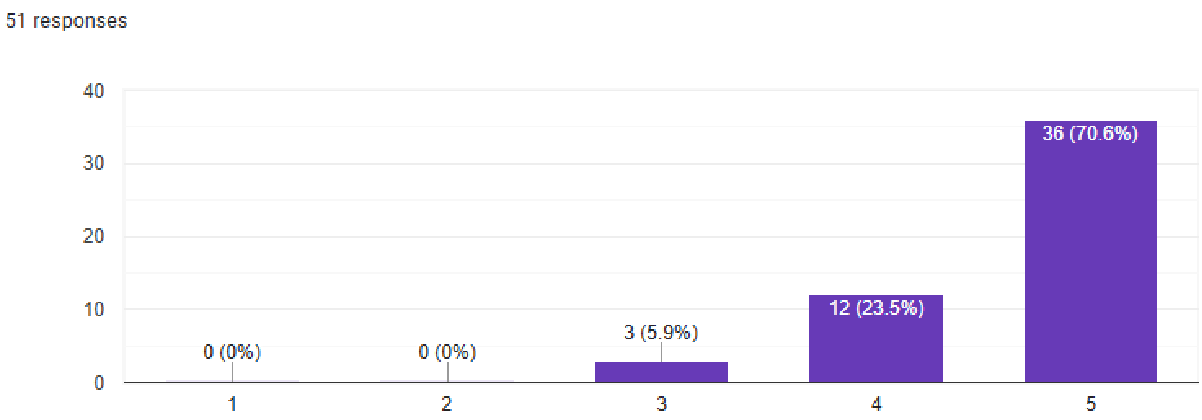


Figure 46 : Data automation reliability

70.6% of respondents rated this feature 5/5, with 23.5% selecting 4/5, and only 5.9% neutral. This indicates that the automation pipeline was stable and minimized the number of people who had to work on it, as well as continuous data availability. The users had a smooth and continuous preprocessing without any interruptions, which proved that the system was stable.

7. The News Service updated consistently and provided relevant headlines.

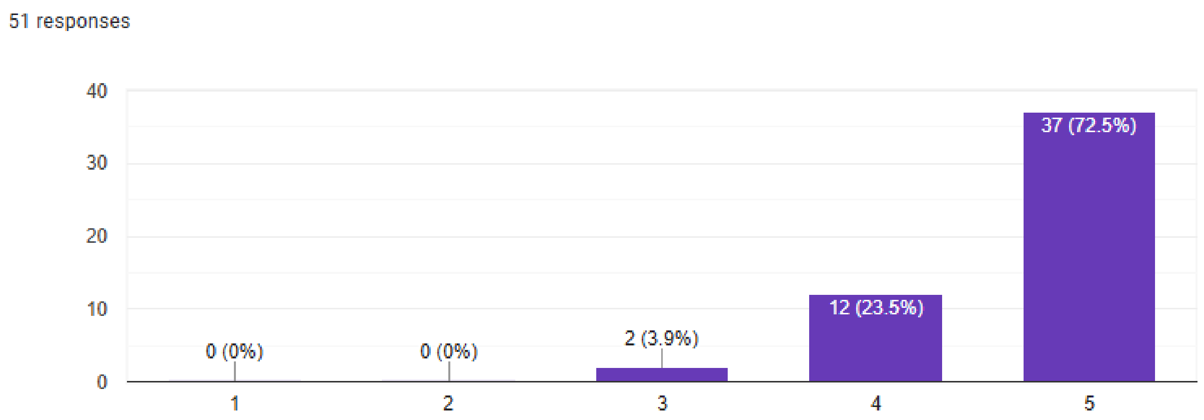


Figure 47 : News service relevance

72.5% rated this 5/5, another 23.5% rated it 4/5, and only 3.9% were neutral. The findings above reveal that the news-fetching module was quite consistent and relevant. The headlines proved useful to the users who were interested in knowing the market conditions, which is a good indicator of API integration and filtering logic.

8. **The system handled special market conditions (e.g., weekends, holidays) logically by pausing or waiting.**

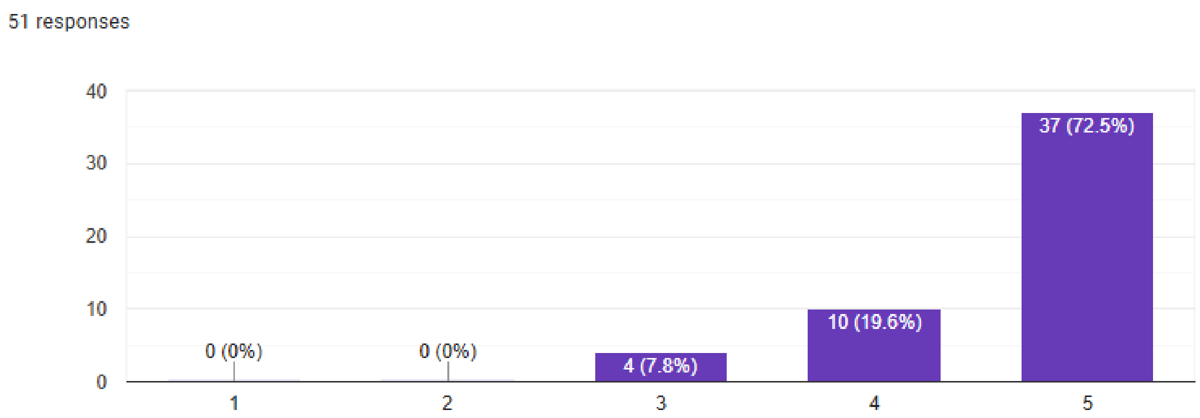


Figure 48 : Handling weekends/holidays

72.5% rated this 5/5, 19.6% rated it 4/5, and 7.8% were neutral. This is an indication that the system has applied the logic of market awareness in the correct way, and that it ensures there is no false alert when its markets are closed. The behaviour raised the trust of the users and provided the simulation flows with more reality.

9. **The simulation ran smoothly without crashing or freezing during the test period.**

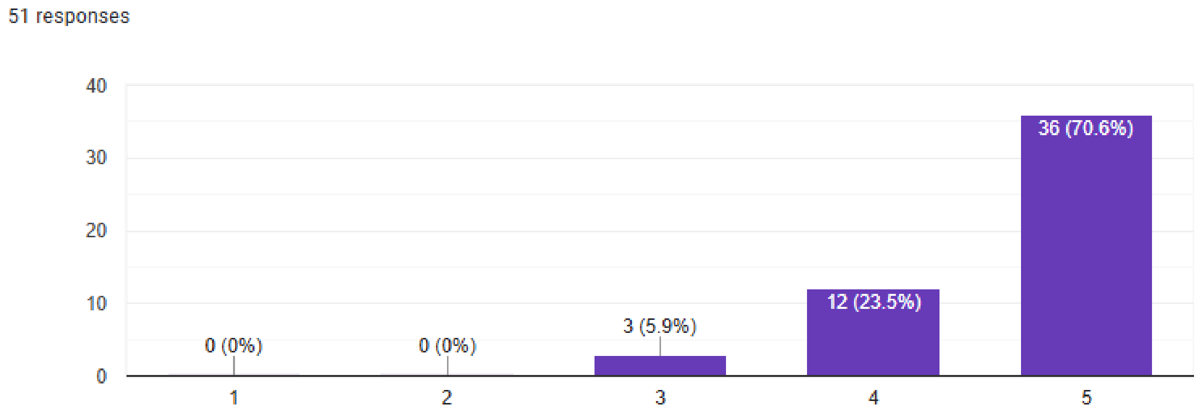


Figure 49 : Simulation stability

70.6% rated system stability 5/5, 23.5% rated it 4/5, and 5.9% were neutral. The results reveal that the platform provided one with high runtime stability. The users also reported low performance problems, which implied that the back-end was highly optimized, and it managed to handle memory efficiently in simulations.

6.3.3 AI Model & Strategy Performance

10. The model's high-precision strategy felt appropriate for gold trading.

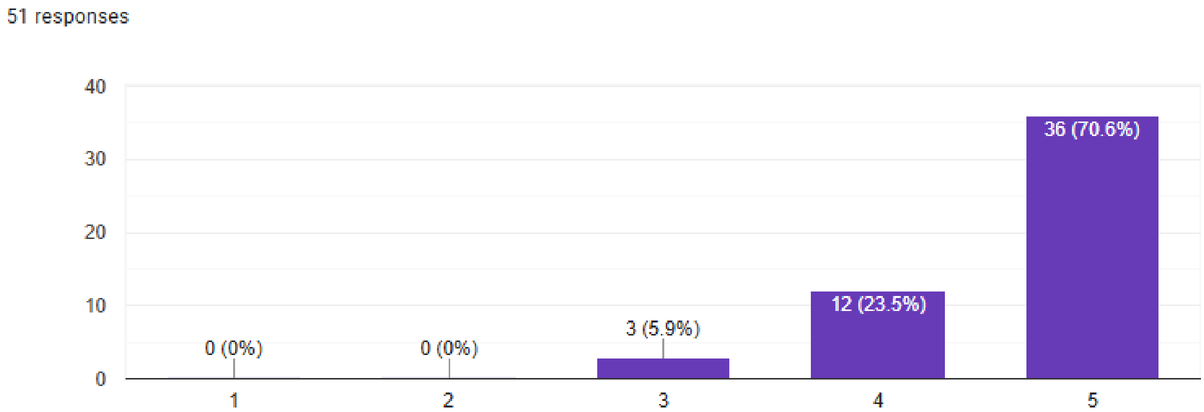


Figure 50 : Strategy appropriateness

70.6% rated this strategy 5/5, 23.5% rated it 4/5, and only 5.9% gave a neutral score. Users perceived the selective trading approach as suitable for gold's volatility profile. This reflects positive reception toward the system's risk-aware behaviour and confirms that conservative trading logic aligns with user expectations.

11. The Dynamic Thresholding mechanism (adjusting buy limits based on news sentiment) seemed logical.

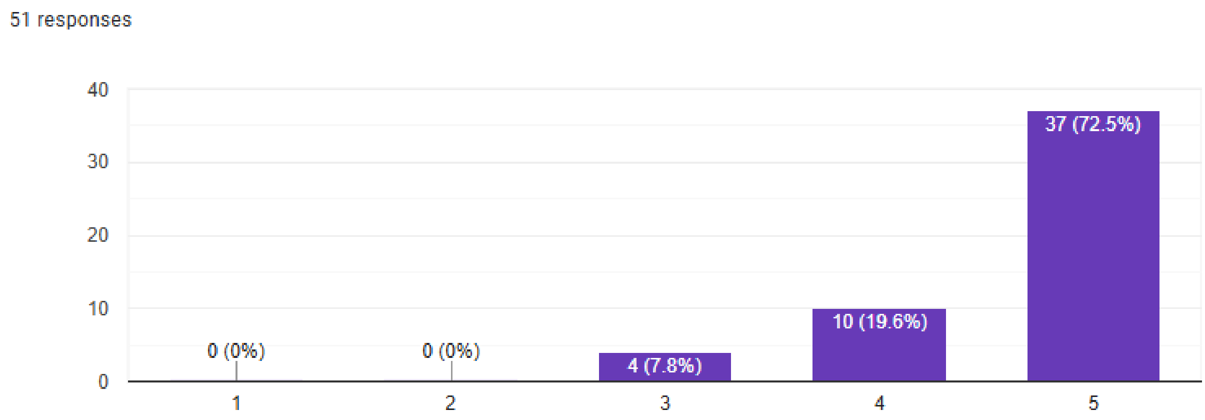


Figure 51 : Dynamic threshold logic

72.5% rated this 5/5, 19.6% rated it 4/5, and 7.8% rated it 3/5. The majority agreed that adapting thresholds using sentiment adds realism and intelligence to the strategy. The favourable results suggest that combining sentiment signals with technical analysis improves decision accuracy

12. The integration of FinBERT (NLP) effectively captured market sentiment (Bullish/Bearish).

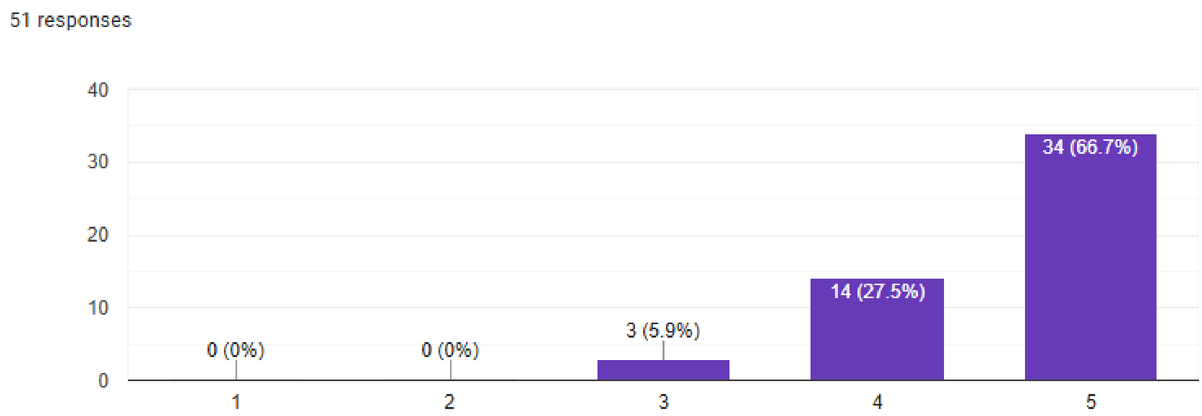


Figure 52 : FinBERT sentiment accuracy

66.7% rated the sentiment module 5/5, 27.5% rated it 4/5, and 5.9% rated neutral. This indicates that users found the sentiment signals meaningful and reliable. The strong positive feedback supports the effectiveness of FinBERT in translating financial news into actionable insights.

13. The backtest results (e.g., Equity Curve, Sharpe Ratio) provided convincing evidence of the strategy's potential.

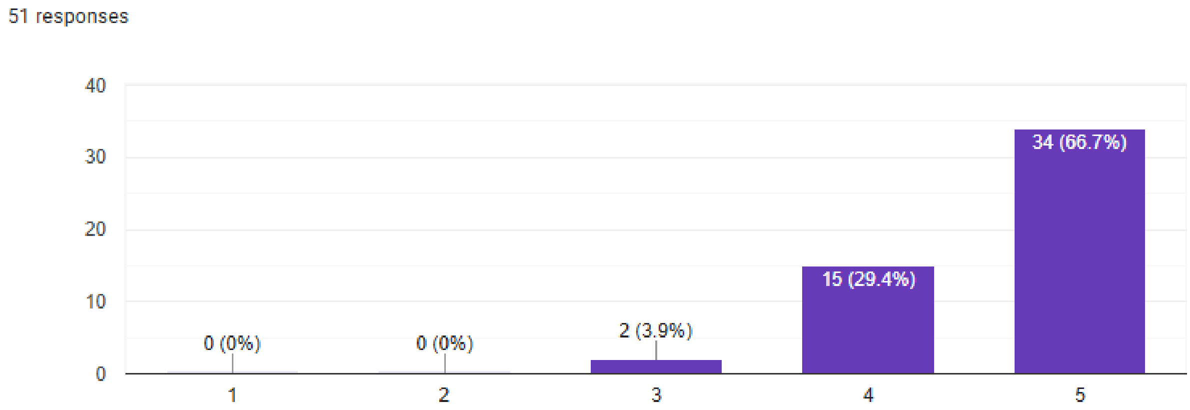


Figure 53 : Backtest outputs credibility

66.7% rated this feature 5/5, 29.4% rated it 4/5, and only 3.9% were neutral. These results indicate that users found the backtesting outputs credible and informative. The strong positive reception suggests that the historical performance metrics successfully demonstrated how the model behaves in real market conditions, increasing user confidence in the strategy's design.

6.3.4 Interpretability & Trust (Explainable AI)

14. The "Factor Analysis" table helped me understand why the AI made a specific decision.

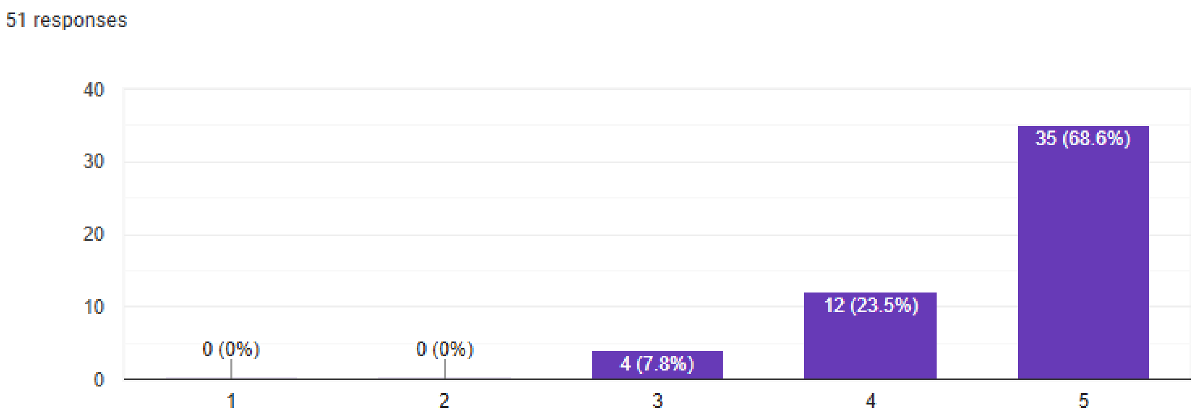


Figure 54 : Factor analysis interpretability

68.6% rated it 5/5, 23.5% rated it 4/5, and 7.8% gave a neutral score. The high agreement shows that users valued the transparency provided by the decision breakdown. This indicates that presenting the underlying reasoning, such as sentiment, trend direction, and indicator values, effectively reduced confusion and improved trust in the model's outputs.

15. The "Feature Importance" chart clearly showed which technical indicators the model values most.

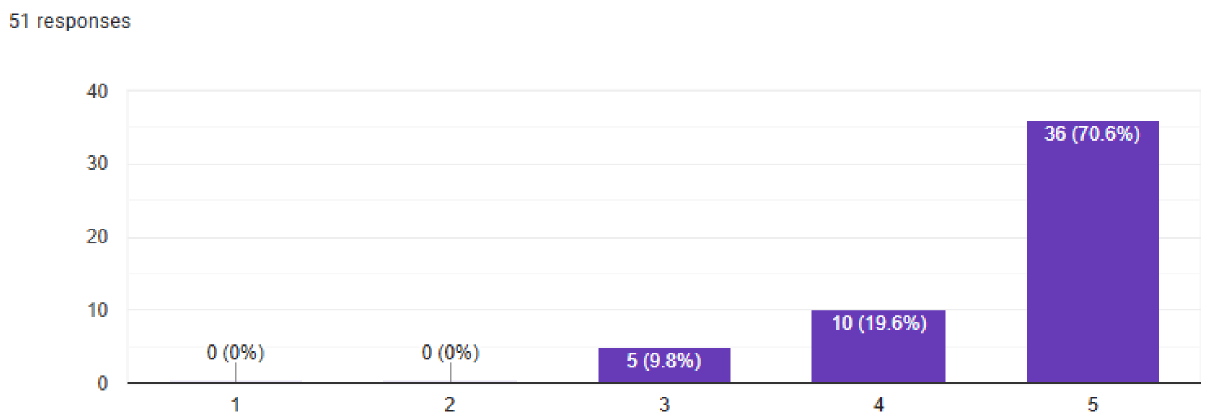


Figure 55 : Feature importance clarity

70.6% rated this 5/5, 19.6% rated it 4/5, and 9.8% rated it 3/5. These results suggest that the feature-importance visualization was helpful for understanding model behaviour. Users found it useful in identifying which technical indicators contributed most strongly to predictions, enhancing transparency and interpretability.

16. I felt the system was transparent, not a "black box" (i.e., I understood the logic behind the trades).

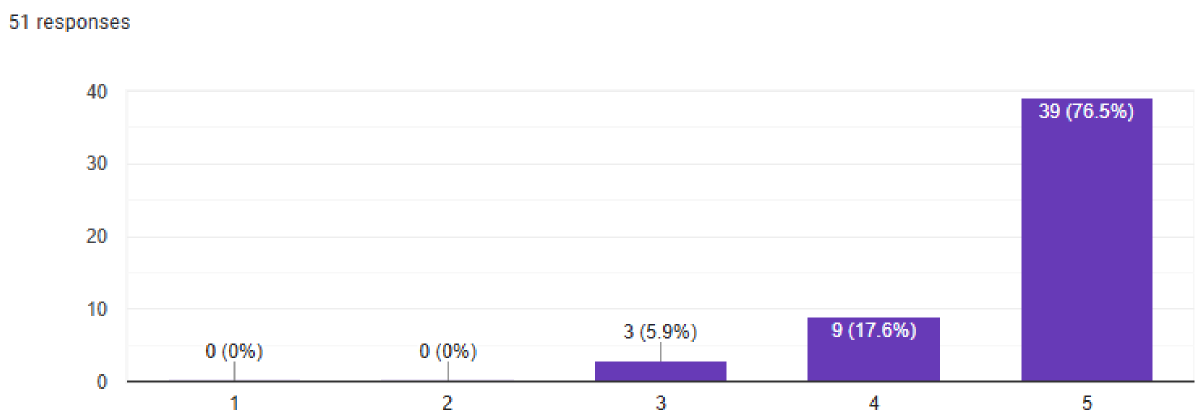


Figure 56 : System transparency

76.5% rated this 5/5, 17.6% rated it 4/5, and only 5.9% were neutral. A very high level of agreement shows that the system succeeded in achieving explainability. The combination of feature importance, factor breakdown, and clear UI indicators helped users feel in control and aware of the model's decision-making logic.

17. I would trust this system's signals to assist in making trading decisions.

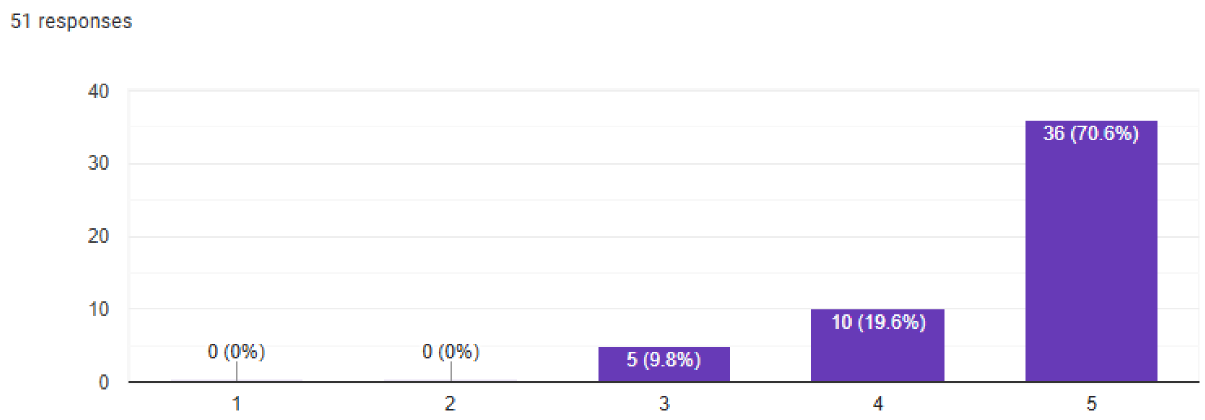


Figure 57 : Trust in system signals

70.6% rated this 5/5, 19.6% rated it 4/5, and 9.8% rated it neutral. The results reflect strong user trust in the system's overall stability, reasoning clarity, and signal accuracy. While users may not rely solely on the system, the majority indicated willingness to use it as a supportive tool for decision-making.

6.3.5 Overall Impact & Value

18. This system improved my understanding of how Machine Learning is applied in algorithmic trading.

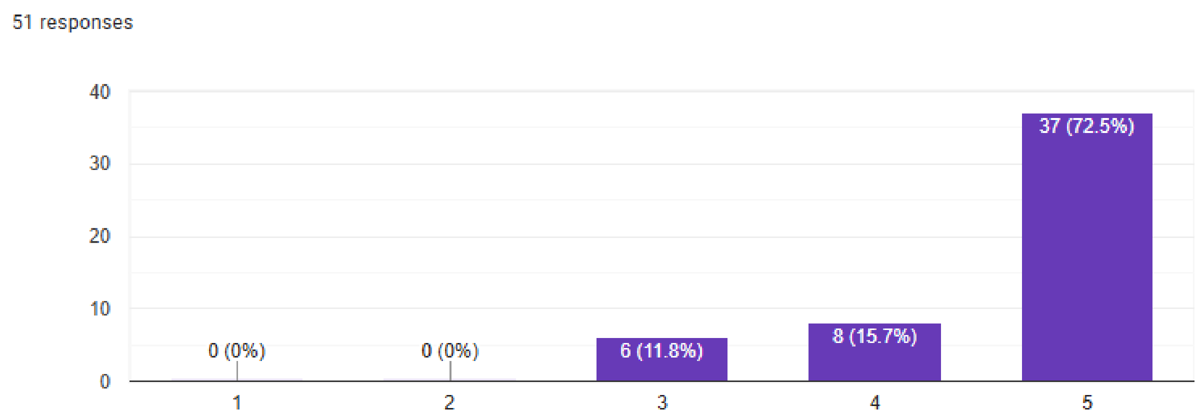


Figure 58 : Understanding ML in trading

72.5% rated this 5/5, 15.7% rated it 4/5, and 11.8% were neutral. Most respondents felt that the system successfully demonstrated key ML concepts such as prediction models, feature engineering, and real-time inference. This shows the project's educational impact and effectiveness in illustrating practical ML-finance workflows.

19. The inclusion of unstructured data (News) significantly added value compared to a pure technical analysis system.

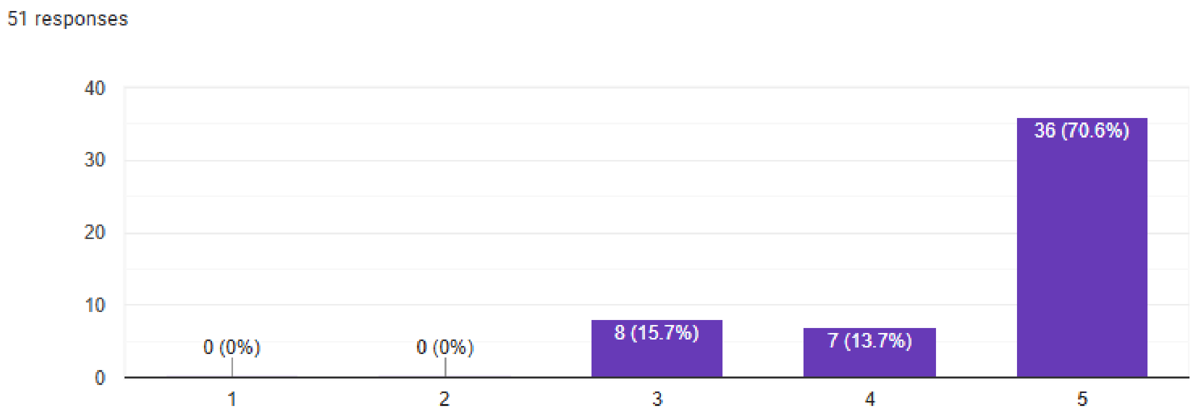


Figure 59 : Value of unstructured news data

70.6% rated it 5/5, 13.7% rated it 4/5, and 15.7% were neutral. These results show strong user support for the integration of NLP-based sentiment analysis. The added value likely comes from combining technical signals with real-world economic context, providing more balanced and intelligent decision-making.

20. This project demonstrates a complete, end-to-end implementation of a financial trading system.

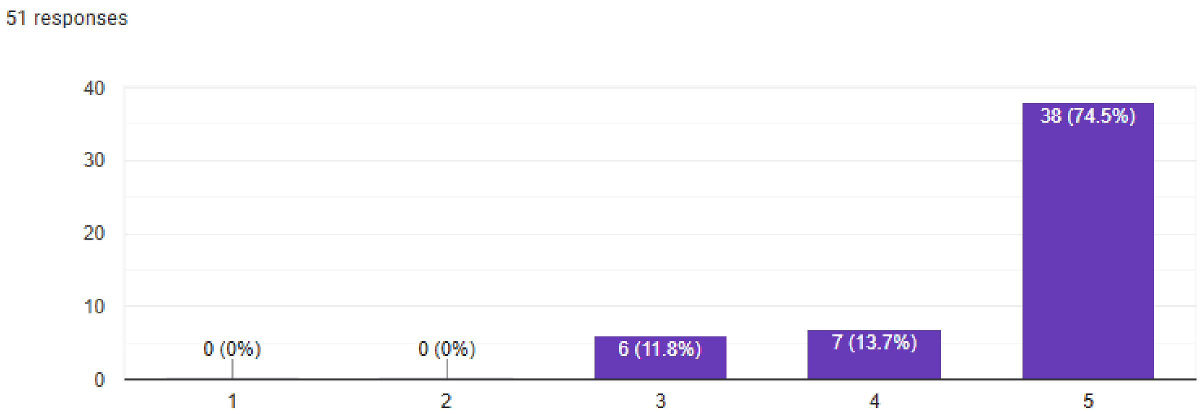


Figure 60 : End-to-end system implementation

74.5% rated this 5/5, 13.7% rated it 4/5, and 11.8% rated 3/5. The high agreement reflects that users recognized the system as a full pipeline from data collection to visualization, modelling, trading logic, and explanation dashboards. This suggests strong completeness and practical relevance of the solution.

21. Overall, I am satisfied with the performance and features of this AI trading system.

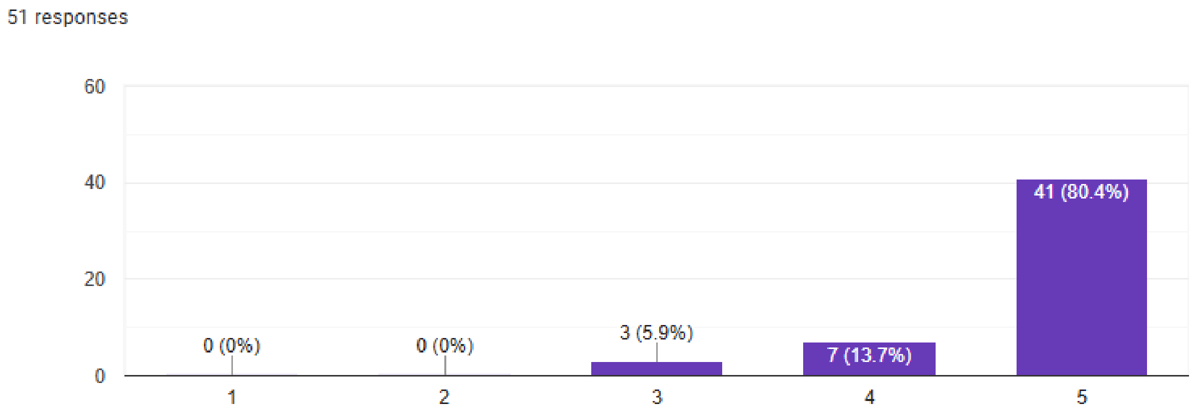


Figure 61 : Overall system satisfaction

This was rated 5/5 by 80.4 percent of the respondents, 4/5 by 13.7 percent and neutral by 5.9 percent. This is a very high positive feedback that most of the users perceived the system effective, stable and informative. The overall experience with the automation, explainability, and trading intelligence was positive to the majority of the participants.

6.3.6 Open-Ended Feedback

22. What was the most impressive feature of the system?

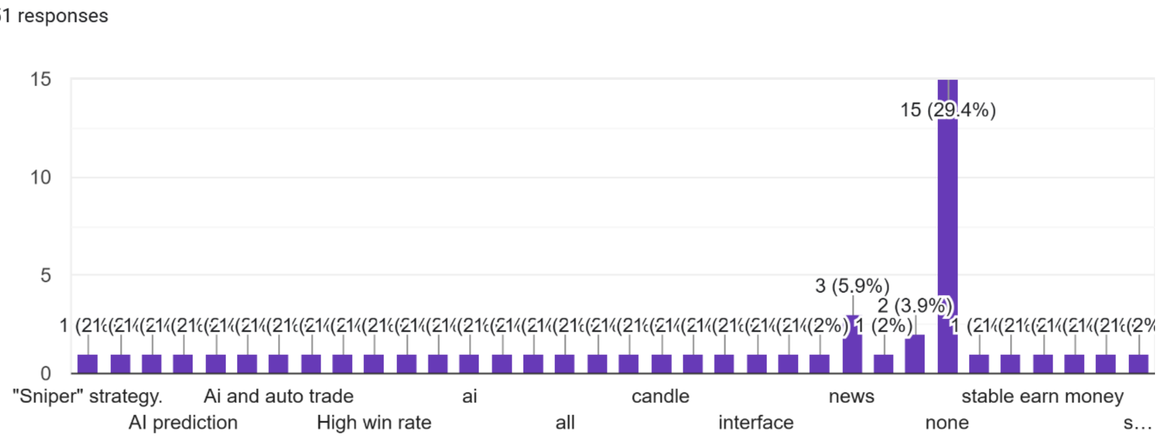


Figure 62 : Most impressive feature

The participants also identified a vast number of features as the most impressive, and the most highly rated ones were the high-precision strategy, AI prediction, auto-trading reasoning, real-time news, FinBERT sentiment analysis, and the user-friendly dashboard interface. Other features like robust stop-loss protection, consistent simulated profit, and live candlestick chart were also valued by a less number of users. Nevertheless, a considerable percentage (29.4) responded with none meaning that they are either neutral or have a hard time trying to single out a feature outstanding.

All in all, users appear to appreciate the smart, automated features of the system, especially the logic behind the strategy and AI features, as the clarity of the UI and the ability to combine data in real-time also helped to improve their experience. The high number of none replies could also suggest that there are users who were not firmly influenced by any particular feature or were light testers who had slight involvement.

23. Did you encounter any bugs or confusing elements? If yes, please describe.

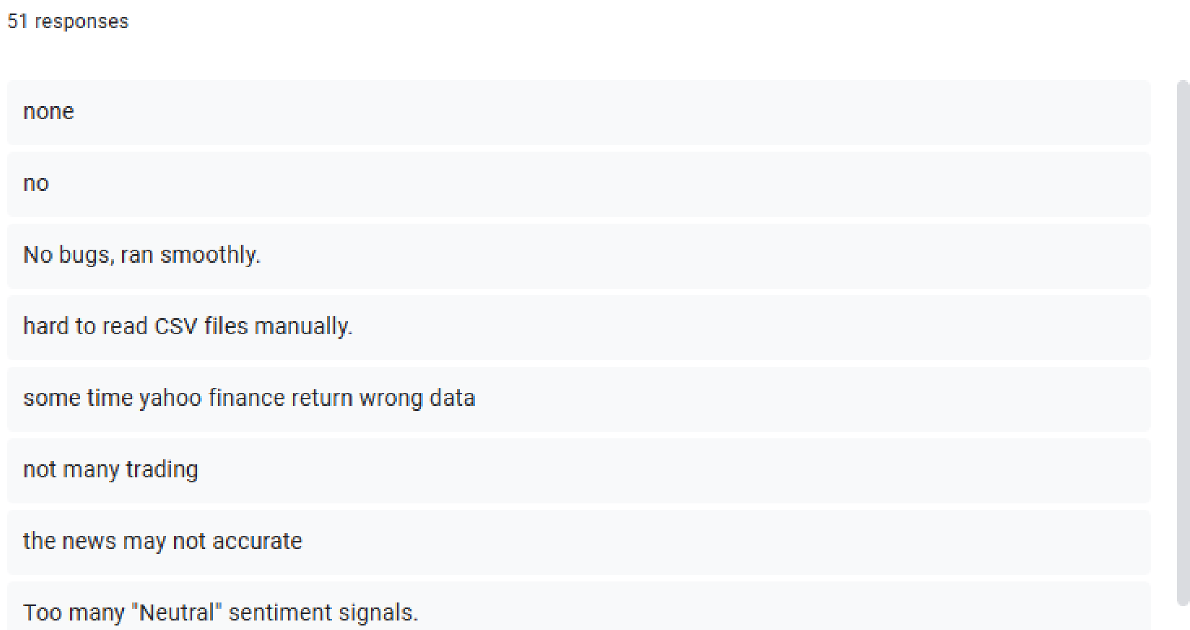
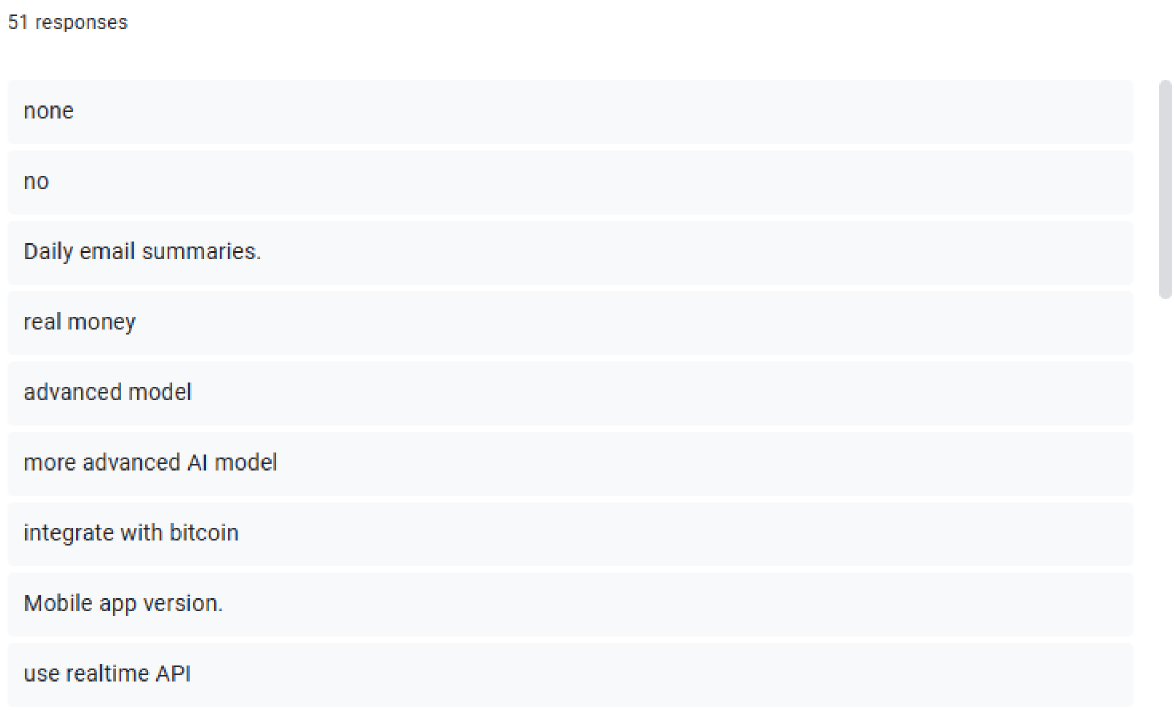


Figure 63 : Bugs or confusing elements

As shown in Figure 63, The majority of users stated that there were no bugs and they stated that the system is not volatile. Some small problems were identified, including being unable to read CSV files manually, some failures of the Yahoo Finance data, frequent Neutral sentiment results of the news model, and the strategy trading less than anticipated.

These remarks suggest that the system was reliable in testing, and minor data-quality or usability issues. The problems described were largely either extrinsic dependencies (e.g., Yahoo Finance accuracy) or model behaviour phenomena (e.g., a large number of neutral headlines), and not system design flaws.

24. If you could add one feature to this system, what would it be?


51 responses

none

no

Daily email summaries.

real money

advanced model

more advanced AI model

integrate with bitcoin

Mobile app version.

use realtime API

Figure 64 : Suggested feature enhancements

Users proposed a variety of enhancements, including:

- Real-money or wallet integration (e.g., crypto wallets, real trading execution)
- Mobile app or notifications (phone alerts, Telegram/SMS signals, app version)
- More advanced or adaptive AI models
- 24/7 operation using real-time APIs
- Support for crypto, stocks, and other markets
- Daily email summaries or improved alerts

These proposals indicate that the users are eager to scale the system to a more production oriented trading tool. The most frequently requested additions were notifications, mobile access, wider support of assets, and real-time execution which are features that one would expect to find in a professional algorithmic trading platform.

6.3.7 Discussion

The results of the evaluation show that the AI-based gold trading system was positively accepted and was effective in terms of usability, functionality, strategy design, and explainability. The interface was easy to use, and the visualizations were clear so that users can easily interpret engineered financial and machine learning outputs. The automated pipeline of the system, constant simulation behaviour and market conscious logic boosted reliability and user confidence. The high-precision trading strategy together with dynamic thresholding and FinBERT-based news sentiment was viewed to be appropriate to gold trading and value added to pure technical analysis. The positive rating of factors analysis and feature importance shows that the system was effective in decreasing the black box aspect of AI models, enhancing responsible and trustworthy trading signals. Generally, the system is proven as a sound end-to-end AI trading system, yet future modifications will be directed to real-time implementation and expanded market coverage.

7.0 Future Improvements

7.1 Enhanced Market Regime & Event Awareness

Current: The system is associated with extensive market volatility (VIX), but is activated in reaction. It does not have clear consciousness of the planned, high-impact economic occurrences, which trigger structural market changes.

Future: The system should be modified to integrate a live Economic Calendar API (e.g. Forex Factory, FRED) so that the system might in advance adjust risk parameters or take a neutral position on important releases such as FOMC decisions, CPI, and Non-Farm Payrolls. This makes this model a predictive instead of a responsive system which helps a lot in terms of stability in high volatility conditions.

7.2 Domain-Specific Sentiment Refinement

Current: The general-purpose financial sentiment analysis (FinBERT) occasionally misinterprets events. As an illustration, the term "Geopolitical Crisis" can be labeled as negative without considering the peculiarities gold has as a safe-haven asset.

Future: It is important to develop a Gold-Specific Sentiment Layer. This may include training the NLP model on a set of gold-market news that are gold-cycle countercyclical, or a rule-based post-processor that re-interprets general news in terms of the gold-cycle (e.g. re-interpreting "Inflation Fears" or "Currency Devaluation" to strong signs that the buy market).

7.3 Infrastructure for Production Reliability

Current: The pipeline uses file-based (csv/json) storage and sequential polling which introduces I/O bottlenecks and cannot tolerate faults to operate 24/7.

Future: Migrating to a robust event-driven microservice architecture is essential for live trading. This includes the utilisation of time-series database (InfluxDB), message broker (Redis) and containerisation (Docker). The utilization of cloud services (AWS/GCP) would guarantee the low-latency access to data, high availability, and the scalability.

7.4 Advanced Sequential Modeling

Current: The ensemble model treats each hourly bar as an independent data point, potentially missing longer-term sequential patterns and regime transitions.

Future: These temporal dependencies could be included by incorporating models that inherently operate on sequences, including Long Short-Term Memory (LSTM) networks or more basic N-BEATS architectures. The main problem will be that the high quality of the past data will have to be sufficient to train such models without overfitting.

7.5 Towards Live Execution & Realism

Current: Backtesting is based on assumed perfect order execution at the past prices without taking into account market microstructure impacts such as slippage and the bid-ask spread.

Future: The last step on the way to a fully automated system is integration with a Broker API (e.g., OANDA, Interactive Brokers). This needs to be done by construction of a powerful order execution system and a realistic slippage model so that the high Sharpe Ratio in the strategy is not lost in the transition between the simulated markets and real market environments. Starting operations would cost little capital in a heavily monitored paper-trading business.

8.0 Conclusion

This project successfully developed and implemented an end-to-end machine learning system for gold price trend prediction and automated trading simulation, demonstrating the practical integration of predictive analytics with real-time financial decision-making.

8.1 Summary of Achievements

The system delivered a high-precision ensemble model combining XGBoost, Logistic Regression, and Gradient Boosting, achieving a precision of 55.3%, recall of 61.4%, and an F1-score of 0.582. Real-time financial news sentiment was effectively integrated using FinBERT, dynamically adjusting trading thresholds. A disciplined risk-aware trading strategy was developed and backtested, generating a +5.48% return with an exceptional Sharpe Ratio of 3.05 and minimal drawdown. The modular microservices architecture and transparent dashboard provided explainability and stability throughout the trading pipeline.

8.2 Evaluation Against Original Objectives

All the goals were achieved in the project. The prediction of gold price variations was created and verified by a strong machine learning framework. The sentiment of financial news was statistically incorporated as a decision modifier of tactic. Lastly, a full automated trading simulation was modeled and tested and demonstrated the feasible translation of model predictions into financially viable signals with high risk-adjusted performance.

8.3 Limitations

Several limitations are acknowledged. The simulation operates without real-world transaction costs, which may affect live performance. Model efficacy may decrease during unforeseen market regimes or "black swan" events. Sentiment analysis is constrained to headline-level interpretation. The scope is specifically tailored to hourly gold trading, limiting direct applicability to other assets. The current architecture is designed for prototyping, not 24/7 production deployment.

8.4 Final Remarks

This initiative may prove conclusively that it is possible to have a transparent risk-conscious AI-driven trading framework. Through its ability to close the divide between machine learning inference and automated implementation, it offers a useful guide to applied computational finance. Still in the simulation setting, the architecture of the system, the methodology and positive outcomes give a solid ground to proceed with the work in the future with the aim of creating a fully functional trading system. The project highlights that there is a powerful potential of explainable responsible Machine Learning to improve financial decision-making.

9.0 References

Acknowledgment: We have used AI tools to polish our original writing in the paper.

We have not used them to generate any figures or manipulate any results.

Ade, M. (2023). Forecasting volatility: Comparative analysis of ARIMA, GARCH, and deep learning models for predicting. . . *ResearchGate*. https://www.researchgate.net/publication/384664982_Forecasting_Volatility_Comparative_Analysis_of_ARIMA_GARCH_and_Deep_Learning_Models_for_Predicting_Stock_Market_Volatility

Bischl, B., Binder, M., Lang, M., Pielok, T., Richter, J., Coors, S., Thomas, J., Ullmann, T., Becker, M., Boulesteix, A., Deng, D., & Lindauer, M. (2023). Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges. *Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery*, *13*(2). https://doi.org/10.1002/widm.1484

Botchkarev, A. (2019). A new typology design of performance metrics to measure errors in machine learning regression algorithms. *Interdisciplinary Journal of Information Knowledge and Management*, *14*, 045–076. https://doi.org/10.28945/4184

Bouoiyour, J., Selmi, R., & Wohar, M. (2018). Measuring the response of gold prices to uncertainty: An analysis beyond the mean. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.1806.07623

Bunnag, T. (2023). The importance of gold and forecasting of the gold price using ARIMA, ARIMA-GARCH, and ARIMA-TGARCH model. *ResearchGate*. https://www.researchgate.net/publication/375284085_The_Importance_of_Gold_and_Forecasting_of_the_Gold_Price_Using_ARIMA_ARIMA-GARCH_and_ARIMA-TGARCH_Model

Cerqueira, V., Torgo, L., & Mozetič, I. (2020). Evaluating time series forecasting models: an empirical study on performance estimation methods. *Machine Learning*, *109*(11), 1997–2028. https://doi.org/10.1007/s10994-020-05910-7

Cheng, G., Peddinti, V., Povey, D., Manohar, V., Khudanpur, S., & Yan, Y. (2017). An Exploration of Dropout with LSTMs. *Interspeech 2022*, 1586–1590. https://www.researchgate.net/publication/319185476

Chio, P. T. (2022). A comparative study of the MACD-base trading strategies: evidence from the US stock market. *ideas.repec.org*. https://ideas.repec.org/p/arx/papers/2206.12282.html

Chong, T. T. L., Ng, W. K., & Liew, V. K. S. (2014). Revisiting the performance of MACD and RSI oscillators. *Journal of Risk and Financial Management, 7*(1), 1–12. https://doi.org/10.3390/jrfm7010001

De Amorim, L. B., Cavalcanti, G. D., & Cruz, R. M. (2022). The choice of scaling technique matters for classification performance. *Applied Soft Computing*, *133*, 109924. https://doi.org/10.1016/j.asoc.2022.109924

E.M. Jordaan, & Smits, G. F. (2002). Estimation of the regularization parameter for support vector regression. *Neural Networks, 2002. IJCNN '02. Proceedings of the 2002 International Joint Conference On*, *3*, 2192–2197. https://doi.org/10.1109/IJCNN.2002.1007481

El Ouardi, A., Er-Raha, B., & Tatane, K. (2024, December 31). *LSTM adaptive hyperparameter tuning for financial time series forecasting using custom gradient-based methods: A comparative study with Bayesian optimization*. *Journal of Theoretical and Applied Information Technology, 102*(24), https://www.jatit.org/volumes/Vol102No24/8Vol102No24.pdf

Elfakhani, S., Baalbaki, I. B., & Rizk, H. (2009). Gold price determinants: empirical analysis and implications. *J for International Business and Entrepreneurship Development*, *4*(3), 161. https://doi.org/10.1504/jibed.2009.029010

Fu, Z., Zheng, Y., & Lin, J. (2025). Adaptive Volatility Analysis: An Integrated ATR Model for Dynamic Market Conditions. *Optimizations in Applied Machine Learning, 5*(1). https://doi.org/10.71070/oaml.v5i1.25

Goldberg, L. R., & Mahmoud, O. (2014, April 29). *Drawdown: from practice to theory and back again*. arXiv.org. https://arxiv.org/abs/1404.7493

Gong, W. (2024). Research on gold price forecasting based on lstm and linear regression. *SHS Web of Conferences*, *181*, 02005. https://doi.org/10.1051/shsconf/202418102005

Griffin, C. H. (2025). Gold, Bitcoin, and central banks in the 21st century: the new dynamic. *International Journal of Finance*, *10*(2), 29–41. https://doi.org/10.47941/ijf.2569

Gunasekaran, K. P. (2023). Exploring Sentiment Analysis Techniques in Natural Language Processing: A Comprehensive review. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2305.14842

He, Q., Guo, Y., & Yu, J. (2020). Nonlinear dynamics of gold and the dollar. *The North American Journal of Economics and Finance*, *52*, 101160. https://doi.org/10.1016/j.najef.2020.101160

Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term memory. *Neural Computation*, *9*(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Hsu, C., Chang, C., & Lin, C. (2003). A Practical guide to support vector Classification Chih-Wei Hsu, Chih-Chung Chang, and Chih-Jen Lin. *ResearchGate*. https://www.researchgate.net/publication/2926909

Jabeur, S. B., Mefteh-Wali, S., & Viviani, J. (2021). Forecasting gold price with the XGBoost algorithm and SHAP interaction values. *Annals of Operations Research*, *334*(1–3), 679–699. https://doi.org/10.1007/s10479-021-04187-w

Jakobsson, N. (2022). *CAN GOLD SERVE AS A HEDGE AGAINST UNCERTAINTY? : A study on the gold price dynamics and its role as a safe haven asset*. DIVA. https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1678264&dswid=-4567

Kaufmann, T. D. &. W. R. A. (1989). The price of gold : A simple model. *ideas.repec.org*. https://ideas.repec.org/a/eee/jrpoli/v15y1989i4p309-313.html

Kiohos, A., & Sariannidis, N. (2010). Determinants of the asymmetric gold market. *ResearchGate*. https://www.researchgate.net/publication/286561533_Determinants_of_the_asymmetric_gold_market

Li, Y. (2024). Predicting the gold price based on XGBoost. *www.scitepress.org*. https://www.scitepress.org/PublishedPapers/2024/132307/

Liu, Y., Zhang, Y., & Peng, X. (2024). Textual analysis and gold futures price forecasting: Evidence from the Chinese market. *Finance Research Letters*, 106116. https://doi.org/10.1016/j.frl.2024.106116

Makala, D., & Li, Z. (2021). Prediction of gold price with ARIMA and SVM. *Journal of Physics Conference Series*, *1767*(1), 012022. https://doi.org/10.1088/1742-6596/1767/1/012022

Man, Y. C., & Kynor, R. (2023). A review of gold pricing: real interest rate and U.S. dollar index. *Journal of Student Research*, *12*(3). https://doi.org/10.47611/jsrhs.v12i3.5028

Nagata, A. B., Hidajat, M. S., Wibowo, D. A., Widyatmoko, W., & Yaacob, N. B. M. (2025). Predicting gold price movement using Long Short-Term Memory model. *Journal of Applied Intelligent System*, *9*(1), 19–28. https://doi.org/10.62411/jais.v9i1.10305

Ok, E., & Emmanuel, M. (2025). XGBOOST Model Optimization Techniques. *ResearchGate*. https://www.researchgate.net/publication/390137875_XGBoost_Model_Optimization_Techniques

Omoseebi, A., Ola, G., & Tyler, J. (2025). Data preparation and feature engineering. *ResearchGate*. https://www.researchgate.net/publication/389860294_Data_Preparation_and_Feature_Engineering

Pillai, G. S., & Mary, M. I. (2024, March 24). *Gold price forecast based on the least square support vector machine*. https://ijisae.org/index.php/IJISAE/article/view/5300

Plevris, V., Solorzano, G., Bakas, N., & Seghier, M. B. (2022). Investigation of performance metrics in regression analysis and machine learning-based prediction models. *8th European Congress on Computational Methods in Applied Sciences and Engineering*. https://doi.org/10.23967/eccomas.2022.155

Putatunda, S., & R, K. (2020). A Modified Bayesian Optimization based Hyper-Parameter Tuning Approach for Extreme Gradient Boosting. *ResearchGate*. https://doi.org/10.48550/arXiv.2004.05041

Qiu, T., Zheng, B., & Chen, G. (2010). Financial networks with static and dynamic thresholds. *New Journal of Physics*, *12*(4), 043057. https://doi.org/10.1088/1367-2630/12/4/043057

Sathyanarayana, S., & Mohanasundaram, T. (2025). The surge in gold price volatility: macroeconomic drivers, geopolitical risk, and market dynamics. *IRA-International Journal of Management & Social Sciences (ISSN 2455-2267)*, *21*(1), 15. https://doi.org/10.21013/jmss.v21.n1.p2

Schäfer, M., Müller, T., & Fernandez, H. (2025). Deep Learning for Stock Performance Prediction: A Sharpe Ratio-Optimized Approach. *Deleted Journal*, *2*(2). https://doi.org/10.62177/apemr.v2i2.210

Schröer, C., Kruse, F., & Gómez, J. M. (2021). A Systematic Literature Review on Applying CRISP-DM Process model. *Procedia Computer Science*, *181*, 526–534. https://doi.org/10.1016/j.procs.2021.01.199

Setyowibowo, S., As'ad, M., Sujito, S., & Farida, E. (2022). Forecasting of Daily Gold Price using ARIMA-GARCH Hybrid Model. *Jurnal Ekonomi Pembangunan*, *19*(2), 257–270. https://doi.org/10.29259/jep.v19i2.13903

Shahare, F. F. (2017, June 15). Sentiment analysis for the news data based on the social media (pp. 1365–1370). https://scispace.com/papers/sentiment-analysis-for-the-news-data-based-on-the-social-zbpcke36hx

Shimaoka, A. M., Ferreira, R. C., & Goldman, A. (2024). The evolution of CRISP-DM for Data Science: Methods, Processes and Frameworks. https://www.researchgate.net/publication/385280269

Simakova, J. (2011). Analysis of the Relationship between Oil and Gold Prices. *ResearchGate*. https://www.researchgate.net/publication/266005958_Analysis_of_the_Relationship_between_Oil_and_Gold_Prices

Smales, L. A. (2014). News sentiment in the gold futures market. *Journal of Banking & Finance*, *49*, 275–286. https://doi.org/10.1016/j.jbankfin.2014.09.006

Srivastava, V., & Sikroria, R. (2024). AI and algorithmic trading: a study on predictive accuracy and market efficiency in fintech applications. *ShodhKosh Journal of Visual and Performing Arts*, *5*(1). https://doi.org/10.29121/shodhkosh.v5.i1.2024.2797

Sun, G., & Deng, S. (2025). Financial time series forecasting: A comparison between traditional methods and AI-driven techniques. *Journal of Computer Signal and System Research, 2*(2), 86–93. https://www.researchgate.net/publication/390275409_Financial_Time_Series_Forecasting_A_Comparison_Between_Traditional_Methods_and_AI-Driven_Techniques

Taj, S., Shaikh, B. B., & Meghji, A. F. (2019). Sentiment Analysis of News Articles: A Lexicon based Approach. *2019 2nd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, 1–5. https://doi.org/10.1109/icomet.2019.8673428

Taneva-Angelova, G., Raychev, S., & Ilieva, G. (2025). A Framework for Gold Price Prediction Combining Classical and Intelligent Methods with Financial, Economic, and Sentiment Data Fusion. *International Journal of Financial Studies*, *13*(2), 102. https://doi.org/10.3390/ijfs13020102

Vuong, P. H., Dat, T. T., Mai, T. K., Uyen, P. H., & Bao, P. (2021). Stock-Price forecasting based on XGBOOST and LSTM. *Computer Systems Science and Engineering*, *40*(1), 237–246. https://doi.org/10.32604/csse.2022.017685

Wang, X., Hu, R., & Chen, X. (2025, February 15). *A comparative study on gold future forecasting based on time series large models and classic models*. https://computerfraudsecurity.com/index.php/journal/article/view/367

Yang, J.-h., & Dou, W. 1 *(Jan. 2014)* - A new prediction method of gold price*: EMD-PSO-SVM - Volume 9 Number -Journal of Software*. (n.d.). https://www.jsoftware.us/show-50-491-1.html

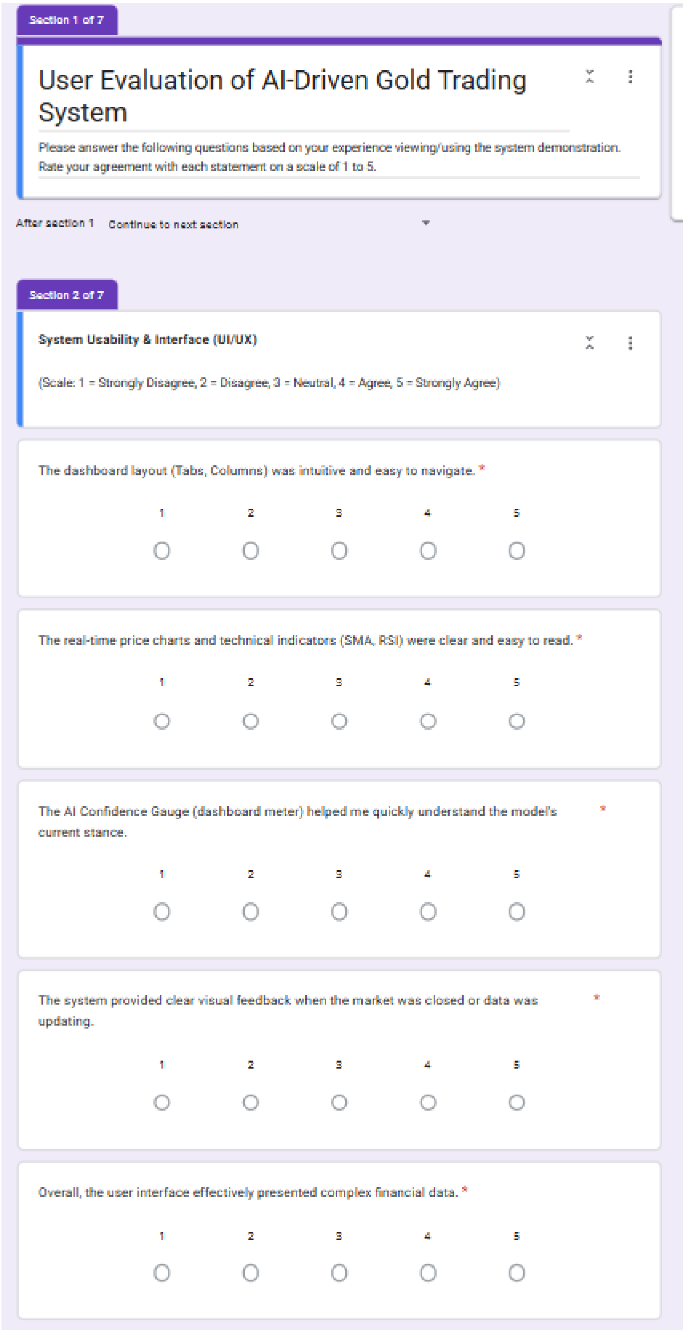
Yousaf, M., Tariq, M., Jabbar, A., & Jalil, S. Q. (2025). A comprehensive survey of cryptocurrency forecasting: Methods, trends, and challenges.

https://www.researchgate.net/publication/391553786_A_Comprehensive_Survey_of_Cryptocurrency_Forecasting_Methods_Trends_and_Challenges

Yousef, I., & Shehadeh, E. (2020). The impact of COVID-19 on gold price volatility. *International Journal of Economics and Business Administration*, *VIII*(Issue 4), 353–364. https://doi.org/10.35808/ijeba/592

Zhang, R. (2024). Gold Price Relative Return Prediction with Machine Learning Models. *www.scitepress.org*. https://www.scitepress.org/publishedPapers/2024/135287/pdf/index.html

APPENDIX A: User Evaluation



Section 1 of 7

User Evaluation of AI-Driven Gold Trading System

Please answer the following questions based on your experience viewing/using the system demonstration. Rate your agreement with each statement on a scale of 1 to 5.

After section 1 Continue to next section

Section 2 of 7

System Usability & Interface (UI/UX)

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

The dashboard layout (Tabs, Columns) was intuitive and easy to navigate. *

1 2 3 4 5

The real-time price charts and technical indicators (SMA, RSI) were clear and easy to read. *

1 2 3 4 5

The AI Confidence Gauge (dashboard meter) helped me quickly understand the model's current stance. *

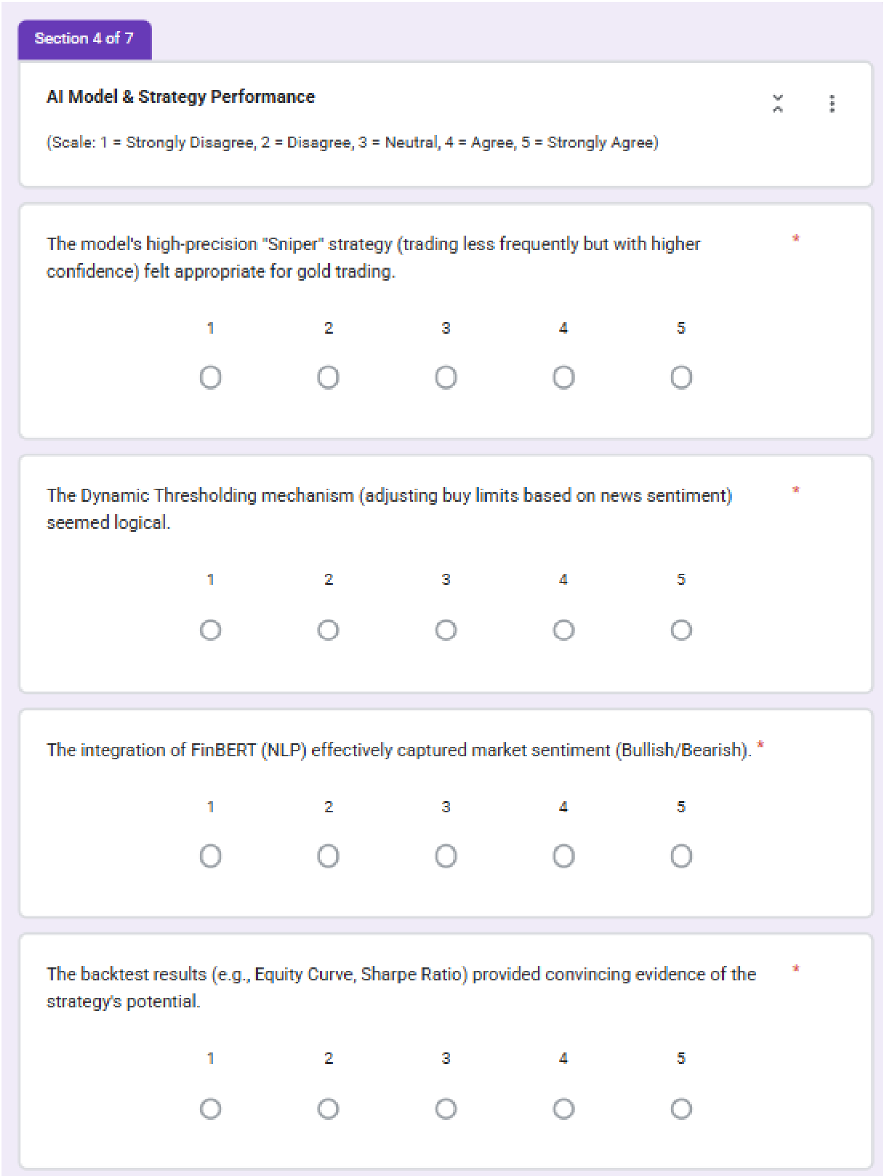
1 2 3 4 5

The system provided clear visual feedback when the market was closed or data was updating. *

1 2 3 4 5

Overall, the user interface effectively presented complex financial data. *

1 2 3 4 5



Section 4 of 7

AI Model & Strategy Performance

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

The model's high-precision "Sniper" strategy (trading less frequently but with higher confidence) felt appropriate for gold trading. *

1 2 3 4 5

The Dynamic Thresholding mechanism (adjusting buy limits based on news sentiment) seemed logical. *

1 2 3 4 5

The integration of FinBERT (NLP) effectively captured market sentiment (Bullish/Bearish). *

1 2 3 4 5

The backtest results (e.g., Equity Curve, Sharpe Ratio) provided convincing evidence of the strategy's potential. *

1 2 3 4 5

Section 5 of 7

Interpretability & Trust (Explainable AI)

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

The "Factor Analysis" table helped me understand why the AI made a specific decision. *

1	2	3	4	5
○	○	○	○	○

The "Feature Importance" chart clearly showed which technical indicators the model values most. *

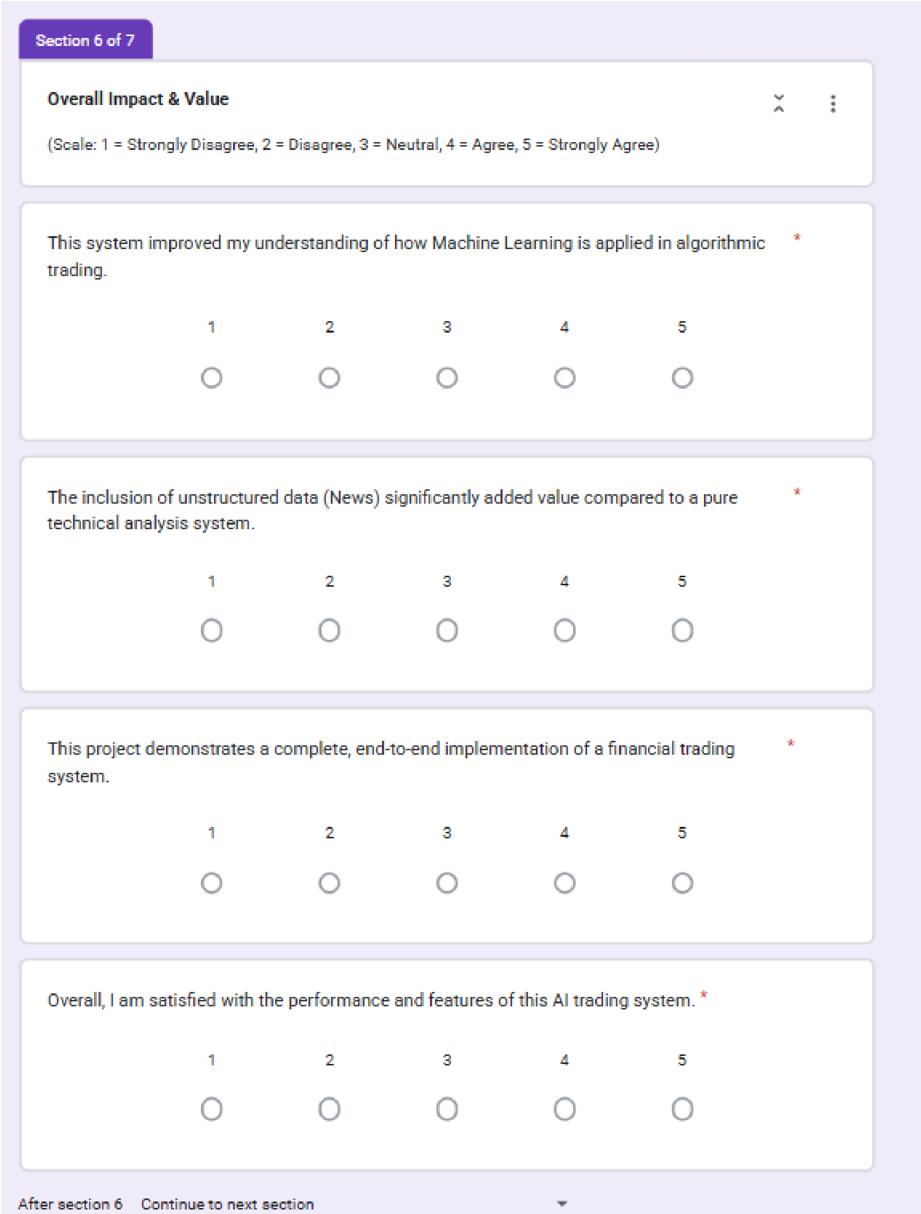
1	2	3	4	5
○	○	○	○	○

I felt the system was transparent, not a "black box" (i.e., I understood the logic behind the trades). *

1	2	3	4	5
○	○	○	○	○

I would trust this system's signals to assist in making trading decisions. *

1	2	3	4	5
○	○	○	○	○


Section 6 of 7

Overall Impact & Value

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

This system improved my understanding of how Machine Learning is applied in algorithmic trading. *

1 2 3 4 5

The inclusion of unstructured data (News) significantly added value compared to a pure technical analysis system. *

1 2 3 4 5

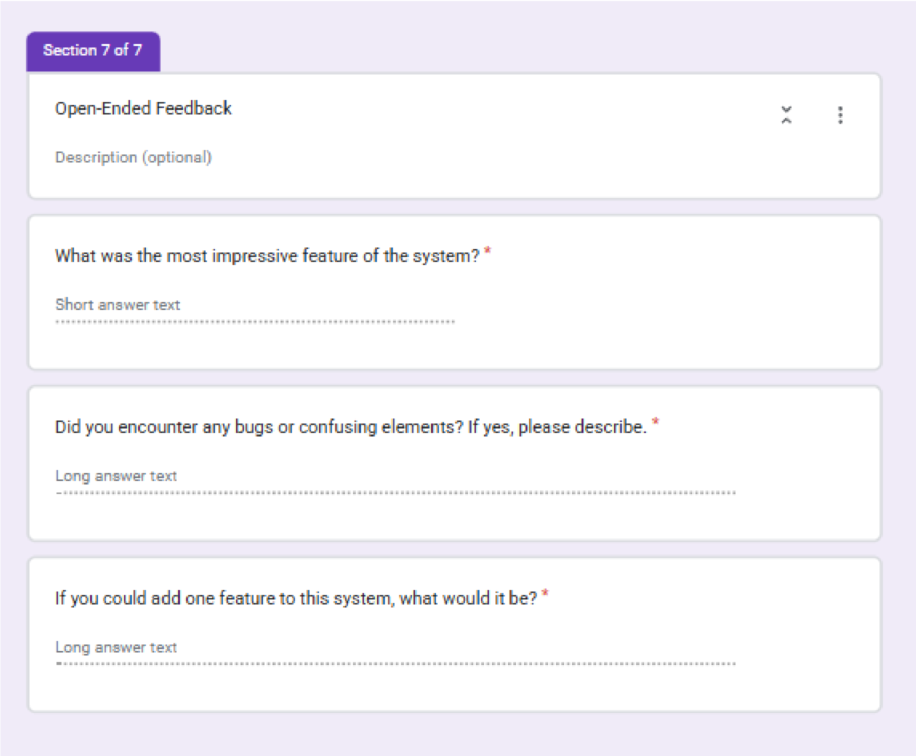
This project demonstrates a complete, end-to-end implementation of a financial trading system. *

1 2 3 4 5

Overall, I am satisfied with the performance and features of this AI trading system. *

1 2 3 4 5

After section 6 Continue to next section


Section 7 of 7

Open-Ended Feedback

Description (optional)

What was the most impressive feature of the system? *

Short answer text

Did you encounter any bugs or confusing elements? If yes, please describe. *

Long answer text

If you could add one feature to this system, what would it be? *

Long answer text

Section 3 of 7

Functionality & Robustness

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

The system successfully automated the data fetching and feature engineering processes without manual intervention. *

1	2	3	4	5
○	○	○	○	○

The News Service updated consistently and provided relevant headlines. *

1	2	3	4	5
○	○	○	○	○

The system handled special market conditions (e.g., weekends, holidays) logically by pausing or waiting. *

1	2	3	4	5
○	○	○	○	○

The simulation ran smoothly without crashing or freezing during the test period. *

1	2	3	4	5
○	○	○	○	○

APPENDIX B: Git Hub Link

https://github.com/jiaxuan10/gold-ml.git